

Kapitel 1

I. Rationally Convex Domains

Domaines convexes par rapport aux fonctions rationnelles

Journal of Science of the Hiroshima University (1936), p. 245–255

Introduction

Despite recent progress in the theory of analytic functions of several variables, several important matters remain more or less obscure, most notably: the kind of domains for which the theorem of Runge or those of Mr. P. Cousin remain valid, the relationship between the convexity of Mr. F. Hartogs and that of Messrs. H. Cartan and P. Thullen¹; and there are intimate relations between these matters. The present memoir and those which will follow are meant to treat these problems.

Now, I have noticed that one can sometimes reduce the difficulty of these problems by raising suitably the dimension of the spaces in which one works. In the present memoir, realising this general idea in a special case, I shall establish a principle which, so to speak, reduces the study of the domains of the title to cylindrical domains of higher dimension. (For the concrete form of this principle, see Problem I of No. 1.)

Once the principle is established, one can deduce that the theorem of Mr. P. Cousin concerning prescribed poles remains valid for the domains of the title. (For the exact form of this result, see Theorem I of No. 5.) The converse is also true. I shall actually prove these theorems simultaneously by a process of induction. Using the above principle, one also recovers immediately the Runge theorem for these domains, which was expounded by Mr. A. Weil².

Thus, I shall be concerned, in the present memoir, with the interior of domains which are convex with respect to rational functions; this will enable me at the same time to investigate, under less restrictive hypotheses than hitherto, some lemmas which are indispensable to me.³

1.1 Definitions

In the space $((x))$ of n complex variables $x_1, \dots, x_n$, let us consider the region Δ defined by

$$\Delta: \quad x_i \in X_i, \quad R_j((x)) \in Y_j \quad (i = 1, 2, \dots, n; \quad j = 1, 2, \dots, v), \quad (\text{A})$$

¹See the book of Messrs H. Behnke and P. Thullen: *Theorie der Funktionen mehrerer komplexer Veränderlichen*, in particular pages 54, 68, 79.

²Sur les séries de polynômes de deux variables complexes, C. R. Acad. Sci. Paris, 1932. *Intégrale de Cauchy et les fonctions de plusieurs variables*, Math. Annalen, 1935.

³As for the generalisation of the theorem of Mr. P. Cousin, I think that one can do this using the integral of Mr. A. Weil cited above.

where X_i, Y_j are univalent (schlicht) bounded domains⁴ in the plane, and $R_j((x))$ are rational functions. For simplicity, we shall say that any region which can be described in such a way belongs to the class $(\mathfrak{Q}_0)$. Given a region of $(\mathfrak{Q}_0)$ in the space $((x))$, we shall call the smallest number of rational functions of the n variables x_i —the functions x_i being excluded—which are needed to define the region, the *order* of the region. The order of the region Δ above is $\leq v$; every region of $(\mathfrak{Q}_0)$ of order 0 is a cylindrical domain.

It is convenient to give here, in concrete form, the problems to be dealt with.

Problem I. Let us introduce new complex variables $y_1, y_2, \dots, y_v$ and consider, in the space $((x, y))$, the variety Σ defined by

$$y_j = R_j((x)), \quad ((x)) \in \Delta, \quad (j = 1, 2, \dots, v). \quad (\Sigma)$$

The boundary of the variety Σ is entirely situated on the boundary of the cylindrical domain

$$(C): \quad x_i \in X_i, \quad y_j \in Y_j \quad (i = 1, 2, \dots, n; j = 1, 2, \dots, v). \quad (C)$$

Let $f((x))$ be a holomorphic function of the n variables x_i in Δ ;⁵ if we regard it as a function of $n + v$ variables x_i, y_j , it is holomorphic at every point $M = ((x^0, y^0))$ on Σ since $((x^0))$ is necessarily a point of Δ .

Under these circumstances, construct a holomorphic function on (C') having the value $f(M)$ for any point M on the portion of Σ lying in (C') , (C') being a given domain in the interior of (C) .

We shall call this *Problem I of order v* , where v is half the difference between the dimension of the space $((x, y))$ and that of the variety Σ . In this case, the order starts with the value 1.

Problem II. We prescribe poles (p) in the region Δ by the usual method, that is to say: we make correspond to any point P of Δ a polycylinder (y) and a function $g((x))$ which is meromorphic on (y) in such a way that on any part common to several of the (y) , the functions $g((x))$ attached to them are mutually equivalent under subtraction (the difference is holomorphic); we define the poles (p) locally as those of the functions $g((x))$. Here is the problem:

Find a meromorphic function with the poles (p) on a region Δ' given a priori in the interior of Δ .

This will be called *Problem II of order μ* , μ being the order of Δ . We know, thanks to Mr. P. Cousin⁶, that Problem II of order zero is solvable.

1.2 Reduction of Problem I

We shall first prove the following:

If Problems I and II are solvable for all orders $< v$, Problem I is solvable for order v .

We suppose that we can solve Problems I and II for any order $< v$. We take Problem I of order v in the exact form in which it is given in the preceding section. In the space $(x_1, x_2, \dots, x_n, y_v)$, consider the region

$$(D): \quad x_i \in X_i, \quad y_v \in Y_v, \quad R_k((x)) \in Y_k \quad (i = 1, 2, \dots, n; k = 1, 2, \dots, v - 1), \quad (D)$$

and the variety

$$(S): \quad y_v = R_v((x)), \quad ((x)) \in \Delta. \quad (S)$$

We find that the boundary of the variety S is supported by that of D .

⁴In what follows, an open set will be called domain or region; no distinction is made between when it is known to be connected and when it is not; it is understood that the regions (domains) of this memoir are, without exception, univalent.

⁵When the region Δ consists of several connected components, $f((x))$ can, naturally, be composed of distinct analytic functions.

⁶Acta, 1895.

To each point of the region D , we attach a function $g((x), y_v)$ of the $n + 1$ variables x_i, y_v in such a way that:

1°. For any point P outside the variety S ,

$$g((x), y_v) = 0.$$

2°. For any point M on S ,

$$g((x), y_v) = \frac{f((x))}{y_v - \frac{P((x))}{Q((x))}},$$

where $P((x))$ and $Q((x))$ are polynomials in x_i , relatively prime to each other (in the algebraic sense) and such that $R_v((x)) = P((x))/Q((x))$.

We are going to restrict each of the functions g to a polycylinder (y) . For any point P outside of S , it is sufficient to take a (y) with center P which does not contain any point of S . Let then M be any point of S ; there are two things to which we must pay attention: 1°. We must choose a polycylinder around M on which the function f is holomorphic. This is possible since f is holomorphic on the region Δ , and, on the other hand, if $((x^0), y_v^0)$ are the coordinates of M , $((x^0))$ belongs to Δ . 2°. Let T be the part of the variety $Q((x))y_v - P((x)) = 0$ in the interior of D ; for any point $M' = ((x'), y'_v)$ on T , there are two cases possible: either M' is a point of S which implies that $((x'))$ belongs to Δ , or $((x'))$ is a point of indeterminacy of $R_v((x))$, when $((x'))$ can never fall inside Δ .

It is necessary to avoid points M' with the second property. This can always be achieved, because, in view of what we have seen, these points are at a positive distance from M . For any point M on S , we attach a polycylinder (y) with center M satisfying these conditions.

The functions $g((x), y_v)$ defined thus are meromorphic in their own polycylinders (y) and satisfy mutually the equivalence condition. These functions therefore define poles (p) in D . These poles all lie on S . Now, the region D of the space $((x), y_v)$ belongs evidently to (Ω_0) and its order is no bigger than $v - 1$. Hence, by hypothesis, there exists a function $G((x), y_v)$, meromorphic in D' and having (p) as poles, D' being a region completely interior to D , whose exact form will be given later.

Let us set

$$\varphi((x), y_v) = G((x), y_v) \cdot [Q((x))y_v - P((x))].$$

It is verified immediately that the function φ so obtained is holomorphic in D' , and that, in addition,

$$\varphi[(x), R_v((x))] = f((x)),$$

for any point on S lying in D' .⁷

For D' we take the region

$$(D'): \quad x_i \in X'_i, \quad y_v \in Y'_v, \quad R_k((x)) \in Y'_k \quad (i = 1, 2, \dots, n; \quad k = 1, 2, \dots, v - 1), \quad (D')$$

X'_i, Y'_j being $(n + v)$ domains completely interior to X_i, Y_j respectively, and this, in such a way that the cylindrical domain

$$x_i \in X'_i, \quad y_j \in Y'_j \quad (i = 1, 2, \dots, n; \quad j = 1, 2, \dots, v)$$

contains the given domain (C') , together with its boundary.

We want to construct a function F of the $(n + v)$ variables x_i, y_j , holomorphic in (C') , such that

$$F[x_1, \dots, x_n, R_1((x)), \dots, R_{v-1}((x)), y_v] = \varphi(x_1, \dots, x_n, y_v)$$

⁷Erratum, taken from Memoir III: The open set D' is not necessarily completely interior to D . This is a phenomenon which arises because of points of indeterminacy. To avoid this, we proceed as follows: 1°. The reasoning of Memoir I is correct for polynomials. 2°. One can therefore verify the theorem in No. 4 and Theorem I in complete analogy with No. 5 of Memoir II; note that Theorem I of Memoir II is obvious for rational functions. 3°. One can then establish Theorem II without modifying the proof given in No. 5.

on the part of the variety $y_k = R_k((x))$ ($k = 1, 2, \dots, v-1$) inside (C') . Such a function $F((x, y))$ certainly exists by our hypothesis; for this is just a Problem I of order $v-1$.

Now, for any point $M = ((x, R))$ on the variety Σ inside (C') , we have

$$F[x_1, \dots, x_n, R_1((x)), \dots, R_v((x))] = \varphi[x_1, \dots, x_n, R_v((x))] = f(x_1, \dots, x_n).$$

The function $F((x, y))$ is therefore the desired solution.

Let us remark, by the way, that Problem I of order 1 is solvable.

1.3 Reduction of Problem II. Conclusion

We now consider Problem II. To apply the classical method of Mr. P. Cousin to the present case, one must overcome the following difficulty. One has to construct a function having a given jump in our regions; more precisely:

Problem A. Consider the region Δ of No. 1. In the domain X_1 of the x_1 -plane, we draw a simple rectifiable Jordan arc L with extremities a and b . Let T be the set of points $((x))$ satisfying $x_1 \in L$, $((x)) \in \Delta$. Let Δ' be an arbitrary region completely interior to Δ , T' the part of T contained in Δ' .

Under these circumstances, given a function $f((x))$ of the n variables x_i , well-defined and holomorphic in the neighbourhood of any point of T , find a function $\varphi((x))$ of the n variables x_i , well-defined and holomorphic at every point of Δ' except on T' , and admitting a jump on T' of the following kind:

1°. If one crosses T' excepting the points $x_1 = a, b$, $\varphi((x))$ can be analytically continued, and this in such a way that

$$\varphi((x)) - \psi((x)) = \pm f((x)),$$

$\psi((x))$ being the function obtained by continuation, the sign $+$ being valid for $((x))$ lying to the left of T' ; by this we mean that the point x_1 lies to the left of the curve L described from a to b .

2°. In the neighbourhood of any point on $x_1 = b$ in Δ' , the analytic continuation of the function

$$\varphi((x)) - \frac{f((x))}{2\pi i} \log(b - x_1)$$

remains holomorphic. For $x_1 = a$, one has only to replace the logarithm by $-\log(a - x_1)$.

We shall call this *Problem A of order μ* when μ is the order of Δ . We shall see that:

If it is possible to solve Problem I of any order $< v$, one can solve Problem A of any order $< v$.

In fact, we shall prove that Problem A formulated above necessarily has solutions under the hypothesis that any Problem I of order $< v$ is solvable. This is sufficient, for, although the order of the problem may be less than v , we can put any Problem A of order v in this form. For the sake of clarity we suppose that the given region Δ' is of the form

$$(A'): \quad x_i \in X'_i, \quad R_j((x)) \in Y'_j \quad (i = 1, 2, \dots, n; \quad j = 1, 2, \dots, v). \quad (A')$$

To the region Δ , we make correspond the cylindrical domain (C) and the variety Σ in the space $((x, y))$ by the method of No. 1. Let (C') , Σ' be those which correspond to Δ' .

The function $f((x))$ being well-defined and holomorphic in the neighbourhood of any point of T' , we can find, given any region completely interior to Δ , a domain G containing L in such a way that $f((x))$ is holomorphic at any common point of the given region and the domain $x_1 \in G$. We suppose, for the sake of simplicity, that this is the case for Δ itself; this does not make it less general, because Δ' and its boundary are contained in Δ . We also suppose that $L \subset G' \subset G \subset X_1$, where G' is again a given domain completely interior to G ; this again can be achieved without loss of generality.

Under these conditions, we want to find a function $F((x, y))$ of the $n + v$ variables x_i, y_j , holomorphic on the common part of (C') and $x_1 \in G'$, and having the value $f((x))$ for any point $((x, y))$ of Σ' belonging to $x_1 \in G'$. This being a Problem I of order v , $F((x, y))$ exists by hypothesis.

Let us now consider the integral

$$\Phi((x, y)) = \frac{1}{2\pi i} \int_L \frac{F(t, x_2, \dots, x_n, y_1, \dots, y_v)}{t - x_1} dt \quad (1.1)$$

taken following the curve L in the x_1 -plane from a to b . The region where the function $\Phi((x, y))$ is defined is just the cylindrical domain

$$x_1 \in G', \quad x_k \in X'_k, \quad y_j \in Y'_j \quad (k = 2, 3, \dots, n; \quad j = 1, 2, \dots, v),$$

where G' contains L . The properties of the integral $\Phi((x, y))$ are therefore well known, thanks to Mr. P. Cousin⁸. They are as follows.

Let V' be the set of points such that $x_1 \in L$, $((x, y)) \in (C')$; the integral $\Phi((x, y))$ has a meaning and represents a holomorphic function at every point of (C') except on V' where the function has a certain property; to state this property, it is sufficient to modify conditions 1°, 2° to be satisfied by the required function $\varphi((x))$ by replacing

$$\Delta', \quad T', \quad \varphi((x)), \quad \psi((x)), \quad f((x))$$

by

$$(C'), \quad V', \quad \Phi((x, y)), \quad \Psi((x, y)), \quad F((x, y))$$

respectively.

Now, when $((x))$ runs over Δ' , the point M of the $((x, y))$ -space having coordinates $x_i = x_i$, $y_j = R_j((x))$ describes the variety Σ' . Let us therefore set

$$\varphi((x)) = \Phi(M).$$

The function $\varphi((x))$ is then well-defined on Δ' except on T' and represents a holomorphic function of the n variables x_i at each of these points. Since, in addition, $f((x)) = F(M)$ near V' for any point M on Σ' , the required properties of $\varphi((x))$ follow immediately from the properties of $\Phi((x, y))$. $\varphi((x))$ is therefore a solution of the problem. Q.E.D.

It remains for us to show that we can solve any Problem II of order $< v$ under the assumption that any Problem A of order $< v$ has solutions. But now, the classical method being applicable almost literally⁹, we shall content ourselves with referring the reader to this¹⁰.

We thus have the proposition that if every Problem I of order $< v$ is solvable, one can solve any Problem II of order $< v$.

From the two auxiliary propositions, we obtain:

Conclusion. Problems I, II of No. 1 are always solvable.

1.4 Remarks on Expansions

In what follows, we shall be concerned with completing the preceding proposition. To do this, we shall start by using some known facts about expansions of functions.

⁸See e.g. Osgood: *Lehrbuch der Funktionentheorie* II (1929), § 22 of Chap. III.

⁹All the modifications necessary are simply of the following form: instead of considering a cylindrical domain (d) : $x_p \in A_p$, $x_q \in \Delta_q$ ($p = 1, 2, \dots, \lambda$; $q = \lambda + 1, \lambda + 2, \dots, n$) in the given cylindrical domain (D) : $x_i \in X_i$ ($i = 1, 2, \dots, n$), one should consider, in the present case, the part common to (d) and the given region Δ , (D) being now an arbitrary cylindrical domain containing Δ "in its interior".

¹⁰Osgood: § 23, 24 of Chap. III.

1°. Let us first consider in the space $((x))$ a domain Δ of the form

$$(\Delta): |x_i| < r_0, |R_j((x))| < 1 \quad (i = 1, 2, \dots, n; j = 1, 2, \dots, v), \quad (A)$$

where the $R_j((x))$ are rational functions of the x_i , r_0 being a positive constant. If the set of points defined by these inequalities is composed of several connected components, we understand by Δ one of these components. For such a domain, one knows by the work of Mr. A. Weil that every holomorphic function can be expanded in a series of rational functions¹¹. This is an immediate consequence of the principle we have just established.

In fact, given a holomorphic function $f((x))$ in the domain Δ , we can find, for any $0 < r < r_0$, $0 < \rho < 1$, a new function $F((x, y))$ holomorphic in the polycylinder

$$(C'): |x_i| < r, |y_j| < \rho \quad (i = 1, 2, \dots, n; j = 1, 2, \dots, v), \quad (C')$$

such that for any point M of (C') having coordinates of the form $x_i = x_i, y_j = R_j((x))$, we have $F(M) = f((x))$.

The function $F((x, y))$ can be expanded in a Taylor series about the origin, valid at least on (C') . From this, if we set $y_j = R_j((x))$ in the expansion, we obtain a series of rational functions converging uniformly to $f((x))$ in the interior of Δ' ,

$$|x_i| < r, |R_j((x))| < \rho, ((x)) \in \Delta \quad (i = 1, 2, \dots, n; j = 1, 2, \dots, v).$$

More precisely, each term of the series is a polynomial in $x_i, R_j((x))$. From this, one obtains an expansion of the same nature for Δ itself.

2°. Let us now consider a bounded domain D in the space $((x))$ which is convex with respect to a class $\mathfrak{R}$ of rational functions of the n variables x_i , holomorphic in D .¹²

while, on the other hand, $|f| < 1$ on D' , let (y) be a polycylinder around M , interior to D , and on which $|f| > 1$. Thus, every point M of the closed set I is the centre of a (y) having the above property. We can therefore cover I by a finite number of these (y) by the Borel-Lebesgue lemma.

Let $f_1, f_2, \dots, f_v$ be the functions of $\mathfrak{R}$ corresponding to these (y) . Consider, in D , the set of points satisfying

$$|f_j((x))| < 1 \quad (j = 1, 2, \dots, v).$$

Among the connected components of this set, there is one which contains D' , which we denote by Δ . The boundary of Δ is still contained in D because the domain Δ cannot extend beyond I .

Thus, to any domain D' completely interior to D , we can make correspond a domain Δ of the above kind. This remains true even for an unbounded domain so long as the domain contains only points of finite distance as is immediately verified. Thus, in view of the preceding considerations, we have the following proposition:

Let D be a domain in the space $((x))$ [in the proper sense, i.e. in the finite part of the space] which is convex with respect to a class of rational functions of the n variables x_i , holomorphic in D . Every function holomorphic in D can be expanded in a series of rational functions converging uniformly in the interior of D , and this, in such a way that the general term of the series is a polynomial in the x_i and functions of $\mathfrak{R}$.

¹¹Work cited above.

¹² D is then necessarily univalent. Let D' be an arbitrary domain completely interior to D , and let δ be the "minimal distance" of D' relative to D , $\delta > 0$. We consider in D the set of points E whose "boundary distance" relative to D is equal to $\delta/2$. To each point M on E , there is always at least one function f of $\mathfrak{R}$ such that $|f(M)| > 1$,

dans C' qui induise, en tout point de $\Sigma \cap C'$, une fonction f donnée, holomorphe dans Σ . On appelle problème II d'ordre v le problème qui consiste à trouver, pour tout ouvert C' relativement compact de C , une fonction méromorphe dans $\Sigma \cap C'$ qui admette des parties principales données dans Σ . (Dans ces formulations, v désigne le nombre des fonctions R_j .) Alors Oka prouve :

(i) si les problèmes I et II ont une solution pour tout entier $v < k$, le problème I a une solution pour $v = k$.

(ii) si le problème I a une solution pour tout entier $v < k$, il en est de même du problème II.

La conclusion est évidemment que les problèmes I et II ont toujours une solution.

Pour prouver (ii), Oka se sert de la méthode de Cousin, qui utilise l'intégrale de Cauchy.

Il reste ensuite à faire un passage à la limite pour obtenir les théorèmes I et II (le lecteur notera que c'est la solution du problème II qui conduit au théorème I, et vice-versa).

Le théorème I avait été annoncé sans démonstration, au moins pour $n = 2$, par H. Cartan dès 1934 (Comptes Rendus Ac. Sciences Paris, 199, p. 925–927).

Le théorème II entraîne aussitôt que toute fonction holomorphe dans Δ est développable en série de fractions rationnelles qui converge uniformément sur tout compact de Δ ; plus précisément, ces fractions rationnelles sont des polynômes en les x_i et les $R_j(x)$. On en déduit : dans tout domaine convexe par rapport à une classe $\mathfrak{K}$ de fractions rationnelles, toute fonction holomorphe est développable en série de polynômes en les coordonnées et les fonctions de $\mathfrak{K}$.

Kapitel 2

II. Domains of Holomorphy

Domaines d'holomorphie

Journal of Science of the Hiroshima University (1937), p. 115–130

Introduction

In the preceding memoir¹, I dealt with the subject of domains convex with respect to rational functions. I shall now examine the same question relative to holomorphic functions; and this will be done by applying the same idea, that is to say, by going over to larger spaces.

Given a bounded univalent region in the space of several complex variables which is convex relative to a finite number of holomorphic functions, we construct from this a variety Σ in a larger space by the procedure adopted previously. It is at this variety Σ and the nature of its convexity that we shall look more precisely.

As a consequence we shall find, with the help of the theorems established in the preceding memoir, that the variety Σ is, in some sense, convex with respect to polynomials. (See Theorem I in No. 4.) In our opinion, this is a fundamental fact about domains of holomorphy. From this, in view of a well-known theorem of Messrs. H. Cartan and P. Thullen², one can easily give an affirmative solution to one of the unsolved problems³ of the theory of the title; namely, that the theorem of Mr. P. Cousin concerning functions with given poles remains valid for univalent, bounded domains of holomorphy.

2.1 Generalities

Consider the space $((x))$ generated by n complex variables $x_1, x_2, \dots, x_n$. We shall first give some definitions. A region is said to belong to the class $(\mathfrak{P})$ if it can be defined as the set of points satisfying

$$|x_i| < r_i, \quad |\Phi_j((x))| < 1 \quad (i = 1, 2, \dots, n; j = 1, 2, \dots, v),$$

the r_i being positive constants and the $\Phi_j((x))$ polynomials in the x_i .

We say that a closed set F belongs to the class $(\mathfrak{P})$, if we can find, in the class $(\mathfrak{P})$, a decreasing sequence of regions having F as its limit.

Given a bounded set E , consider $\mathfrak{S}$, the set common to all sets of class $(\mathfrak{P})$ which contain E . We shall call this the smallest set of class $(\mathfrak{P})$ containing E , because $\mathfrak{S}$ belongs also to the class $(\mathfrak{P})$.

¹This journal 6, (1936).

²Paper cited previously.

³See the book of Messrs H. Behnke and P. Thullen, 68, cited previously.

In fact, $\mathfrak{S}$ is necessarily a closed set. Let (C) be a polycylinder containing the given set E and its points of accumulation; let F be a set in the class $(\mathfrak{P})$ which is contained in (C) and which contains E , but otherwise arbitrary; $\mathfrak{S}$ is then the set common to all these F . For any positive number ρ , let us form the set A consisting of all points of the polycylinder (C) and its boundary for which the distance to $\mathfrak{S}$ is $> \rho$; A is necessarily a closed set.

Let M be an arbitrary point of A ; since M is an exterior point for a certain F , one can find a polynomial $P((x))$ such that we have: $|P((x))| > 1$ in a sufficiently small polycylinder (y) of centre M , and $|P((x))| < 1$ on $\mathfrak{S}$; then, by means of the Borel-Lebesgue lemma, one can cover A by a finite number of such polycylinders (y) ; let $\Phi_j((x))$, $(j = 1, 2, \dots, v)$, be the corresponding polynomials. Consider a region D of the form

$$((x)) \in (C), \quad |\Phi_j((x))| < 1 \quad (j = 1, 2, \dots, v).$$

The region D , which clearly belongs to the class $(\mathfrak{P})$ covers $\mathfrak{S}$ without containing any point having distance $> \rho$ from $\mathfrak{S}$; and this holds whatever be ρ . The set $\mathfrak{S}$ belongs therefore to the class $(\mathfrak{P})$. Q.E.D.

Let E be an arbitrary bounded set in the space $((x))$, and $\mathfrak{S}$ the smallest set of class $(\mathfrak{P})$ containing E . Let (σ_t) be a family of characteristic surfaces⁴ of the form

$$(\sigma_t) \quad f((x), t) = 0, \quad ((x)) \in U, \quad 0 < t < 1,$$

U being a domain in the space $((x))$ and $f((x), t)$ a single-valued function of the variables x_i and t defined for $((x)) \in U$, $0 < t < 1$, holomorphic at every point of this set, and such that for any value of t in the interval, $f((x), t)$ does not vanish identically.⁵

These surfaces can never be situated in such a manner as to have the following properties:

- 1°. For each characteristic surface of the family, its boundary does not enter into a neighbourhood V of $\mathfrak{S}$;
- 2°. None of the characteristic surfaces passes through a point of E or one of its points of accumulation;
- 3°. The characteristic surface (σ_0) passes through a point of $\mathfrak{S}$, while (σ_1) remains outside $\mathfrak{S}$.

Suppose, in fact, that a family (σ_t) of this nature exists. We can then find a domain T in the t -plane, containing the segment $(0, 1)$ and belonging to the class $(\mathfrak{P}_0)$ in the plane in such a way that all the above circumstances remain unchanged when we pass from the segment $(0, 1)$ to the domain T , at least if we shrink a little the given regions U and V . We may also suppose that the region V belongs to $(\mathfrak{P}_0)$ in the space $((x))$ since $\mathfrak{S}$ is a set in $(\mathfrak{P})$. The region (U, T) so constructed belongs to $(\mathfrak{P}_0)$ in the space $((x), t)$, and one can find on it a meromorphic function $G((x), t)$ of the variables x_i and t having as its poles $1/f((x), t)$, thanks to Theorem I of the preceding memoir.

Let (α) be the last characteristic which passes through points of $\mathfrak{S}$, when t traces the segment from 0 to 1 and M one of the points of $\mathfrak{S}$ on this characteristic. When t tends to α along the segment $(\alpha, 1)$, the value $G(M, t)$ becomes arbitrarily large since the function necessarily has the point (M, α) as a pole; on the other hand, the function $G((x), t)$, being regular at every point of the set $[E, (0, 1)]$ as well as its points of accumulation, is bounded on this set; one can therefore choose a value β near α on the segment $(\alpha, 1)$ in such a way that

$$\max_E |G(E, \beta)| < |G(M, \beta)|,$$

the first term meaning the upper bound of $|G((x), \beta)|$ on E .

⁴For characteristic surfaces (*charakteristische Flächenstücke*), see the book of Messrs Behnke and Thullen, cited previously, p. 2, 4.

⁵Erratum taken from Memoir III: One should add to the proposition the condition that $f((x), t)$ be a function which is single-valued and continuous on $((x)) \in \bar{U}$, $0 \leq t \leq 1$, where $\bar{U}$ denotes the closed domain consisting of U and its boundary; we are consequently always assuming that U is bounded.

Now, as the function $G((x), \beta)$ is regular at every point of the closed set $\mathfrak{S}$ (belonging to the class $(\mathfrak{P})$ in the space $((x))$), we can expand it in a series of polynomials in the neighbourhood of $\mathfrak{S}$ because of our studies in the preceding memoir⁶; from this follows the existence of a polynomial $\psi((x))$ in the x_i such that $\max_{\mathfrak{S}} |\psi(E)| < |\psi(M)|$, which is a contradiction since $\mathfrak{S}$ is minimal. A family of characteristics of this nature cannot therefore exist. Q.E.D.

Remark. We can replace polynomials by rational functions in the above considerations and define classes $(\mathfrak{R}_0)$ and $(\mathfrak{R}_1)$ and the smallest set in the class $(\mathfrak{R}_1)$. One can also check, without changing our method of argument, that this set also has the above property.

2.2 Class (H_0) . Functions $R_j(x_1)$

Let us consider in a region $\mathcal{G}$ of the space $((x))$ [assumed univalent and bounded] v holomorphic functions $f_j((x))$ of the variables x_i , with the help of which we form a set Δ of points as follows:

$$(\Delta): \quad |x_i| < r_i, \quad |f_j((x))| < 1 \quad (i = 1, 2, \dots, n; j = 1, 2, \dots, v), \quad (\Delta)$$

$$((x)) \in \mathcal{G},$$

the r_i being positive constants. We assume also that no point of accumulation of Δ lies on the boundary of $\mathcal{G}$; in other words that Δ is a closed set. We shall say that this closed set $\bar{\Delta}$, which corresponds to the region Δ of the preceding memoir, belongs to the class (H_0) .

As in the previous case, we introduce v new complex variables y_j and construct, in the $((x, y))$ -space, a closed set Σ of the form

$$(E): \quad y_j = f_j((x)), \quad ((x)) \in \bar{\Delta}, \quad (j = 1, 2, \dots, v); \quad (E)$$

one finds then that the whole boundary of the variety Σ lies on that of the polycylinder $|x_i| < r_i$, $|y_j| < 1$.

In this new space $((x, y))$ we consider the smallest set $\mathfrak{A}$ in the class $(\mathfrak{P})$ containing Σ , and we shall apply the general proposition we have just proved to this set $\mathfrak{A}$; in the first place, it is obvious that $\mathfrak{A}$ is contained in the closed polycylinder $|x_i| < r_i$, $|y_j| < 1$.

Let $\mathcal{B}$ be the image of $\mathfrak{A}$ projected onto the space $((x))$ ⁷, and $\mathfrak{C}$ that of projection on the x_1 -plane. $\mathfrak{A}$ being a closed set, so are $\mathcal{B}$ and $\mathfrak{C}$. We form the section of $\mathcal{B}$ by the characteristic plane $x_1 = x$ (constant) which we consider as a set of points in the space $(x_2, x_3, \dots, x_n)$ and denote by $\mathcal{B}_x$.⁸ $\mathcal{B}_x$ is a closed set. Let G_x be the section of $\mathcal{G}$; G_x is an open set. In the x_1 -plane, let us mark the set $\mathcal{Q}$ consisting of those points x for each of which $\mathcal{B}_x$ has at least one point $(x_2, x_3, \dots, x_n)$ outside of G_x . $\mathcal{B}$ being a closed set and $\mathcal{G}$ an open set, the set $\mathcal{Q}$, if it is not empty, is necessarily closed. Let (ω) be the region of the x_1 -plane having the points of $\mathcal{Q}$ and the point at infinity as exterior or boundary points, but containing all other points.

In the region (ω) so obtained, we shall define real functions $R_j(x_1)$ of the variable x_1 ($j = 1, 2, \dots, v$) as follows: If x_1 is not in $\mathfrak{C}$, we set $R_j(x_1) = 0$.

Let then x_1^0 be a point of $\mathfrak{C}$ in (ω) . Let $(x_2^0, \dots, x_n^0; \eta_1^0, \eta_2^0, \dots, \eta_v^0)$ be a point of $\mathfrak{A}$ for which $x_1 = x_1^0$, but otherwise arbitrary. Since x_1^0 is a point of (ω) , $(x_2^0, \dots, x_n^0)$ lies necessarily on $\mathcal{G}$ by very definition of (ω) . Consequently, the functions $f_j((x))$ being well-defined at this point, we can set

$$R_j(x_1^0) = \max |\eta_j - f_j(x_1^0, x_2^0, \dots, x_n^0)|, \quad (x_1^0, x_2^0, \dots, x_n^0; \eta_1, \eta_2, \dots, \eta_v) \in \mathfrak{A}.$$

⁶See No. 4, where we expounded this proposition on a domain, but since we did not use the connectedness hypothesis in the proof, the proposition remains valid for regions.

⁷This means that to each point $((x'), \eta')$ there corresponds at least one point $((x), \eta)$ of $\mathfrak{A}$ such that $((x)) = ((x'))$, and conversely.

⁸This means that if $(x_2, x_3, \dots, x_n)$ is a point of $\mathcal{B}_x$, the point $((x'))$ belongs necessarily; and in the contrary case, $((x'))$ does not belong to $\mathcal{B}_x$.

We shall show that the functions $R_j(x_1)$ have the following property:

Proposition. $R_j(x_1)$ are logarithmically subharmonic functions⁹.

Let us take the function $R_1(x_1)$, to fix our ideas. We start by showing that $\log R_1(x_1)$ is upper semi-continuous; it is sufficient to check this for $R_1(x_1)$ itself. To do this, it will be enough to affirm the semi-continuity on $\mathfrak{C}'$, $\mathfrak{C}'$ being the portion of $\mathfrak{C}$ contained in the region (ω) . Let, therefore, x_1^0 be an arbitrary point of $\mathfrak{C}'$, and let $x_1^{(1)}, x_1^{(2)}, \dots, x_1^{(p)}, \dots$ be a sequence of points of $\mathfrak{C}'$ tending towards x_1^0 and such that $\lim R_1(x_1^{(p)}) = \alpha$, where α is a constant; the sequence is otherwise arbitrary. It is sufficient to prove that $\alpha < R_1(x_1^0)$.

Now, to each $x_1^{(p)}$ in the sequence, there corresponds at least one point $M_p = ((x^{(p)}, \eta^{(p)}))$ of $\mathfrak{A}$ such that $R_1(x_1^{(p)}) = |\eta^{(p)} - f_1((x^{(p)}))|$; and we have a new sequence of points $M_1, M_2, \dots, M_p, \dots$; let $M_0 = ((x^0, \eta^0))$ be one of its limit points. We then have $x_1^0 = x_1^0$; in addition, $\mathfrak{A}$ being closed, M_0 is part of $\mathfrak{A}$. Letting $x_1^{(p)}$ tend towards x_1^0 , it follows from this that $\alpha = |\eta^0 - f_1(x_1^0, x_2^0, \dots, x_n^0)| < R_1(x_1^0)$; $R_1(x_1)$ is therefore upper semi-continuous.

We must now look at harmonic majorants of the function $\log R_1(x_1)$, the logarithm being real. Suppose that we have a circle (γ) , $|x_1 - x_1^0| < \rho$ in the interior¹⁰ of the domain of existence such that the arithmetic mean of the function $\log R_1(x_1)$ taken over the circumference in the sense of Mr. Lebesgue is smaller than its value at the centre. It will be sufficient to show that this hypothesis leads to a contradiction. Now, under the above hypothesis, by using the Poisson integral, one can easily find a holomorphic function $\psi(x_1)$ of the variable x_1 in the circle, whose absolute value is single-valued, continuous and non-zero on $|x_1 - x_1^0| < \rho$, and such that $R_1(x_1) |\psi(x_1)| < 1$ for any x_1 on the circumference, while, at the centre, $R_1(x_1^0) |\psi(x_1^0)| = 1$.

With this function $\psi(x_1)$, let us form the family of characteristic surfaces

$$(\sigma_t): [y_1 - f_1((x))] \psi(x_1) = e^{i\theta} (1 + t), \quad |x_1 - x_1^0| < \rho, ((x)) \in \mathcal{G}, \quad 0 < t < +\infty, \quad (\sigma_t)$$

θ being a certain argument which we shall determine later, and $i = \sqrt{-1}$. We see that both sides of the equation are holomorphic functions of the $n+1$ variables x_i and t at the points indicated. For this family (σ_t) we shall now examine successively the conditions formulated in No. 1.

1°. Let us look at the boundaries of the characteristics; there are two kinds of boundary points, one over the boundary of $\mathcal{G}$ and the other over $|x_1 - x_1^0| = \rho$; let us begin with the first kind. The closed circle $|x_1 - x_1^0| < \rho$ has been traced in the region (ω) , the part of $\mathcal{B}$ in this circle is contained in $\mathcal{G}$, $\mathcal{B}$ being the image of $\mathfrak{A}$ under projection on the space $((x))$; consequently, if we take a region V containing $\mathfrak{A}$ and sufficiently close to $\mathfrak{A}$, the boundary of $\mathcal{G}$ will lie entirely exterior to V ; the same is true of the boundary points of (σ_t) under consideration. As for the second kind of boundary points, since we have $|y_1 - f_1((x)) \psi(x_1)| < 1$ for every point $((x), \eta)$ of $\mathfrak{A}$ over $|x_1 - x_1^0| = \rho$, if we choose V sufficiently close to $\mathfrak{A}$, we can ensure that all the boundary points in question remain outside V .

2°. For any t in the interval $0 < t < +\infty$, the characteristic (σ_t) never passes through a point of $\bar{\Delta}$ since $\bar{\Delta}$ lies entirely on the surface $y_1 - f_1((x)) = 0$.

3°. Since $R_1(x_1^0)$ is necessarily positive because of our hypothesis, the centre x_1^0 lies on $\mathfrak{C}$. Consequently, we can find a point $((x^0, y^0))$ of $\mathfrak{A}$ for which we have $|y^0 - f_1((x^0))| = R_1(x_1^0)$, and therefore $|y^0 - f_1((x^0)) \psi(x_1^0)| = 1$; we can therefore choose the argument θ so that the characteristic (σ_0) passes through the point $((x^0, y^0))$ of $\mathfrak{A}$. On the other hand, in the equation for (σ_t) we may regard the function $f_1((x))$ as being bounded on the region under consideration,

⁹That is to say $\log R_j(x_1)$ are subharmonic functions, the logarithm being natural. We call subharmonic a function $\varphi(z)$ of the complex variable which is real, upper semi-continuous, bounded above in the interior of its domain of existence and such that if we draw a circle in the interior of the domain of existence, the arithmetic mean of $\varphi(z)$ taken in the sense of Mr. Lebesgue on the circumference is bigger than or equal to the value at the centre. We shall count the constant $-\infty$ among these functions. For subharmonic functions, see the book of Mr. G. Julia: *Principes géométriques d'analyse*, II.

¹⁰We say that a condition is fulfilled in the interior of a certain region if it is fulfilled on every closed set contained in the region.

since this can always be achieved by taking, if necessary, a region a little smaller than $\mathcal{G}$. Then, from a certain value of the parameter on, none of the characteristics (σ_t) will enter $\mathfrak{A}$.

We can thus construct a family of characteristics fulfilling the conditions formulated earlier. This contradicts the general proposition, and the function $R_1(x_1)$ is therefore logarithmically convex, and so also are the other functions. Q.E.D.

Let us decompose the region (ω) into connected components $(\omega_1), (\omega_2), \dots, (\omega_\lambda), \dots$, on each of which the functions $R_j(x_1)$ are logarithmically subharmonic. Suppose that one of these domains, (ω_1) for example, contains a point exterior to $\mathfrak{C}$. Then, all the functions $R_j(x_1)$ vanish in the neighbourhood of this point by very definition, and consequently vanish identically on (ω_1) because they are logarithmically subharmonic.

Let us also remark that if, for a point of $\mathfrak{C}$ contained in (ω) , all v functions R_j are zero, we necessarily have $E_{x_1^0} = \Delta_{x_1^0}$, where E_x and Δ_x represent respectively the sections of $\bar{\Delta}$ and Δ by $x_1 = x$. In fact, if this is not the case, the set $\Delta_{x_1^0}$ would contain at least one point $(x_2, x_3, \dots, x_n, y_1, \dots, y_v)$ exterior to $E_{x_1^0}$. Now, x_1 being in (ω) , the point $((x^0))$ belongs to $G_{x_1^0}$. On the other hand, this point is outside Δ because if $((x^0))$ were a point of Δ , the point $(x_2, \dots, y_v)$ would belong to E_x because all the functions R_j vanish at x_1^0 . Under these conditions, it follows from the definition of Δ that at least one of the functions, let us say $f_1((x))$, has an absolute value bigger than 1 at $((x^0))$. Since $R_1(x_1^0) = 0$, we have $y = f_1((x^0))$ and consequently $|y| > 1$. This is absurd, because, as we remarked above, $\mathfrak{A}$ is necessarily contained in $|x_i| < r_i$, $|y_j| < 1$; we must therefore have $E_x = \Delta_x$.

2.3 Preliminary Proposition

In what follows, we are going to look at the boundary of (ω) , and to do this, we shall first prove the following proposition.

Let $\varphi(x)$ be a subharmonic function of the complex variable x . If we let the point x run along a simple Jordan arc L in the interior of the domain of existence and terminating at a point ξ , we always have as limit

$$\lim \varphi(x) = \varphi(\xi), \quad (x \rightarrow \xi).$$

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We draw two concentric circles (γ) and (C) around ξ in the domain of existence of $\varphi(x)$; of these (γ) is inside (C) . Let us suppose that when the point x describes the curve L starting at ξ , this point meets the circumference γ for the last time at the point b , and the outside circumference C for the first time at the point B . Let I be the arc of the curve from b to B , and D the domain having I and C as boundary. This domain is simply connected; let us map it conformally on $|y| < 1$ in such a way that the point ξ corresponds to the origin in the y -plane. This transformation will be denoted by $x = F(y)$.

To simplify the notation, let us suppose that (C) is the unit circle and let (γ) be of radius ρ . The function $\frac{F(y)}{y}$ is holomorphic and non-zero in the unit circle and takes the value $F'(0)$ at the origin. In addition, the absolute value of the function remains single-valued and continuous up to the circumference and does not vanish there either. Consequently, the arithmetic mean of the real function $\log |F(y)|$ on $|y| = 1$ equals $\log |F'(0)|$. If we denote by $2\pi\alpha$ the length of the arc on $|y| = 1$ corresponding to the arc I in the x -plane, we obtain from this $\rho^\alpha < |F'(0)|$.

On the other hand, by the theorem of Mr. Koebe, we have $\rho > k|F'(0)|$, k being a certain positive constant; we therefore have $\rho^{(1-\alpha)} > k$; we find therefore that when ρ tends to 0, α tends necessarily to 1.

¹¹We have supposed here that the curve is a simple Jordan curve. But this is just for the sake of simplicity. The proposition remains valid for any continuous curve which converges uniquely to ξ . The proof of this can be given similarly, using well-known theorems of Messrs Fatou and Lebesgue.

On considère, dans $\mathbb{C}^{n+v}$, la sous-variété Σ définie par $x \in \bar{\Delta}$, $y_j = f_j(x)$. Alors Σ appartient à la classe $(\mathfrak{P})$ de $\mathbb{C}^{n+v}$, i.e. Σ est intersection d'une suite décroissante d'ouverts définis par des inégalités (strictes) portant sur des polynômes.

La démonstration du théorème I utilise des fonctions dont le logarithme est sous-harmonique, et procède par récurrence sur n , en se servant des résultats du Mémoire I.

Pour déduire le théorème II du théorème I, on procède ainsi : si D est un domaine d'holomorphic, tout domaine D' dont l'adhérence est compacte et contenue dans D est contenu dans un $\bar{\Delta}$ du type précédent.

Remarques du commentateur. On sait aujourd'hui que le théorème II résulte immédiatement de deux faits : (1) cohérence du faisceau des germes de fonctions holomorphes (résultat prouvé par Oka dans son Mémoire VII) ; (2) application du théorème B à ce faisceau.

Quant au théorème I, on a plus généralement le résultat suivant : dans un polycylindre compact $D \subset \mathbb{C}^N$, soit Σ un sous-ensemble analytique fermé ; alors Σ est intersection d'une suite décroissante d'ouverts dont chacun est défini par des inégalités en nombre fini de la forme $|P(x)| < 1$, où P est un polynôme. Ceci peut se démontrer comme suit : d'après le théorème B, il existe, pour tout point $a \in D - \Sigma$, une f holomorphe dans D telle que $f(x) = 0$ pour tout $x \in \Sigma$, $|f(a)| > 1$. Comme f est limite uniforme de polynômes, il existe un polynôme P tel que $|P(x)| < 1/2$ pour tout $x \in \Sigma$, $|P(a)| > 1$. Étant donné un voisinage ouvert V de Σ dans D , la compacité de $D - V$ entraîne l'existence d'un nombre fini de polynômes P_k tels que l'ensemble des $x \in D$ tels que $|P_k(x)| < 1$ pour tout k soit contenu dans V et contienne Σ .

Kapitel 3

III. The Second Cousin Problem

Deuxième problème de Cousin

Journal of Science of the Hiroshima University (1939), p. 7–19

Introduction

P. Cousin was the first to pose the problem of finding meromorphic functions with given poles and points of indeterminacy, and that of finding holomorphic functions with given zeros. We shall call these, with Mr. H. Cartan¹, the first and second Cousin problem respectively. These two problems are intimately related but differ more profoundly than may appear at first sight. For example, the first problem is always solvable on univalent domains of holomorphy without points at infinity as we have seen in the preceding memoir, but one cannot expect the same result to hold for the second problem; for there effectively exist domains whose character is contrary to this². In the present memoir, we shall attempt to extract that which characterizes the second problem.

Here is an example from which the author started. Consider, in the space of three real variables x, y and z , the cylinder D and the segment L given by

$$(D): \quad x^2 + y^2 < 1, \quad |z| < 2,$$

$$(L): \quad x = y = 0, \quad |z| < 1;$$

there are infinitely many continuous functions $F(x, y, z)$, well-defined on D , which vanish at every point of L , without having zeros outside. But a purely geometric method such as this is not the method defining zeros in the second Cousin problem. Let us therefore add the following condition:

$$F(x, y, 0) = (x + iy) \psi(x, y) \quad \text{for } x^2 + y^2 < \rho^2,$$

i being the imaginary unit, ψ a function which is continuous and non-vanishing, and ρ a positive number, be it ever so small; and one finds that such an F no longer exists.

For the bibliography of the problem of this paper, we shall content ourselves with referring the reader to the following:

H. Behnke and P. Thullen. *Theorie der Funktionen mehrerer komplexer Veränderlichen*, 1934.

H. Behnke and K. Stein: *Analytische Funktionen mehrerer Veränderlicher zu vorgegebenen Null- und Polstellenflächen*, Jahresb. Deutsch. Math. Vereinig. 47, 1937, p. 1–22.

In consideration of this paper, the author must confess that Theorem I expounded in Memoir I should be attributed to him.

¹H. Cartan: *Les problèmes de Poincaré et de Cousin pour les fonctions de plusieurs variables complexes*, C. R. Acad. Sci. Paris, 1934.

²One will find a concrete example in the following paper: T.H. Gronwall: *On the expressibility of a uniform function of several complex variables as the quotient of two functions of entire character*, Amer. Math. Soc. Trans., 1917. We shall demonstrate a very simple new example in what follows.

Non-analytic Solutions

3.0.1 Definitions

We shall extend the second Cousin problem to the domain of non-analytic functions.

Consider a domain D in the space of several real variables and two continuous functions f_1 and f_2 on it³. If these functions satisfy a relation of the form $f_1 = f_2\lambda$ on D , λ being a continuous non-vanishing function, they will be called *equivalent* on the domain⁴. Two functions, continuous at a point P of finite space will be called equivalent at P if we can find a hypersphere of centre P on which the functions are equivalent.

Consider a mechanism of the following type on D . To each point P of the domain, there corresponds a hypersphere (γ) centred at P and a continuous function f on (γ) , and this, in such a way that for each pair of contiguous hyperspheres, the corresponding functions are equivalent on the part common to the two hyperspheres. We shall call f the function attached to P . Under these conditions we ask to find a continuous function F on D equivalent at every point P of the domain to the function attached to P . We shall say that the function F , if it exists, has the given zeros.

Now, the problem so generalised is not necessarily solvable even on a sphere⁵. To avoid this behaviour, we abandon, once for all, all cases in which the given zeros fill out a whole portion of the space of the variables.

Let us make a remark regarding this convention. Let F_1, F_2 be two continuous functions on the domain D which are equivalent at each point of D . These functions are not, in general, equivalent on the domain⁶; but they are certainly equivalent if the set of points on which they vanish has no interior points. In fact, F_1 and F_2 being equivalent at a point P of D , one finds a relation $F_1 = F_2\lambda$ in the neighbourhood of P , λ being continuous and non-vanishing. Now, because of the hypothesis, λ is uniquely determined in the neighbourhood of P and this for every P in D . The global equivalence follows from this.

Let us consider zeros (β) in a domain D , and let us map D onto another domain in the same space by a bijective, bicontinuous transformation. The mechanism defining (β) can be transported to D' , and generates new zeros in D' , say (β') . Under these circumstances, if our problem is solvable for D and (β) , it is certainly also solvable for D' and (β') , and conversely. Thus, the generalised problem is invariant under topological transformations.

3.0.2 An Example

Let us consider, in the space of the two complex variables x, y , a cylindrical domain (r, r') of the form $r < |x| < 1$, $r' < |y| < 1$, where the sum $r + r'$ of the radii will be assumed to be greater than 1. The characteristic surface $y = x - 1$ decomposes, in (r, r') , into two pieces, one situated above the real axis and the other below them. Let (α) be the first piece.

Suppose that we had a continuous function $F(x, y)$ in (r, r') equivalent to $f(x, y) = y - x + 1$ in the neighbourhood of (α) and equivalent to 1 in the exterior of (α) . In the annulus r , let us consider points ξ_1 and ξ_2 of which ξ_1 is on the real axis and positive, and ξ_2 is on the real axis

³In the present memoir domains will be understood to be univalent and in the finite part of space unless the contrary is explicitly indicated.

⁴We shall apply the same definition to two functions continuous on an arbitrary set of points in finite space.

⁵In fact, let us consider, in the space of three real variables x, y, z , two spheres of centre $(0, 0, 1)$ and $(0, 0, -1)$ and of radius 1. We can find an infinity of functions, continuous and real on all of finite space, which vanish on these two spheres without vanishing anywhere in the exterior of the two. Let $\psi(x, y, z)$ be one of these functions. We attach to each point (x, y, z) of space a sphere (itself) around this point of radius ρ and a function $f(x, y, z')$, in such a way that $f(x, y, z) = \psi(x, y, z)$ for $|z| \geq 1$, $f(x, y, z) = (x + iy)\psi(x, y, z)$ for $|z| < 1$, i being the imaginary unit. If the radius ρ is sufficiently small, the equivalence conditions are satisfied, but there is no continuous function having the given zeros in the sphere of centre (0) and radius 3.

⁶The two functions $\psi(x', y')$ and $(x + iy)\psi(x', y')$ are equivalent at every point of the circle $x^2 + y^2 < 2$; but they are not on the whole circle.

and negative. Let us draw in the annulus r' a circumference C' about the origin. Since $F(\xi_1, y)$ does not vanish on C' , we can calculate the variation of the argument of F without ambiguity when y describes C' once in the positive sense; we shall denote this by $V(F, \xi_1)$. We shall use this notation in the general case. We can similarly calculate $V(F, \xi_2)$. Let x' be a point of r situated below the real axis, but otherwise arbitrary; $F(x', y)$ is never zero in r' ; from this it follows that $V(F, \xi_1) = V(F, \xi_2)$.

On the other hand, let A be the set of points of r situated on or above the real axis. Since the functions $F(x, y)$ and $f(x, y)$ are equivalent at every point of (A, r') , there is a function $\lambda(x, y)$, continuous and non-vanishing on (A, r') such that one has, identically $F = f\lambda$. For this λ , we have $V(\lambda, \xi_1) = V(\lambda, \xi_2)$.

As for f , since it equals $y - x + 1$, we have $V(f, \xi_1) = 2\pi$, $V(f, \xi_2) = 0$, so that $V(F, \xi_1) = V(F, \xi_2) + 2\pi$.

This contradicts the equality established earlier; therefore F does not exist. We have thus seen that the generalised problem is not necessarily solvable in the domain (r, r') .

Remark. We have found, at the same time, that the second Cousin problem is not always solvable in a domain of holomorphy.

3.0.3 The Generalised Problem in a Cylindrical Domain

We shall start by studying the generalised problem in cylindrical domains to keep things from getting complicated from the beginning.

Consider a cylindrical domain $(X_1, X_2, \dots, X_n)$ in the space of n complex variables $x_1, x_2, \dots, x_n$, X_i being a domain in the x_i -plane. We shall denote this simply by $((X))$. From the preceding example, it follows immediately that if at least two of the X_i are multiply connected, the generalised problem is not always solvable on $((X))$. We shall therefore suppose, in what follows, that all the X_i are simply connected, excepting at most one of them. We now ask if the problem is solvable on $((X))$.

1°. On the plane X_1 , we draw a rectangle R which we decompose into equal small rectangles (W_i) by two systems of straight lines parallel to the sides of R . Let A_1 be a closed domain consisting of one or more of the (W_i) and their boundaries, and A_2 a closed domain consisting of one of the (W_i) , contiguous to A_1 , and its boundary. We shall denote the common part of the boundaries of A_1 and A_2 by L , and the closed domain consisting of A_1 and A_2 by A . Let us take a domain B in the space $(x_2, x_3, \dots, x_n)$.

In what follows, it will be understood that the point $(x_2, \dots, x_n)$ lies in the domain B . Let f_1, f_2 be two continuous functions of the n variables x_i in neighbourhoods of A_1 and A_2 respectively⁷, which are globally equivalent in a neighbourhood of L . Under these conditions, one can find a continuous function of the n variables x_i , defined in a neighbourhood of A , and equivalent to at least one of the two given functions f_1, f_2 at any point of A , if L does not coincide with the entire boundary of the rectangle A_2 and if B is simply connected.

In fact, the functions f_1 and f_2 being globally equivalent, satisfy an equation $f_1 = f_2\lambda$, where λ is a function of the n variables x_i , continuous and non-vanishing in a neighbourhood of L . We propose to find a function μ of the n variables, continuous and non-vanishing in a neighbourhood of the closed rectangle A_2 and equal to λ in a neighbourhood of L .

This would in general be impossible if L coincided with the boundary of A_2 or if B were multiply connected⁸. But, in the case actually before us, this is immediate because any determination of $\log \lambda$ is single-valued in a neighbourhood of L . Let us set $F((x)) = f_1((x))$ in a neighbourhood of A_1 and $F((x)) = f_2((x))\mu((x))$ in a neighbourhood of A_2 . The function F has, obviously, the required properties. Q.E.D.

⁷That is, in certain domains containing A_1, A_2 .

⁸The first case is obvious; for the second, one can find an example immediately using the results of the preceding section.

2°. Given zeros (β) in the cylindrical domain $((X))$, there exists, on any domain Δ completely interior to $((X))$ ⁹, a continuous function having the zeros (β) on $\bar{\Delta}$.

This is an immediate consequence of the preceding proposition because all the X_i are supposed to be simply connected with at most one exception, and in addition, the zeros are supposed not to fill out a whole portion of space.

3°. We can always find a continuous function having given zeros on the cylindrical domain $((X))$.

It is the following kind of extension of non-vanishing continuous functions which will be important in the proof of this proposition: Let $\lambda(z)$ be a function of the variable z which is continuous and non-vanishing in $1 < |z| < 2$; extend $\lambda(z)$ to $0 < |z| < \infty$ without ceasing to be continuous and non-vanishing. This can be done by extending the function along each half-line from the origin without changing the original values.

Let us consider in the interior of each X_i a domain X'_i bounded by one or several simple closed Jordan curves (without common points) such that each of these curves, if it is surrounded by another one of them, contains in its interior a point not belonging to X_i . We can construct a cylindrical domain $((X'))$ so as to contain a domain completely interior to $((X))$ given a priori. We thus obtain a sequence $((X^p))$ of cylindrical domains of the kind indicated ($p = 1, 2, \dots$) tending towards $((X))$; we can assume that each is completely interior to the following.

Let (β) be given zeros in $((X))$. By virtue of the preceding proposition, there exists, for each p , a continuous function $F_p((x))$ having (β) as its zeros in a neighbourhood of the closed domain $((X^p))$. We must construct a continuous function $F((x))$ having (β) as its zeros on the domain $((X))$, which we shall do successively as follows.

For the closed domain $((X^1))$, let us set $F((x)) = F_1((x))$.

Let us pass to $((X^2))$. Since the two functions F_1 and F_2 are equivalent on $((\bar{X}^1))$, they satisfy a relation $F_1 = F_2\lambda$, λ being a function which is continuous and non-vanishing on the closed set $((X^1))$. We can certainly extend this function λ to all of $((X))$ without losing these properties. Let μ be the extension; let us set $F((x)) = F_2((x))\mu((x))$.

The new $F((x))$ is a continuous extension of the old one to $((\bar{X}^2))$ and has the given zeros in a neighbourhood of the closed domain $((\bar{X}^2))$. Repeating the same procedure with $((X^3)), ((X^4)), \dots$, we obtain the required function. Q.E.D.

Lemma I. Let $((X))$ be a cylindrical domain in the space $((x))$. For the generalised problem to be solvable on $((X))$, always assuming that the given zeros never fill out a whole portion of the space $((x))$, it is necessary and sufficient that all the X_i , with at most one exception, be simply connected.

Analytic Solutions

3.0.4 Definitions

From now on we shall study the second Cousin problem in domains of holomorphy with the help of what we have just seen.

Consider a domain D in the space of the n complex variables x_i . Zeros (β) in D will be called *analytic* if to every point of D there corresponds a holomorphic function (of the variables x_i) having (β) as its zeros in the neighbourhood of that point. The second Cousin problem consists in finding a holomorphic function having given analytic zeros in D ; solutions of this problem will sometimes be called *analytic* to distinguish them from solutions of the generalised problem.

Let us introduce the following auxiliary notion. Given zeros (β) in D , we shall say that they can be *swept out* if, to each point P of D there corresponds a hypersphere (γ) and a continuous function $f((x), t)$ on $[(\gamma), 0 < t < 1]$ such that:

1°. $f((x), 0)$ has (β) as its zeros and $f((x), 1)$ is non-vanishing.

⁹This means that $\bar{\Delta}$ is contained, together with its boundary, in $((X))$.

2°. $f((x), t)$ is not identically zero on a whole portion of $(2n + 1)$ dimensions.

3°. For each pair $(\gamma_1), (\gamma_2)$ of contiguous hyperspheres, the corresponding functions $f_1((x), t)$ and $f_2((x), t)$ are equivalent on $[(\delta), 0 < t < 1]$, (δ) being the part common to (γ_1) and (γ_2) .

The second condition is essential, since, without it, all zeros could be swept out¹⁰. Let $F_1((x), t), F_2((x), t)$ be continuous functions on $[J, 0 < t < 1]$, where J is an arbitrary domain in D , which are equivalent to at least one of the functions $f((x), t)$ when $((x))$ is in the neighbourhood of any point of J , and t on the closed interval. Then, because of the same second condition, it follows that F_1 and F_2 are globally equivalent on $[J, 0 < t < 1]$.

If there exists a continuous function having given zeros in D , these zeros can necessarily be swept out.

In fact, let $F((x))$ be the solution. Let us set $f((x), t) = (1 - t)F((x)) + t$; we find that the three conditions are fulfilled.

It is the converse of this statement with which we shall be occupied in what follows.

3.0.5 A Preliminary Proposition

The example given in the introduction leads us to a proposition which we shall use frequently in what follows.

Let D be a domain in the space $((x))$ and $\lambda((x))$ a function continuous and non-vanishing in D . We shall say that $\lambda((x))$ satisfies condition (a) in the domain if there exists a function $\Lambda((x), t)$ on $(D, 0 < t < 1)$, continuous and non-vanishing on this set, and such that $\Lambda((x), 0) = \lambda((x))$ and $\Lambda((x), 1) = 1$.

If $\log \lambda((x))$ is single-valued in D , condition (a) is necessarily satisfied; to see this, it is enough to set $\Lambda((x), t) = e^{(1-t)\log \lambda((x))}$.

Suppose, conversely, that condition (a) is fulfilled for the function $\lambda((x))$ in the domain D ; let $\Lambda((t), (x))$ be one of the functions realising this. Let us draw a closed curve C in D . When $((x))$ describes C in a definite sense, and t is fixed in $0 < t < 1$, the variation of the argument of $\Lambda((x), t)$ can be calculated unambiguously since $\Lambda((x), t)$ is never zero. We shall denote the variation by $V(t)$. This is a continuous function of the variable t on $0 < t < 1$; the values it can take are discrete; it is therefore constant, and, in particular $V(0) = V(1)$.

Now, for $t = 1$, we have $\Lambda = 1$ identically; $V(1)$ is therefore equal to zero; hence $V(0)$ is also zero. This is true whatever be C . Hence, any determination of $\log \lambda((x))$ is single-valued on D .

We have thus established the following proposition:

Proposition. Every determination of $\log \lambda((x))$ is single-valued in D if and only if $\lambda((x))$ fulfills condition (a) on D .

3.0.6 Zeros Which Can be Swept Out

1°. Let us consider once again the configuration of No. 3 in the x_1 -plane, consisting of the rectangles R , (W_i) , the closed domains A , A_1 , A_2 and the common part L of the boundaries of A_1 and A_2 . Let us consider also the closed set A in finite $((x))$ space.

In what follows, it will be understood that $((x))$ remains in a neighbourhood of A and t in the interval $(0, 1)$. Let f_1 be a continuous function of the variables x_i, t ($i = 1, 2, \dots, n$) in a neighbourhood of A_1 , analytic for $t = 0$ and identically 1 for $t = 1$. Let f_2 be a function having these properties in a neighbourhood of A_2 . Suppose that f_1 and f_2 are globally equivalent in a neighbourhood of L as functions of $n + 1$ variables. We shall prove the following:

We can find a continuous function of the $n + 1$ variables having the properties indicated above in a neighbourhood of A and equivalent to at least one of the given functions at every

¹⁰In fact, let $f((x))$ be any function attached to a point (defining the zeros) and let us set $f((x), t) = (1 - 3t)f((x))$ for $0 \leq 3t \leq 1$, $= 0$ for $1 \leq 3t \leq 2$, $= (3t - 2)$ for $2 \leq 3t \leq 3$. The functions $f((x), t)$ so defined satisfy the first and the third condition.

point $((x))$ for any t in $(0, 1)$ if L does not coincide with the entire boundary of the rectangle A_2 , and if, in addition, A belongs to the class $(\mathfrak{P})$.

In fact, as f_1 and f_2 are globally equivalent, we have a relation $f_1((x), t) = f_2((x), t) \lambda((x), t)$, where λ is a function which is continuous and non-vanishing in a neighbourhood of L , from which it follows that $\lambda((x), 0)$ is analytic and $\lambda((x), 1) = 1$.

Hence, by the preceding proposition, $\log \lambda((x), 0)$ is single-valued, the determination being arbitrary. Thus $\log \lambda((x), 0)$ is a holomorphic function in a neighbourhood of L . A is a set in the class $(\mathfrak{P})$. In view of the results of No. 3 in Memoir I, we can therefore find two holomorphic functions $\varphi_1((x))$, $\varphi_2((x))$ in neighbourhoods of A_1, A_2 respectively, such that in a neighbourhood of L we have the relation $\varphi_1((x)) - \varphi_2((x)) = -\log \lambda((x), 0)$.

Let us set

$$\Pi = f_1((x), t) e^{(1-t)\varphi_1} / f_2((x), t) e^{(1-t)\varphi_2},$$

k being a suitable integer depending on the domains. The function Π is continuous and non-zero in a neighbourhood of L and reduces to 1 for $t = 0$ and $t = 1$. We find a function $v((x), t)$ having the same properties as Π in a neighbourhood of the rectangle A_2 , and identical with Π in a neighbourhood of L . By virtue of the preliminary proposition, every determination of $\log \Pi((x), t)$ is single-valued, and therefore v certainly exists. We set

$$F((x), t) = f_1((x), t) e^{(1-t)(\varphi_1 + 2k\pi i)}$$

or

$$F((x), t) = v((x), t) f_2((x), t) e^{(1-t)\varphi_2}$$

according as x is situated on A_1 or on A_2 . The function F satisfies the condition imposed.

2°. Let us consider a domain of holomorphy D in the space $((x))$. We construct a sequence of closed sets:

$$(S): \quad \bar{\Delta}_1, \bar{\Delta}_2, \dots, \bar{\Delta}_p, \dots, \quad (S)$$

tending to D from the interior such that each $\bar{\Delta}_p$ is of the form $|x_i| < r_{p,i}$, $|g_{p,j}((x))| < 1$ ($i = 1, 2, \dots, n$; $j = 1, 2, \dots, p'$), where the $r_{p,i}$ are positive constants and the $g_{p,j}((x))$ functions holomorphic on D . Thanks to the work of Messrs. H. Cartan and P. Thullen, (S) always exists. We shall also suppose that $\bar{\Delta}_p$ consists of points interior to $\bar{\Delta}_{p+1}$.

Let us consider in the domain D zeros (α) which are analytic and can be swept out. There then exists, for every point P of the domain, a hypersphere (γ) of centre P , a holomorphic function $\varphi((x))$ having (α) as its zeros on (γ) and a function $f((x), t)$ satisfying the three conditions formulated in No. 5 on $[(\gamma), (0, 1)]$. Now, we can always suppose that f is analytic for $t = 0$ and reduces to 1 for $t = 1$. In fact, $\varphi((x))$ and $f((x), 0)$ are equivalent in (γ) and $f((x), 1)$ is never zero. Hence, if we put $\lambda((x)) = \frac{f((x), 0)}{\varphi((x)) f((x), 1)}$, λ is continuous and non-vanishing on (γ) ; it satisfies condition (a) on (γ) because (γ) is simply connected. Let $\lambda((t), (x))$ be a function realising condition (a), and let us set $\varphi((x), t) = \frac{f((x), t)}{\lambda((x), t) f((x), 1)}$. The function $\varphi((x), t)$ has the properties required, and we shall denote it again by $f((x), t)$.

We shall prove the following: To each $\bar{\Delta}_p$ of the sequence S there corresponds a continuous function $F_p((x), t)$ in the neighbourhood of $\bar{\Delta}_p$ which is equivalent to at least one of the functions $f((x), t)$ at every point of $\bar{\Delta}_p$, t being in $(0, 1)$, and such that it is holomorphic for $t = 0$ and reduces to 1 for $t = 1$.

When the $\bar{\Delta}_p$ belong to $(\mathfrak{P})$, this is an immediate consequence of the preceding proposition.

Let us consider the general case. In view of Theorem I in Memoir II, the closed set E , $y_j = g_{p,j}((x))$, $((x)) \in \bar{\Delta}_p$, ($j = 1, 2, \dots, p'$), always belongs to the class $(\mathfrak{P})$ in the space $((x, y))$. We pose the present problem in the space $((x, y))$ by regarding the function $f((x), t)$ as being given in a neighbourhood of E . This being the case above, there exists a function

$\Phi((y), (x), t)$ in a neighbourhood of E and having the properties indicated above. Substituting $y_j = g_{p,j}((x))$ in Φ , we obtain the required function.

3°. From the sequence of functions obtained thus $F_1((x), t), F_2((x), t), \dots, F_p((x), t), \dots$, we shall construct a holomorphic function $\Phi((x))$ having the zeros (α) in the domain D .

Let ε_p ($p = 1, 2, \dots$) be a sequence of positive numbers such that the series $\sum \varepsilon_p$ converges. We first set $\Phi_1((x)) = F_1((x), 0)$; Φ_1 is a holomorphic function having the zeros (α) in a neighbourhood of $\bar{\Delta}_1$.

The functions $F_1((x), t), F_2((x), t)$ are globally equivalent in a neighbourhood of $\bar{\Delta}_1$ because they have the same zeros which do not fill out any portion of $((x), t)$ space; hence, if we set $F_1((x), t) = F_2((x), t) \Lambda_1((x), t)$, the function Λ_1 is continuous and non-vanishing in a neighbourhood of $\bar{\Delta}_1$, t being in $(0, 1)$. It is holomorphic for $t = 0$ and reduces to 1 for $t = 1$. Hence $\log \Lambda_1((x), 0)$ is holomorphic in a neighbourhood of $\bar{\Delta}_1$, the determination being arbitrary. In view of the results expounded in No. 5 of Memoir II, we can therefore find a holomorphic function $\psi_1((x))$ in the domain, satisfying, in a neighbourhood of $\bar{\Delta}_1$, the condition $|\psi_1((x)) - \log \Lambda_1((x), 0)| < \varepsilon_1$. We set $\Phi_2((x)) = F_2((x), 0) e^{\psi_1}$. Φ_2 is a holomorphic function having the zeros (α) in a neighbourhood of $\bar{\Delta}_2$; we see that $e^{-\varepsilon_1} < \Phi_2((x))/\Phi_1((x)) < e^{\varepsilon_1}$ in a neighbourhood of $\bar{\Delta}_1$. Moreover, the logarithms of Φ_2/F_3 with $t = 0$ are holomorphic in a neighbourhood of $\bar{\Delta}_2$.

From this last property of Φ_2 , we form a function $\Phi_3((x))$ holomorphic in a neighbourhood of $\bar{\Delta}_3$ and satisfying the condition $e^{-\varepsilon_2} < \Phi_3((x))/\Phi_2((x)) < e^{\varepsilon_2}$ in a neighbourhood of $\bar{\Delta}_2$, and such that the logarithms of Φ_3/F_4 are holomorphic in a neighbourhood of $\bar{\Delta}_3$ when $t = 0$; and so on. Every limit of the sequence of functions so obtained gives us the required function. We have thus established the following proposition.

Lemma II. Given analytic zeros which can be swept out on a univalent domain of holomorphy without points at infinity, we can find a holomorphic function having the given zeros on the domain.

Remark. Let us mention the corresponding proposition for the generalised problem. Given zeros which can be swept out on a univalent domain in the finite space of several complex variables, we can find a continuous function having the given zeros on the domain¹¹. One can prove this in exactly the same way, and in fact even more simply.

3.0.7 Theorem I

Theorem I. Given analytic zeros in a univalent domain of holomorphy without points at infinity, if there exists a continuous solution, an analytic solution also exists.

This is an immediate consequence of Lemma II since zeros admitting a continuous solution can necessarily be swept out.

3.0.8 Theorem II

Theorem II. Let Γ be a univalent cylindrical domain in the finite space of several complex variables, all of whose projections on the coordinate planes, with at most one exception, are simply connected. The second Cousin problem is always solvable on a univalent domain of holomorphy in the same finite space if it can be mapped onto Γ by a topological transformation.

For, by Lemma I, the generalised problem is always solvable on Γ ; consequently the same is true on the given domain of holomorphy.

¹¹Whether the zeros fill out a portion of space or not.

Kapitel 4

IV. Domains of Holomorphy and Rationally Convex Domains

Domaines d'holomorphie et domaines rationnellement convexes
Japanese Journal of Mathematics 17 (1941), p. 517–521

Introduction

As we have said before, the subject of the present researches consists of the following problems: the problems of Cousin, representations of holomorphic functions, and the classification of domains¹. In the present memoir, we shall be concerned with the last problem².

4.1 Classification of Domains

We shall consider domains in the space $((x))$ described by the complex variables $x_1, x_2, \dots, x_n$. In the present research, we shall always suppose that they are univalent without points at infinity to keep matters from becoming complicated from the beginning. If we classify domains of the type discovered by F. Hartogs using the method of Mr. H. Cartan, and simplify the classification, we obtain the following types of domains:

- (I) Cylindrical domains.
- (II) Rationally convex domains.
- (III) Domains of holomorphy.
- (IV) Pseudoconvex domains.

A domain D will be called rationally convex if it is convex with respect to rational functions which are holomorphic on D , or if D can be approximated from the interior by domains of the above class.

Let us consider a bounded domain D . Let E be the set consisting of points of finite space $((x))$ which do not belong to D . If, in the neighbourhood of an arbitrary boundary point P of D , the

¹For these problems, see, above all G. Julia, *Sur les familles de fonctions analytiques de plusieurs variables*, 1926 (Acta Mathematica).

²The 3 preceding memoirs are the following: I. Rational convex domains, 1936. II. Domains of holomorphy, 1937. III. The second Cousin problem, 1939 (Journal of Science of the Hiroshima University.)

set E satisfies the continuity theorem³, and if this property remains invariant under all bijective pseudoconformal transformations in the neighbourhood of P , D will be called *pseudoconvex*.

More generally, we shall call pseudoconvex any domain which can be approximated from the interior by domains of the above character.

These 4 fundamental types satisfy the following relations:

$$(I) \subset (II) \subset (III) \subset (IV).$$

In other words, these types are nested in each other. The two last distinctions are but provisional⁴. It is possible that there exist only 2 types (I) and (II). Here then are two problems: $(II) = (III)$? and $(III) = (IV)$? The second of these will be treated later. In the present memoir, we shall treat the first.

4.2 The Di-annulus Example

Consider the di-annulus

$$(r, r') : \quad r < |x| < 1, \quad r' < |y| < 1$$

in the space of two complex variables x, y , and, at the same time, the characteristic plane

$$y = x - 1.$$

The part of this characteristic in the di-annulus decomposes into two (continuous) pieces if the radii satisfy the relation $r + r' > 1$.

Let us consider this case. One of the pieces consists of points (x', y') such that x' as well as y' lie in the upper half-plane. We denote this by (u) ⁵. Let us construct a function $G(x, y)$, meromorphic in (r, r') , having $\frac{1}{y - x + 1}$ as poles on (u) , and holomorphic outside (u) ; $G(x, y)$ certainly exists, by Cousin's theorem. With this G , we construct the set of points of the form

$$(A): \quad r < |x| < 1, \quad r' < |y| < 1, \quad |G(x, y)| < M, \quad (A)$$

where M is a positive number which we shall determine later. The set A is open; it consists of domains of holomorphy.

We choose a positive number d as follows: $6d < 1 - r'$; we draw in the x -plane a circumference C around the origin of radius half of $(1 + r)$, and in the y -plane, the annulus

$$(r') : \quad r' - d < |y| < 1 - d.$$

Let (x', y') be any point on (u) . On the characteristic plane $x = x'$, we describe a circle $(\gamma_{x'})$ of centre y' and radius d . (γ_x) is uniquely determined, if it is defined, by x . Let E be the set of points consisting of all the circles $(\gamma_{x'})$. Let (x', y') be now a point belonging to the set

$$|x| = \frac{1+r}{2}, \quad r' - d < |y| < 1 - d.$$

³This means that there exists a hypersphere S of centre P having the following property: Consider an arbitrary point $((a))$ and a circumference of the form $x_1 = a_1, x_2 = a_2, \dots, x_{n-1} = a_{n-1}, |x_n - a_n| = r$, r being an arbitrary radius. If this circumference is exterior to E but the point $((a))$ is not, we can find a positive number d such that to each point $(x_1^0, x_2, \dots, x_{n-1})$ in $|x_k - a_k| < d$ ($k = 1, 2, \dots, n-1$) there corresponds at least one point x_n in $|x_n - a_n| < r$ such that the point $((x^0))$ belongs to E .

⁴In Memoir I, we looked for a method to pass from (II) to (I); and in Memoir II for the passage from (III) to (II).

⁵This is the configuration used once before for the second Cousin problem.

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Paper IV (Continued)

without belonging to E . The set consisting of all these (x', y') will be denoted by F ; F is a continuum contained in (Γ, r') . At any point of F , $G(x, y)$ is holomorphic. Hence A contains F as long as M is sufficiently large. Let A_0 be a domain of the form A and containing F . We shall prove that the domain of holomorphy A_0 is not rationally convex.

Let us suppose that A_0 is rationally convex. Then, thanks to the theorem of Weil explained in No. 4 of Memoir I, we can represent the function $G(x, y)$ in the neighbourhood of F^1) as the limit of a sequence of holomorphic rational functions:

$$(S) \quad f_1(x, y), f_2(x, y), \dots, f_n(x, y), \dots$$

converging uniformly.

Let ξ be a point on the circumference C such that $\Gamma(\eta)$ is contained in the annulus r ; ξ certainly exists, because the distance between the two pieces of the boundary of r is bigger than $4d$, and, on the other hand, the radius of $\Gamma(\eta)$ equals d . Let ξ_1 be the centre of this circle Γ . Then, in the sequence

$$(S') \quad f_1(\xi, y), f_2(\xi, y), \dots, f_n(\xi, y), \dots$$

one can find at least one function having poles in the circle $\Gamma(\eta)$ in the y -plane. In fact, if not, every function of (S') would be holomorphic in $\Gamma(\eta)$. Now, each of these functions is a priori holomorphic on the circumference since the circumference, considered in (x, y) space, belongs to F , and for this reason, the sequence (S') converges uniformly to $G(\xi, y)$ on the circumference. $G(\xi, y)$ would therefore be holomorphic in the circle, in contradicting its having the pole $1/(y - \eta + 1)$ at the centre. Let then $f_m(\xi, y)$ be a function of (S') having poles in $\Gamma(\eta)$; let us set

$$\psi(x, y) = f_m(x, y) \cdot \varphi(x, y),$$

φ and ψ being two polynomials without common factors.

Let (x', y') be a point having the same properties as (ξ, η) but otherwise arbitrary. Consider the equation

$$\psi(y; x) = 0,$$

in which we consider x to be an independent parameter. Since $\psi(y; x)$ is non-zero on F , no root of this equation can leave the circle $\Gamma(\eta)$ when x describes the arc of C from ξ to x' . Consequently, the equation $\psi(x; y') = 0$ has at least one root in the circle $\Gamma(y')$ of the y -plane, x' being arbitrary. The number of these roots is independent of x' , and we shall denote it by A .

Let us draw, in the y -plane, a circumference C' around the origin, of radius $10d$, and let us consider the equation $\psi(x; y) = 0$ in the circle (C') . When x describes C once positively, A roots of the algebraic equation $\psi = 0$ in (C') will leave this circle; this is, of course, absurd. Thus, A_0 is not rationally convex.

Theorem 3.1. *A domain of holomorphy is not necessarily rationally convex, even if it is univalent (and without points at infinity).*

Chapter 4

The Cauchy Integral

L'intégrale de Cauchy

Japanese Journal of Mathematics 17 (1941), pp. 523–531

In the present memoir, we shall be concerned with the problem of representations of holomorphic functions in a univalent domain of holomorphy. Now, as we have seen, it is in general impossible to do this by means of series of rational functions. We shall therefore go over to Cauchy integrals, more precisely to the integral of Mr. A. Weil.

4.1 The Weil Integral

We shall begin by describing the integral of Mr. Weil by translating a part of his article¹.

“Let $\chi_1, \chi_2, \dots, \chi_N$ be N holomorphic functions (non-constant) in a domain D (univalent and bounded) in the space of two complex variables x, y ; let Δ_i be a closed (univalent bounded) domain in the χ_i plane, whose boundary C_i consists of a finite number of analytic arcs. Let Δ be a connected component, or the sum of several connected components, of the set in D defined by the conditions: $\chi_i(x, y) \in \Delta_i$. We suppose that Δ is completely interior to D , and, denoting by S_i the set of boundary points of Δ for which $\chi_i \in C_i$, that no two of the S_i have common elements of more than two dimensions (it is always sufficient, in order to fulfil this condition, to make infinitesimal displacements of the contours C_i as needed). Let σ_{ij} be the sum of the two dimensional elements of the intersection of S_i and S_j , the orientation being suitably defined by the sequence $S_i \cdot S_j \times S_j$.”

We shall say that Δ is of type (α) , D being a bounded open set containing the closed set Δ , but otherwise arbitrary.

“Suppose, on the other hand, that we can make correspond to each $\chi_i(x, y)$, two functions $P_i(x, y; x_0, y_0)$, $Q_i(x, y; x_0, y_0)$ which are holomorphic when $(x, y) \in D$ and $(x_0, y_0) \in D$, and such that, we have

$$\chi_i(x, y) - \chi_i(x_0, y_0) = (x - x_0)P_i + (y - y_0)Q_i$$

identically.”

We shall call this condition (β) .

“Under these conditions, let $f(x, y)$ be a single-valued function holomorphic at every point of Δ . Consider the sum of integrals:

$$I = \frac{1}{4\pi^2} \sum_{i,j} \iint_{\sigma_{ij}} \frac{P_i Q_j - P_j Q_i}{[\chi_i(x, y) - \chi_i(x_0, y_0)][\chi_j(x, y) - \chi_j(x_0, y_0)]} f(x, y) dx dy,$$

the summation being over all pairs of indices (i, j) ; we have

$$I = f(x_0, y_0), \quad \text{or} \quad I = 0,$$

¹L'intégrale de CAUCHY et les fonctions de plusieurs variables, 1935 (Math. Annalen).

according as (x_0, y_0) is a point interior to Δ , or is exterior to Δ ."

This is the result of Mr. Weil. Put briefly, if the "polyhedron" Δ is of type (α) and satisfies condition (β) , then Δ possesses a Cauchy integral $I(x_0, y_0)$. The integral is taken along the σ_{ij} , intersection of the "faces" S_i and S_j .

4.2 The New Condition (γ)

We propose the following new condition (γ) :

Suppose that we can make correspond to each $\chi_i(x, y)$ 2 functions $P_i(x, y; x_0, y_0)$, $Q_i(x, y; x_0, y_0)$, holomorphic when $(x, y) \in D$ and $(x_0, y_0) \in D$ and such that we have

$$(\chi_i - \chi_i^0)R = (x - x_0)P_i + (y - y_0)Q_i$$

identically, where χ_i^0 stands for $\chi_i(x_0, y_0)$, and $R(x, y; x_0, y_0)$ is a new function, holomorphic when $(x, y) \in D$ and $(x_0, y_0) \in D$, independent of χ_i and such that $R(x_0, y_0; x_0, y_0) = 1$.

We construct a Cauchy integral for the "polyhedron" Δ of type (α) under this new hypothesis. Let us form the integral of the same form as I with the new functions P_j , Q_j :

$$J = \frac{1}{4\pi^2} \sum_{i,j} \iint_{\sigma_{ij}} \frac{P_i Q_j - P_j Q_i}{[\chi_i(x, y) - \chi_i(x_0, y_0)][\chi_j(x, y) - \chi_j(x_0, y_0)]} f(x, y) dx dy.$$

One finds, exactly as before, that

$$J = f(x_0, y_0), \quad \text{or} \quad J = 0$$

according as (x_0, y_0) is an interior point of Δ , or is exterior to Δ .

The set of the integrals J contains the integrals I . We shall still call this integral J the Weil integral.

We shall study condition (γ) .

4.3 Remarks on Theorem I of Memoir II

We first make a remark on Theorem I of Memoir II.

Consider a closed set Δ belonging to the class (H_0) in the space of 2 complex variables of the form:

$$(\Delta) \quad |x| < r, \quad |y| < r', \quad |f_j(y; x)| < 1, \quad (x, y) \in G, \quad (j = 1, 2, \dots, \nu),$$

where G is a bounded open set in the space (x, y) , the f_j are holomorphic on G and r and r' are positive numbers. It is understood that Δ is contained in the interior of D so as to be a closed set. We raise the dimension of the space in which we work by introducing ν new complex variables $z_1, z_2, \dots, z_\nu$, and consider a closed set Σ on a characteristic variety of dimension 4 in the space $[x, y, (z_i)]$ as follows:

$$(\Sigma) \quad z_j = f_j(x, y), \quad (x, y) \in \Delta, \quad (j = 1, 2, \dots, \nu).$$

Then Σ belongs to the class $(\mathcal{P})$, that is to say, one can find a sequence of open sets, convex with respect to polynomials in x, y and z_i , which tends towards Σ from the exterior. This is Theorem I of Memoir II.

We are going to make the characteristic variety on which Σ lies algebraic by a sufficiently small deformation. Since Σ is of class $(\mathcal{P})$, we can choose a suitable open set V containing Σ and expand the holomorphic functions $z_j - f_j(x, y)$ ($j = 1, 2, \dots, \nu$) in a series of polynomials

in x, y, z_j converging uniformly on V . We can therefore find polynomials $F_j[x, y, (z)]$ sufficiently close to the initial functions.

We shall now look at the system of roots of the simultaneous equations

$$(1) \quad F_j(x, y, z_1, z_2, \dots, z_\nu) = 0 \quad (j = 1, 2, \dots, \nu), \quad (4.1)$$

in which we consider x and y as independent parameters.

We want to start with the functions $F_j[x, y, (z)]$ instead of the $z_j - f_j(x, y)$.

Let (x, y) be an arbitrary point of Δ . In the space (z) , we consider the polycylinder

$$(\gamma) \quad |z_j - f_j(x, y)| < \rho, \quad (j = 1, 2, \dots, \nu),$$

choosing ρ sufficiently small so that the open set V contains the set of points $[\Delta(x, y)]$ completely in its interior. We then take the F_j sufficiently close to the $(z_j - f_j)$ respectively for the following three conditions to be satisfied on the open set V :

1°. Let ε be a positive number, given a priori, so that $\varepsilon < \rho$ and let p be any one of $1, 2, \dots, \nu$; if $z_p = f_p(x, y)$, we have $|F_p[x, y, (z)]| < \varepsilon$.

2°. Let, moreover, ρ' be a positive number given a priori so that $\rho' < \rho$; if $|F_p| < 1$, we have $|z_p - f_p(x, y)| < \rho$; if $|F_p| = 1$, we have $|z_p - f_p(x, y)| > \rho'$.

3°. $J = \frac{\partial(F_1, F_2, \dots, F_\nu)}{\partial(z_1, z_2, \dots, z_\nu)} \neq 0$.

These are the 3 conditions. The first two conditions are certainly realisable; as for the third, we have only to remark that $J \rightarrow 1$.

Under these conditions, we shall prove the following concerning the system of roots

$$z_j = \varphi_j(x, y) \quad (j = 1, 2, \dots, \nu)$$

of the simultaneous equations (1): there is one, and only one, such system in (γ) ; moreover, the functions $\varphi_j(x, y)$ are holomorphic when $(x, y) \in \Delta$.

We consider the auxiliary system of simultaneous equations

$$(2) \quad F_j[x, y, (z)] = u_j, \quad (j = 1, 2, \dots, \nu),$$

where the u_j are independent complex variables. Let us take an arbitrary point (x_0, y_0) in Δ , which we consider as being fixed. The only variables then are z_j and u_j ; we denote the polycylinder (γ) corresponding to the point (x_0, y_0) by (γ_0) . In (γ_0) , for $|u_j| < \sigma$ ($j = 1, 2, \dots, \nu$), there exists one and only one system of holomorphic solutions of (2), by virtue of 3°. For $u_j = 0$, this system reduces to $z_j = f_j(x_0, y_0)$.

We have thus, in $|u_j| < \sigma$:

$$z_j = \varphi_j(x_0, y_0; u_1, \dots, u_\nu), \quad (j = 1, 2, \dots, \nu),$$

holomorphic in the u_j , and satisfying:

$$z_j = f_j(x_0, y_0) \quad \text{for} \quad u_1 = \dots = u_\nu = 0.$$

In $|z_j - f_j(x, y)| < \rho$ ($j = 1, 2, \dots, \nu$); the functions $\varphi_j(x, y)$ are holomorphic for $(x, y) \in \Delta$.

4.4 Construction of the Functions P_i, Q_i, R

Now, on the variety defined by (1), we have $F_j = 0$ ($j = 1, 2, \dots, \nu$), and we have

$$(z_k - z_k^0)N = (x - x_0)a_k + (y - y_0)b_k,$$

Chapter 5

Pseudoconvex Domains

Domaines pseudoconvexes

Tohoku Mathematical Journal 49 (1942), pp. 15–52

Introduction

In 1906, F. HARTOGS discovered that domains of holomorphy are subject to a very curious restriction¹; I believe that it was with this very discovery that the recent development of the theory of analytic functions of several variables was begun.

The same restriction was found at the foundations of different branches of the theory successively by E. E. LEVI, G. JULIA, W. SAXER and the author². We shall call a domain restricted in this way pseudoconvex³.

Convexity of this sort can be analysed locally. Now, in 1932, H. CARTAN and P. THULLEN found that domains of holomorphy are even, in a certain sense, globally convex⁴. We established several global theorems concerning domains of holomorphy because of this property⁵.

We thus have several classes of pseudoconvex domains about which we know almost nothing, except on domains of holomorphy⁶. We are thus brought back to F. HARTOGS, to ask whether or not conversely every pseudoconvex domain is a domain of holomorphy. And if these 2 types of domains coincided, we would have a local criterion for domains of holomorphy. Among the different classes of pseudoconvex domains, those defined by G. JULIA have been shown to consist of domains of holomorphy, when they are in finite space, by H. CARTAN, P. THULLEN,

¹F. HARTOGS: Einige Folgerungen aus der CAUCHYschen Integralformel bei Funktionen mehrerer Veränderlichen, 1906 (Münch. Berichte).

²E. E. LEVI: Studi sui punti singolari essenziali delle funzioni analitiche di due o più variabili complesse, 1910 (Annali di Matematica).

G. JULIA: Sur les familles de fonctions analytiques de plusieurs variables, 1926 (Acta Mathematica).

W. SAXER: Sur les familles de fonctions méromorphes de plusieurs variables, 1931 (Comptes Rendus, Paris).

K. OKA: Note sur les familles de fonctions analytiques multiformes etc., 1934 (Journal of Science of the Hiroshima University).

³For univalent domains in finite space, we have defined them in Memoir IV; see No. 10 of the present memoir.

⁴As well as the converse. See H. CARTAN–P. THULLEN: Regularitäts- und Konvergenzbereiche, 1932 (Math. Annalen). For the idea, see H. CARTAN: Sur les domaines d'existence des fonctions de plusieurs variables complexes, 1931 (Bull. Société mathématique de France).

⁵Previous memoirs in the present research: I. Rationally convex domains, 1936; II. Domains of holomorphy, 1937; III. The second COUSIN problem, 1939 (Journal of Science of the Hiroshima University); IV. Domains of holomorphy and rationally convex domains, 1941; V. The CAUCHY integral, 1941 (Japanese Journal of Mathematics).

⁶It is also known, for example, that if in a univalent finite domain in the space of 2 complex variables, the first COUSIN problem is always solvable, the domain is necessarily a domain of holomorphy. See H. CARTAN: Les problèmes de Poincaré et de Cousin pour les fonctions de plusieurs variables, 1934 (Comptes Rendus, Paris).

H. BEHNKE and K. STEIN⁷. But outside this case, the problem remains more or less open up until now⁸.

In the present memoir, we shall treat this problem. We limit ourselves to the space of 2 complex variables for simplicity, but the conclusion, I think, applies to spaces of any number of variables. We shall see that finite univalent, pseudoconvex domains are domains of holomorphy⁹.

The problem we have just explained is the last of those which formed the subject of the present researches¹⁰.

5.1 The Main Problem

5.1.1 The Problem

We shall continue to suppose that all domains in the present memoir are univalent and lie in the finite part of space.

In our researches, the first Cousin problem plays a fundamental role nearly everywhere. The essential part of this consists in the following problem.

Let us consider a bounded domain Δ in the space of two complex variables x, y . The upper and lower parts of this domain relative to the hyperplane $X_2 = 0$, where $x = X_1 + iX_2$, i being the imaginary unit, will be denoted by A_1 and A_2 respectively. Given a holomorphic function $f(x, y)$ “in the neighbourhood” of the common boundary of A_1, A_2 , find a pair of functions $F_1(x, y), F_2(x, y)$, holomorphic in A_1 and A_2 respectively, which remain holomorphic at every point of Δ on $x_2 = 0$, in such a way as to have

$$F_1(x, y) - F_2(x, y) = f(x, y)$$

identically.¹¹

In 1895, P. COUSIN solved this problem for cylindric domains¹². In 1934, H. CARTAN indicated that this problem is solvable for rationally convex domains by applying the Weil integral¹³. The same method can be used for domains of holomorphy because of what we have seen in Memoir V. And, more recently, we solved this problem by a different method for domains of holomorphy in Memoirs I, II.

We shall solve this problem for a pseudoconvex domain composed of two domains of holomorphy which overlap on each other, of which one contains A_1 , the other A_2 ; the domain will be otherwise more or less arbitrary.

5.1.2 The Hypotheses

In the present section, we shall treat this problem with the following hypotheses:

⁷This is called the problem of JULIA. See H. CARTAN–P. THULLEN cited above.
H. BEHNKE–K. STEIN: *Konvergente Folgen von Regularitätsbereichen und die Meromorphiekonvexität*, 1938 (Math. Annalen); *Approximation analytischer Funktionen in vorgegebenen Bereichen des Raumes von n komplexen Veränderlichen*, 1939 (Göttinger Nachrichten); *Die Konvexität in der Funktionentheorie mehrerer komplexen Veränderlichen*, 1940 (Mitteilungen der Mathematischen Gesellschaft Hamburg).

⁸For this problem, see: the book of H. BEHNKE–P. THULLEN: *Theorie der Funktionen mehrerer komplexen Veränderlichen*, 1934, pages 54–56; also H. BEHNKE–K. STEIN, 1940, cited above.

⁹The author has published this result in the note: *Sur les domaines pseudoconvexes*, 1941 (Proc. Imp. Acad. Tokyo).

¹⁰The author undertook these researches thanks to the interest won by the work of H. BEHNKE–P. THULLEN cited above.

¹¹We say that a phenomenon occurs in the neighbourhood of a set E if it does on a certain open set containing E and its points of accumulation.

We say that a phenomenon occurs “in the interior of” an open set $\mathcal{D}$ if it does on any bounded open set which is contained, together with its boundary, in $\mathcal{D}$.

¹²Acta Mathematica.

¹³H. CARTAN, cited above.

1°. Let Δ be a bounded domain such that the part above $X_2 = 0$ and that below $X_2 = 0$ consist of domains of holomorphy. Let $\chi_j(x, y)$ ($j = 1, 2, \dots, \nu$) be holomorphic functions in a domain $\mathcal{D} \supset \Delta$, and let $\mathcal{D}'$ be a domain completely interior to $\mathcal{D}$. Let us consider the domain defined by:

$$|x_j(x, y)| < 1 \quad (j = 1, 2, \dots, \nu), \quad (x, y) \in \mathcal{D}'.$$

Δ is supposed to be a connected component of this domain. The domain Δ is thus pseudoconvex.¹⁴

2°. Let x_0, y_0 be new variables. We suppose that we have, identically,

$$(X_j - X_j^0)R = (x - x_0)P_j + (y - y_0)Q_j \quad (j = 1, 2, \dots, \nu),$$

where X_j^0 stands for $X_j(x_0, y_0)$, P_j , Q_j and R are holomorphic functions of the variables x, y, x_0, y_0 when $(x, y) \in \mathcal{D}$, $(x_0, y_0) \in \mathcal{D}$ and the function R reduces to 1 when $x = x_0$ and $y = y_0$.

Concerning this hypothesis, we refer to the lemma of Memoir V.

3°. We suppose that the derivative $\partial\chi_j/\partial y$ does not vanish identically for any j . Let L_j be the analytic variety defined in $\mathcal{D}$ by:

$$x_2 = 0, \quad |\chi_j| = 1;$$

by our hypothesis, L_j , if it is non-empty, consists of connected components which are domains of holomorphy.¹⁵

$\partial\chi_j/\partial y$ is non-zero on the variety L_j except, perhaps, at a finite number of points; let us take a point P on L_j different from the exceptional points but otherwise arbitrary. Near P , we can represent the analytic variety L_j by means of real parameters u, v in the form

$$x = x(u, v), \quad y = y(u, v),$$

where the right hand terms are power series in u, v , in such a way the correspondence between (x, y) and (u, v) is one-to-one. We can, for instance, take (x_1, θ_j) as parameters, θ_j being the argument of χ_j . We regard (u, v) as being a point in the plane given by rectangular axes in cartesian geometry. To the boundary of L_j correspond a finite number of analytic arcs in the (u, v) plane.

5.1.3 The Cauchy Integral

We set

$$\langle \chi_j \rangle(x, y; x_0, y_0) = \frac{Q_j}{\chi_j - \chi_j^0} \cdot \varphi(x, y);$$

denote by Γ the projection on the y -plane of the intersection of L_j with $x = \xi$. The function $Q_j/(\chi_j - \chi_j^0)$ is holomorphic with respect to (x_0, y_0) when $(x_0, y_0) \in L_j$ and $(x, y) \in \Gamma$ because $|\chi_j| = 1$. $\varphi(x, y)$ is holomorphic in a neighbourhood of S ; $\langle \chi_j \rangle(x, y; x_0, y_0)$ is therefore holomorphic at every point of the set

$$x = \xi, \quad y \in \Gamma; \quad x_0 = \xi, \quad y_0 = \eta.$$

In the y -plane, we consider a circle $\Gamma(y')$ around η and an open set E containing Γ . In order that $\langle \Phi \rangle(y; x; x_0, y_0)$ be holomorphic on the open set:

$$(I) \quad x \in \Gamma(y), \quad y \in E; \quad x_0 \in \Gamma(y'), \quad y_0 \in E,$$

¹⁴More or less arbitrary; the same remark applies in what follows.

¹⁵More precisely, it consists of connected components which are domains of holomorphy.

it is sufficient that $\chi_j(x, y) \neq \chi_j(x_0, y_0)$ on this open set.

Two cases are possible. Either this relation holds identically, in which case $\mathcal{P}$ consists of a finite number of points since $\partial\chi_j(x', y)/\partial y$ vanishes for any point of $\langle\chi\rangle$; or this relation is an equation representing a finite number of analytic curves in the y -plane; in this generic case, the figure $\langle\chi\rangle$ is composed of a finite number of analytic arcs (and of points).

The intersection of the analytic varieties L_j, L_k ($j \neq k$; $j, k = 1, 2, \dots, \nu$), if it is non-empty, is of dimension 1; consequently, the analytic variety given by

$$|\chi_j(x', y)| = 1, \quad |\chi_k(x', y)| = 1$$

can contain analytic curves only. The intersections of these curves with $\mathcal{P}$, if they exist, are isolated points (and their number is finite for each x').

We have, by CAUCHY'S theorem:

$$\frac{1}{2\pi i} \int \frac{R(\xi, x_0, y; x_0, y_0)}{\chi_j(x, y) - \chi_j(x_0, y_0)} \varphi(x_0, y) dy = \varphi_*(x_0, y_*),$$

the integral being taken in the positive sense, because $R = 1$ when $x = x_0, y = y_0$. From this, it immediately results that

$$I_2(x_0, y_0) = \frac{1}{2\pi i} \int \varphi \frac{X - x_0}{\chi_j(x, y) - \chi_j(x_0, y_0)} dx,$$

where the integral is taken positively since $\partial(x_1, \theta_j)/\partial(u, v) > 0$.

Therefore, the integral $I(x_0, y_0)$ is a solution of the problem for the domain Δ .

5.1.4 The Case of Pseudoconvex Domains

We shall modify the integral to construct similarly functions B_j ($j = 1, 2, \dots, \nu$) with respect to $\mathcal{D}_2$, and consider the following integrals:

$$J_1(x_0, y_0) = \frac{1}{4\pi^2} \sum_j \iint_{\sigma_j} (\psi_j + A_j) \varphi(x, y) dx dy, \quad (5.1)$$

$$J_2(x_0, y_0) = \frac{1}{4\pi^2} \sum_j \iint_{\sigma_j} (\psi_j + B_j) \varphi(x, y) dx dy, \quad (5.2)$$

($j = 1, 2, \dots, \nu$), where $\varphi(x, y)$ is an arbitrary function holomorphic in a neighbourhood of S .

When $(x, y) \in U_j$, the function $\psi_j + A_j$, being equal to $\Phi_j - \eta_j$, is holomorphic relative to (x_0, y_0) in A_1 ; hence $J_1(x_0, y_0)$ is holomorphic in A_1 . Similarly, $J_2(x_0, y_0)$ is holomorphic in A_2 .

The functions $J_1(x_0, y_0)$, $J_2(x_0, y_0)$ and the corresponding integrals for other domains can be used to solve the Cousin problem in the pseudoconvex case.

5.2 Convergence of the Sequence of Domains of Holomorphy

5.2.1 Bounded Domains

I think that this theorem has been proved precisely only for bounded domains. We shall therefore examine the case where the limit domain $\mathcal{D}$ is not bounded. Taking a point M of the domain $\mathcal{D}$ as centre we draw a polycylinder $(\mathcal{P}_\rho)$ of radius ρ , ρ being an arbitrary positive number. We denote by $\mathcal{D}_\rho$ the connected component of the part of $\mathcal{D}$ in $(\mathcal{P}_\rho)$ which contains M . Now, each $\mathcal{D}_\rho$, being bounded and of the same nature as $\mathcal{D}$, is a domain of holomorphy.

Consequently, taking into account the form of $\mathcal{D}_\rho$, $\mathcal{D}_\rho$ is convex with respect to functions holomorphic on $\mathcal{D}_\rho$.

5.2.2 Property (α)

Let us consider the interior of $\mathcal{D}$ which contains $\mathcal{D}(X, r)$ and has property $(\mathcal{P})$. Every set $\mathcal{D}(1, \rho)$, ρ being any positive number, has property (α) by CARTAN–THULLEN; a set F as above therefore certainly exists. Consider the intersection F_0 of all the F . F_0 is a closed set containing $\mathcal{D}(X, r)$. Any intersection F_1 of finitely many of the sets F is a set of the same nature. We can find an F_1 arbitrarily close to F_0 . F_0 therefore has property $(\mathcal{P})$. Consequently, it is sufficient to prove that F_0 consists of $\mathcal{D}(X, r)$ and its points of accumulation. Suppose that this were not so.

5.2.3 Properties of the Sets A, B, C

We have:

1°. If $\chi = 0$, we have

$$(a) = (b) = (c)$$

since $|X_1| = |X_2| = 1$.

2°. If $\chi > 0$, since $|X_1| > 1 > |X_2|$, hence $\rho|X_1| > \rho > \rho|X_2|$, we have

$$(a) < (c) < (b).$$

More precisely, if (a) is non-empty, then so is (c) and contains (a) completely in its interior; similarly, if (c) is non-empty, then so is (b) and contains (c) completely in its interior.

3°. If $\chi < 0$, we have, similarly,

$$(b) < (c) < (a).$$

Let P, Q be two fixed points of such that one of them is below $\mathcal{E}$ and holomorphic elsewhere. Since A_1 consists of domains of holomorphy and the given poles do not lie near $\mathcal{E}$, $\Phi(y; x)$ certainly exists. This function Φ has M as a pole and is holomorphic at every point of F ; here M is an arbitrary point of the common boundary of G_1 and B_2 . Thanks to CARTAN and THULLEN, we can therefore find an open set consisting of domains of holomorphy, which is contained in G_1 and contains F . Since F is an arbitrary closed set in G_1 , G_1 itself consists of domains of holomorphy in view of the theorem of BEHNKE–STEIN. Similarly for G_2 . We also remark that the set G_1 , as well as G_2 , is the limit of a decreasing sequence of open sets.

5.2.4 Hypothesis 2°

Let us pass now to hypothesis 2°. We have seen the following in Memoir V:

“If $\mathcal{V}$ is a domain of holomorphy in the space (x, y) , then, given a positive number ε and a domain $\mathcal{V}_0$ contained completely in the interior of $\mathcal{V}$, we can find a holomorphic function R of the complex variables $x, y; x_0, y_0$ in $(\mathcal{V}_0, \mathcal{V}_0)$, which reduces to 1 for $x = x_0, y = y_0$ in such a way that to every holomorphic function $f(x, y)$ in $\mathcal{V}$ there corresponds a holomorphic function $\varphi \dots$ ”

We choose the numbers $\chi_1, \chi_2, \dots, \nu$ successively, each time avoiding a finite number of values; the three earlier properties of Δ are not changed by this modification, so long as one takes the χ_j sufficiently close to 1.

5.2.5 Hypothesis 3°

We have now to make Δ satisfy the condition that $\partial\chi_j/\partial y$ be non-zero on the variety L_j , except, perhaps, at a finite number of points, j being arbitrary. Now, the characteristic variety $\partial\chi_j/\partial y = 0$ may contain characteristic planes of the form $x = \xi$ where ξ is a point on the real axis. To avoid this in G_3 , it is sufficient to take the functions

$$\chi_j(x + \beta y, y)$$

instead of the $\chi_j(x, y)$, β being a suitable sufficiently small purely imaginary number.

5.2.6 Construction of the Meromorphic Function

Let (p) be poles given in a neighbourhood of Δ ; there exists a meromorphic function $\varphi_1(x, y)$ having the poles (p) in a neighbourhood of Δ_1 . Similarly, let $\varphi_2(x, y)$ be a function corresponding to Δ_2 . Let $f = \varphi_1 - \varphi_2$; f is a holomorphic function in a neighbourhood of the common boundary S of Δ_1 and Δ_2 . We can find the solution $F_1(x, y), F_2(x, y)$ of the problem of Section I on Δ corresponding to the function f . Let us set

$$\varphi = \varphi_1 - F_1 \quad \text{or} \quad \varphi = \varphi_2 - F_2$$

according as $(x, y) \in \Delta_1$ or $\in \Delta_2$. $\varphi(x, y)$ is meromorphic in Δ and has the poles (p) . We have thus seen that if we are given poles in the neighbourhood of Δ , we can construct a meromorphic function having these poles. The interior of Δ is a domain of holomorphy; and the same is true of $\mathcal{D}$ ¹⁶.

Theorem 5.1. *Given a bounded domain $\mathcal{D}$ and 2 hyperplanes of the form $x_2 = a_1, x_2 = a_2, a_1 < a_2, x_2$ being the imaginary part of x , if the part of $\mathcal{D}$ above $X_2 = a_1$ and that below $x_2 = a_2$ consist of domains of holomorphy, then $\mathcal{D}$ is a domain of holomorphy.*

5.3 Domains Pseudoconvex in the Sense of H. Cartan

As we have often mentioned, it was H. CARTAN who introduced the notion of global convexity in our theory¹⁷. Now, in the same memoir, the characteristic plane $x = \xi$ in the neighbourhood of (ξ, η) , excepting (ξ, η) itself, then, to any given positive number ρ , there exists a sufficiently small positive number r such that for any x' in $|x - \xi| < r$, there exists at least one y' in $|y - \eta| < \rho$ for which (x', y') belongs to E .

The principal properties of pseudoconvex domains were studied by F. HARTOGS and E. E. LEVI, and will be found treated very systematically in the memoir of G. JULIA cited earlier¹⁸.

Let us consider in the space (x, y) a bounded domain Δ such that its boundary S is given by $\varphi(X_1, X_2, Y_1, Y_2) = 0$, where $x = X_1 + iX_2$ and $Y = Y_1 + iY_2$, in such a way that to each point (x, y) , there corresponds a value of φ , finite or infinite but only one¹⁹.

5.4 The Hartogs Radius

Let $R_y(x)$ be the Hartogs radius of holomorphy of F. Hartogs. Hartogs has proved that

$$-\log R_y(x)$$

is a subharmonic function with respect to x , (the determination of the logarithm being real).

Let us make a remark about the definition of subharmonic functions. Let A be a domain in the x -plane. We call a function $\varphi(x)$ subharmonic with respect to x on A if it is real, well-defined on A and satisfies the following conditions:

1°. $\varphi(x)$ is finite and upper semicontinuous.

2°. Let Γ be a circumference contained, together with its interior, in A ; let ξ be an arbitrary point of the interior of Γ . If we denote by $\chi(z)$ a function, holomorphic on Γ and in its interior, satisfying $|\chi(z)| \geq \varphi(z)$ on Γ , we also have $|\chi(\xi)| \geq \varphi(\xi)$.

¹⁶The corresponding result due to H. CARTAN (Note 6, Introduction) cannot be applied to the present case.

¹⁷Bull. Soc. Math. France, 1931.

¹⁸Acta Math. 1926. In what follows, the author will often content himself with referring the reader to this memoir.

¹⁹That is to say, to each point (x, y) , there corresponds a value of φ , finite or infinite but only one.

Let $\mathcal{D}$ be a pseudoconvex domain in the space (x, y) , and let $R_y(x)$ be the Hartogs radius with respect to $\mathcal{D}$. The function

$$\Phi(x, y) = -\log R_y(x)$$

is pseudoconvex in $\mathcal{D}$ (the determination of the logarithm being real).

First, $1/R_y(x)$ is a well-defined finite function on $\mathcal{D}$. To examine the semicontinuity, let us first suppose that $\mathcal{D}$ is bounded. Let (ξ, η) be an arbitrary point of $\mathcal{D}$, and (ρ_p, η_p) ($p = 1, 2, \dots$) a sequence of points of $\mathcal{D}$ tending to (ξ, η) . Then

$$\limsup_{p \rightarrow \infty} [-\log R_{\eta_p}(\rho_p)] \leq -\log R_{\eta}(\xi),$$

from which the upper semicontinuity results.

To examine $\varphi(0, y)$ near $y = 0$: $R_y(0)$ represents the distance of the point y to the boundary of $\mathcal{D}(0)$ and $\mathcal{D}(0)$ contains the origin $y = 0$. Hence $1/R_y(0)$ is continuous. This function reduces to the constant 0 if $\mathcal{D}(0)$ coincides with the y -plane. Let us therefore suppose that $\mathcal{D}(0)$ has (finite) boundary points; let η be an arbitrary one of these points. Then, we obviously have

$$\varphi(0, y) = \max[-\log |y - \eta|],$$

where the symbol “max” represents the upper bound of the quantities inside the brackets, and the determination of the logarithm is real. The functions $-\log |y - \eta|$ are harmonic with respect to y , from which it follows easily that $\varphi(0, y)$ is subharmonic.

5.5 Special Properties of Pseudoconvex Functions

The function $\varphi(x, y) = -\log R_y(x)$ has the following special properties:

- 1°. It is continuous in $\mathcal{D}$, so long as $\mathcal{D}$ does not coincide with (finite) (x, y) -space.
- 2°. If the point (x, y) of $\mathcal{D}$ approaches the boundary of $\mathcal{D}$ in an arbitrary way, this function always tends to $+\infty$.

5.6 Differential Condition

Suppose that we are given a real continuous function $\varphi(X_1, X_2, Y_1, Y_2)$ in an arbitrary domain in the space (x, y) , which has continuous partial derivatives up to second order; we shall look for a condition that φ be pseudoconvex in x, y .

We take an arbitrary point P in $\mathcal{D}$, which we shall suppose to be the origin to simplify the notation. Let us consider a characteristic plane of the form

$$y = \alpha x, \quad \alpha = \alpha_1 + i\beta$$

on $\mathcal{D}$.

For φ to be subharmonic in x on every characteristic plane, it is first necessary that $D > 0$, since otherwise $m = -\infty$. Moreover:

- 1° if $D = 0$, we must have $B = C = 0$ and $A > 0$;
- 2° if $D > 0$, we must have $mD > 0$. Thus,

$$A > 0, \quad D > 0, \quad AD - (B^2 + C^2) > 0 \tag{5.3}$$

gives the necessary condition.

Suppose, conversely, that condition (4) is satisfied. Then, there are only two possibilities: either $D = 0$, which implies $m = A > 0$, since then, necessarily $B = C = 0$; or $D > 0$, and we then have $mD > 0$ by (3), and consequently $m > 0$. The condition (4) is thus sufficient.

Let us come back to our original problem, and let us suppose that the function φ is pseudoconvex in $\mathcal{D}$. Φ being then subharmonic with respect to x at the origin, we necessarily have, setting:

$$\begin{aligned} a &= \frac{\partial^2 \varphi}{\partial X_1^2} + \frac{\partial^2 \varphi}{\partial X_2^2}, & d &= \frac{\partial^2 \varphi}{\partial Y_1^2} + \frac{\partial^2 \varphi}{\partial Y_2^2}, \\ b &= \frac{\partial^2 \varphi}{\partial X_1 \partial Y_1} + \frac{\partial^2 \varphi}{\partial X_2 \partial Y_2}, & c &= \frac{\partial^2 \varphi}{\partial X_1 \partial Y_2} - \frac{\partial^2 \varphi}{\partial X_2 \partial Y_1}, \end{aligned}$$

we have

$$L(\varphi) = Ad + Da - 2Bb - 2Cc.$$

In what follows, we shall understand by a, b, c, d and A, B, C, D , their values at the point P . By hypothesis, at least one of a, d is non-zero. Suppose, to fix our ideas, that

$$d > 0.$$

We have

$$ad - (b^2 + c^2) \geq 0.$$

If we set

$$\alpha = \frac{-b - c}{d}, \quad \beta = \frac{-b + c}{d},$$

we therefore have

$$L(\varphi) = A + 2B\alpha + 2C\beta + D(\alpha^2 + \beta^2).$$

Let us look at the form $U(\alpha, \beta)$ of No. 12. If $D > 0$, the lower bound m is given by $mD = \dots$ Consequently, $a \dots$

5.6.1 Condition for Characteristic Planes

If the plane $y = \alpha x$ does not pass through any point of $\varphi < 0$ near the origin, excepting the origin, it is sufficient that $\varphi > 0$ in a sufficiently small circle of the x -plane around the origin, except at the centre. And for this, it is sufficient that

$$\mu_1 > 0, \quad \mu_1 \mu_3 - \mu_2^2 > 0. \quad (5.4)$$

By a simple calculation we have

$$\begin{aligned} \mu_1 &= \frac{\partial^2 \varphi}{\partial Y_1^2} \alpha^2 + \frac{\partial^2 \varphi}{\partial Y_2^2} \beta^2 + 2 \frac{\partial^2 \varphi}{\partial Y_1 \partial Y_2} \alpha \beta + 2 \frac{\partial^2 \varphi}{\partial X_1 \partial Y_1} \alpha + 2 \frac{\partial^2 \varphi}{\partial X_1 \partial Y_2} \beta + \frac{\partial^2 \varphi}{\partial X_1^2}, \\ \mu_3 &= \frac{\partial^2 \varphi}{\partial Y_1^2} \alpha^2 - 2 \frac{\partial^2 \varphi}{\partial Y_1 \partial Y_2} \alpha \beta + \frac{\partial^2 \varphi}{\partial Y_2^2} \beta^2 + 2 \frac{\partial^2 \varphi}{\partial X_2 \partial Y_2} \alpha - 2 \frac{\partial^2 \varphi}{\partial X_2 \partial Y_1} \beta + \frac{\partial^2 \varphi}{\partial X_2^2}, \\ \mu_2 &= \frac{\partial^2 \varphi}{\partial Y_1 \partial Y_2} (\alpha^2 - \beta^2) - \left(\frac{\partial^2 \varphi}{\partial Y_1^2} - \frac{\partial^2 \varphi}{\partial Y_2^2} \right) \alpha \beta + \frac{\partial^2 \varphi}{\partial X_1 \partial Y_2} \alpha + \frac{\partial^2 \varphi}{\partial X_2 \partial Y_1} \alpha - \frac{\partial^2 \varphi}{\partial X_1 \partial Y_1} \beta + \frac{\partial^2 \varphi}{\partial X_2 \partial Y_2} \beta. \end{aligned}$$

$\mu_2 = 0$ represents either a hyperbola or 2 perpendicular lines. We then have:

$\mu_2 = 0$ represents a hyperbola, μ_1 and μ_3 do not change sign on each of the branches of the hyperbola, and if $\mu_2 = 0$ represents a pair of lines, this is so on the whole curve.

Under these conditions, we shall prove the existence of a point (α, β) satisfying condition (1). This is sufficient.

We shall denote by

$$A_1(\alpha, \beta), \quad A_2(\alpha, \beta), \quad A_3(\alpha, \beta)$$

the terms of second degree in α, β of μ_1, μ_2, μ_3 respectively.

Consider the transformation

$$\alpha = \beta', \quad \beta = -\alpha',$$

which is a rotation of the plane about the origin by $\pi/2$. We have

$$A_2(\alpha', \beta') = -A_2(\alpha, \beta).$$

est > 0 . Si cette forme est partout définie positive, on dit que φ est strictement plurisous-harmonique.

On doit à E. E. LEVI les résultats suivants: si D est un domaine qui, au voisinage de chacun des points P de la frontière ∂D , est défini par une inégalité $\varphi(z) < \varphi(P)$ (avec φ strictement plurisous-harmonique au voisinage de P), alors D est “pseudo-convexe” au sens de Levi.

On peut montrer: pour que D soit pseudo-convexe au sens de Hartogs, il faut et il suffit que D soit pseudo-convexe en chaque point P de ∂D . Nous dirons désormais simplement: D est pseudo-convexe.

Ces définitions étant posées, revenons, avec OKA, au cas $n = 2$. On veut montrer que tout domaine D pseudo-convexe est un domaine d’holomorphie. En vertu du théorème de BEHNKE–STEIN, il suffit de l’éprouver lorsque D est pseudo-convexe et borné.

Soit D un domaine de $\mathbf{C}^2$, et soit $(x_0, y) \in D$. Pour chaque x assez voisin de x_0 , on a $(x, y) \in D$, et on peut considérer la borne supérieure des $r > 0$ tels que D contienne les (x, y') satisfaisant à $|y' - y| < r$. Soit $R_y(x)$ ce nombre (qui peut être infini si D n’est pas borné). HARTOGS a montré que si D est pseudoconvexe, $-\log R_y(x)$ est une fonction plurisous-harmonique de (x, y) . OKA montre que, réciproquement, si $-\log R_y(x)$ est plurisous-harmonique, D est un domaine d’holomorphie.

Une dernière remarque: le raisonnement d’OKA (n°14) est incorrect, car l’enveloppe d’holomorphie d’un domaine univalent n’est pas nécessairement univalente.

Chapter 6

On Some Arithmetical Notions

Sur quelques notions arithmétiques

Bulletin de la Société Mathématique de France 78 (1950), pp. 1–27

Introduction

We are now embarked on thinking hard about the subject; among other things, understanding the nature of the difficulties we have met on the path that we have followed¹, and studying the form of the difficulties we shall meet in continuing. We present here one of the results of this reflection.

It will be noticed that there are already certain arithmetical notions in Theorem II of Memoir I (first Cousin problem) and in Theorem II of the same memoir (second Cousin problem). This memoir is thus, so to speak, an amplification of the two theorems in question. We shall present, besides, some problems which we have been unable to solve^{2,3}.

6.1 Ideals of Holomorphic Functions

6.1.1 Preliminary Notions

Let us consider two holomorphic functions f, φ in a domain D in the space of n complex variables $X_1, X_2, \dots, X_n$; let $(F_1, F_2, \dots, F_p)$ be a finite collection of holomorphic functions on D . The space of the variables will be denoted briefly by (x) ; the function f will also be denoted by $f(x)$.

Suppose that we have a relation of the form

$$f - \varphi = \alpha_1 F_1 + \alpha_2 F_2 + \dots + \alpha_p F_p,$$

the α_i ($i = 1, 2, \dots, p$) being holomorphic functions on D ; we shall say that f is congruent to φ modulo (F) , and we shall write this

$$f \equiv \varphi \pmod{(F)}.$$

If $\varphi = 0$, we shall say that f belongs to the ideal (F) .

We shall call a domain of the form $|x_i - x_i^0| < r_i$ ($i = 1, 2, \dots, n$) a polycylinder if all its components are circles.

¹See the note at the end of Memoir VI.

²The first draft of this memoir was published in Japanese in 1940, pp. 1–26 (Journal of Mathematics, vol. 19); we owe much to this memoir.

³The author would like to express here his sincere thanks to HU JU-KAI for his help since around the time of Memoir VI.

Let us now consider two finite collections of holomorphic functions on a domain D in the space (x) ; we denote these by $(f_1, f_2, \dots, f_p)$, $(\varphi_1, \varphi_2, \dots, \varphi_q)$.

6.1.2 Equivalence of Systems

Suppose that we have the two relations

$$\varphi_i \equiv 0 \pmod{(f)}, \quad f_j \equiv 0 \pmod{(\varphi)}, \quad (i = 1, 2, \dots, p; j = 1, 2, \dots, q)$$

globally on D . We shall then say that the systems of functions (f) , (φ) are equivalent in D ; we denote these simultaneous relations by

$$(f) \sim (\varphi).$$

If we have only one of the two relations, for instance the first, we shall write this

$$(\varphi) \subset (f).$$

We shall call the systems (f) , (φ) equivalent at a point P of D , if they are equivalent in some neighbourhood of P ; similarly with the relation $(\varphi) \subset (f)$.

Regarding equivalence, we have the following problems:

Problem (C_1) . Given a finite system (f) of holomorphic functions in a domain D in the space (x) , find a finite system (φ) of holomorphic functions equivalent to (f) at every point of D .

Problem (C_2) . Given a finite system (f) of holomorphic functions in a domain D , find, for each point P of D , a neighbourhood V_P and a finite system (φ) of holomorphic functions in V_P equivalent to (f) at P , in such a way that any two of the systems (φ) are equivalent in the intersection of their domains of definition.

6.1.3 Cartan's Theorem and Cousin Integral

We can overlap two by two. We then consider the closed polycylinder Δ_j ($j = 1, 2, 3$) defined by having as components the closed circles (a_i) , (b_i) and suppose given a finite system (f) of holomorphic functions in the neighbourhood of Δ_j in such a way that any two of the systems (f) are globally equivalent in the neighbourhood of the corresponding intersection.

We shall try to apply CARTAN'S theorem under these conditions. This is certainly possible for $\Delta = \Delta_1 \cup \Delta_2$, and we find a finite system (f) of functions holomorphic in a neighbourhood of Δ such that $(f) \sim (f_j)$ globally in the neighbourhood of Δ_j ($j = 1, 2$). But we cannot always apply the theorem to $\Delta \cup \Delta_3$ because, depending on the geometric configuration, it is not obvious whether ...

We shall find holomorphic functions $a_i(x)$ in a neighbourhood of Δ' , $b_i(x)$ in one of Δ'' in such a way that

$$a_i(x) - b_i(x) = \alpha_i(x) \quad (i = 1, 2, \dots, p),$$

identically; let us form the functions

$$\begin{aligned} \psi' &= f' - (a_1 F_1 + \dots + a_p F_p), \\ \psi'' &= f'' - (b_1 F_1 + \dots + b_p F_p); \end{aligned}$$

ψ' is then holomorphic in a neighbourhood of Δ' , ψ'' in a neighbourhood of Δ'' . Now, for Δ_0 , we have

$$\psi' - \psi'' = \varphi - (\alpha_1 F_1 + \dots + \alpha_p F_p) = 0;$$

ψ' and ψ'' are therefore parts of the same function $f(x)$ which is holomorphic in a neighbourhood of Δ ; we have $f \equiv \psi' \equiv f' \pmod{(F)}$ in a neighbourhood of Δ' , and similarly for Δ'' . Q.E.D.

From this we immediately deduce:

Theorem 6.1. *If the problem (C_1) is solvable for any closed polycylinders, so is problem (C_2) .*

6.1.4 Properties of Ideals with Indeterminate Domains

Problems (C_1) , (C_2) ... We say that a function belongs to the ideal (I) with indeterminate domains at a point P if it belongs to (I) on a neighbourhood of P .

Let us consider the following properties of an ideal (I) with indeterminate domains:

Property (T_1) . If $(f, b) \in (I)$, $(f, b') \in (I)$, we necessarily have $f \in (I)$ on $b \cup b'$.

Property (T_2) . If $b_1 \subset b_2 \subset \dots$ and $(f, b_1) \in (I)$, $(f, b_2) \in (I)$, ... and b_0 is the limit of the sequence $b_1, b_2, \dots$, then, we necessarily have $(f, b_0) \in (I)$.

Let us examine these notions.

Example 6.2. Let us consider the circle (C) around the origin, of radius 1, in the plane of a complex variable x ; we consider in (C) an arbitrary domain δ of diameter d (which is the upper bound of the distances between two points of δ). Let (I) be the set of (f, b) for which $d < 1$. (I) is an ideal having property (T_1) .

We shall sometimes call (F) a local pseudobasis.

We can then pose also the following:

Problem (J) . Given an ideal (I) with indeterminate domains in the space (x) and a point P in (x) , find a finite pseudobasis of (I) at P .

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Problem (J) is a special case of problem (I); there is however the following relation between these two problems.

Consider, in the space (x) , a problem (I) relative to an ideal (I) and a closed set E which is a closed polycylinder S . Suppose that, for any point P of E , the problem (J) for the ideal (I) at the point P is solvable. Suppose also that problem (C_1) is always solvable for a closed polycylinder. One then finds that problem (I) is solvable for the ideal (I) and the polycylinder E .

In fact, consider any point P of E ; the problem (J) being solvable at P for the ideal (I) , one can find a polycylinder (γ) around P and a finite system (f) of holomorphic functions on (γ) such that (f) gives a pseudobasis of (I) on (γ) . Let us examine a pair $[(\gamma), (\gamma')]$ of contiguous polycylinders. The corresponding systems $(f), (f')$ of functions are equivalent at each point of the intersection $(\gamma) \cap (\gamma')$ since they both give pseudobases. We want to find a finite system (F) of holomorphic functions in a neighbourhood of E such that $(F) \sim (f)$ at every point of E . Now, this is a problem (E) for the closed polycylinder E ; because of our hypothesis that problem (C_1) is always solvable for a closed polycylinder, the system (F) exists as we have seen. Since $(F) \sim (f)$ at any point of E , (F) gives a pseudobasis at any point of E , thus is a pseudobasis for E . Q.E.D.

Problem (J) is thus the part of problem (I) not covered by problem (C_1) . However, it is not always solvable, as seen by the following example.

Example 6.1. Let us consider two hyperspheres $(C), (\gamma)$ about the origin in the space of two complex variables x, y , such that $(C) \supset (\gamma)$. We denote by L_0 the part of the characteristic curve $x = y$ which is either between the two hyper-surfaces C, γ or on Γ . [C, Γ denote the boundaries of $(C), (\gamma)$ respectively.]

Consider the set (I) of pairs (f, δ) , where $\delta \subset (C)$ and such that $f/(x - y)$ is holomorphic at every point of the intersection on δ . (I) is an ideal (having properties (T_1) and (T_2)), but since the set of zeros of $(f, \delta) \in (I)$ contains L_0 on δ , this ideal cannot have finite local pseudobases at any point of Γ .

One cannot therefore always solve problem (J) without further conditions. We shall see further diverse kinds of counterexamples later.

We shall see that problem (J) on closed polycylinders can be solved for two kinds of ideals with indeterminate domains. One of them will be dealt with in the present memoir.

The other kind consists of geometric ideals with indeterminate domains (which correspond to the ideal of a variety in the space of polynomials). These are indispensable when we have to deal with domains admitting points of ramification. The proof that problem (J) is solvable on closed polycylinders for this kind of ideal requires, besides the results of the present memoir, some notions concerning domains with ramification. We shall therefore treat this question in a later memoir.

3. Homogeneous Linear Functional Equations and Formulae for Their Solutions

We shall scrutinize the circle of questions around problem (C_1) using the notions introduced above.

1°. Homogeneous linear functional equations.

Let $F_i(x)$ ($i = 1, 2, \dots, p$) be p holomorphic functions, not identically zero, in a domain D in the space (x) . Consider the equation

$$A_1 F_1 + A_2 F_2 + \dots + A_p F_p = 0, \quad (1)$$

where the A_i are unknown functions. A system of holomorphic functions $[A_1(x), A_2(x), \dots, A_p(x)]$ on a domain (connected or not) contained in D which satisfies equation (1) identically in (ξ) will be called a (holomorphic) solution of the equation on ξ . We call any equation of this nature a homogeneous linear functional equation.

Given equation (1), consider for every solution $(A_1, A_2, \dots, A_p)$ on δ the pair (A_1, ξ) ; let (I_1) be the set of pairs so obtained. I say that (I_1) is an ideal.

For, if $(A_1, \xi) \in (I_1)$ and $X(x)$ is holomorphic on a domain ξ' (connected or not), we have $(XA_1, \xi \cap \xi') \in (I_1)$; if $(A_1, \xi) \in (I_1)$, $(A'_1, \xi') \in (I_1)$, we have $(A_1 + A'_1, \xi \cap \xi') \in (I_1)$.

We do not see immediately whether (I_1) satisfies properties (T_1) , (T_2) . This was the kind of ideal we had in mind at the end of the preceding section. Problem (J) for these ideals is as follows :

Problem (K). Given a homogeneous linear functional equation in a domain D in the space (x) and a point P of D , find a finite pseudobasis of the corresponding ideal (I_1) at P .

2°. Formulae for solutions.

We propose now to find a representation of the solutions of equation (1). We considered the ideal (I_1) relative to the equation (1). Let us now consider the equation

$$A_2F_2 + A_3F_3 + \dots + A_pF_p = 0, \quad (2)$$

and the set (I_2) of the corresponding pairs (A_2, ξ) . (I_2) is an ideal by what we saw above. We continue this process; we have finally to consider the equation $A_pF_p = 0$. The corresponding ideal $(I_p) = (0)$ [consists of the single element 0] since the function F_p is not identically zero.

We now pose the following composite problem :

Problem (L). Find finite pseudobases of the ideals $(I_1), (I_2), \dots, (I_p)$ on any closed set E contained in the domain D consisting of functions which belong to the corresponding ideal globally in a neighbourhood of E .

Let us suppose that a solution $[A_1(x), A_2(x), \dots, A_p(x)]$ of the equation (1) at P (i.e. in a neighbourhood of P), (φ) being a pseudobasis of (I_1) , the function $A_1(x)$ can be represented, near P , in the form

$$A_1 = C_1\varphi_1 + C_2\varphi_2 + \dots + C_q\varphi_q,$$

the C_i ($i = 1, 2, \dots, q$) being holomorphic functions.

Let us next consider $A_2(x)$. Since each φ_i belongs globally to the ideal (I_1) in a neighbourhood of E , the Equation (1) has q solutions of the form $(\varphi_i, \psi_{2i}, \dots, \psi_{pi})$ ($i = 1, 2, \dots, q$) in a neighbourhood of E . Let us set

$$B_j = C_1\psi_{1j} + C_2\psi_{2j} + \dots + C_q\psi_{qj} \quad (j = 2, 3, \dots, p),$$

and one obtains a solution of equation (1) of the form $(A_1, B_2, \dots, B_p)$ near P . If we now set $A_j = A_j - B_j$, $(A_2, A_3, \dots, A_p)$ is a solution of equation (2) near P ; consequently, in a neighbourhood of P , the function $A_2(x)$ will be represented in the form

$$A_2 = B_2 + C_1\varphi'_1 + C_2\varphi'_2 + \dots + C_r\varphi'_r,$$

where the C_j ($j = 1, 2, \dots, r$) are holomorphic functions, and B_2 can be represented as above. And we can continue thus.

We see now that the solution $[A_1(x), A_2(x), \dots, A_p(x)]$ of equation (1) can be represented, near P , as follows :

$$A_i = C_{i1}\pi_{i1} + C_{i2}\pi_{i2} + \dots + C_{iN}\pi_{iN} \quad (i = 1, 2, \dots, p) \quad (3)$$

with certain identities of the form $C_{ij} = C_{kl}$ ($k = 1, \dots, p; l = 1, \dots, N$); here π_{ij} ($i = 1, 2, \dots, p; j = 1, 2, \dots, N$) are certain well-determined functions holomorphic in a neighbourhood of E , and the C_{ij} are holomorphic functions in a neighbourhood of P which depend on the solution $(A_1, A_2, \dots, A_p)$. For example, we have for A_1 :

$$\pi_{1i} = \varphi_i, \quad \pi_{1j} = 0 \quad (i = 1, 2, \dots, q; j = q+1, q+2, \dots, N);$$

for A_2 we have

$$\pi_{2i} = \psi_{2i}, \quad \pi_{2j} = \varphi'_{j-q}, \quad \pi_{2k} = 0$$

($i = 1, 2, \dots, q; j = q+1, \dots, q+r; k = q+r, \dots, N$), and

$$C_{1i} = C_{2i} \quad (i = 1, 2, \dots, q).$$

Let us consider, conversely, any point P in a neighbourhood of E and an arbitrary collection (C) of functions holomorphic at P as in (3) (respecting, naturally, the identities indicated). To this collection (C) , we make correspond a system (A) of holomorphic functions at P by (3); this system obviously satisfies the functional equation (1).

Suppose, in general, that to the equation (1), there corresponds a collection of functions π_{ij} ($i = 1, 2, \dots, p; j = 1, 2, \dots, N$) holomorphic on a domain D' (connected or not) given in D , and identities of the form $C_{ij} = C_{kl}$ as indicated.

2° (continued).

If C_{ij} are functions holomorphic at a point P of D and satisfying the indicated identities, then the expression (3) gives a solution (A) of equation (1) at P . Under these conditions we shall say that the expression (3) is a (linear) formula for the solutions of the functional equation (1) on the domain D' . We shall call the collection (π) the kernel of this formula.

Thus, the problem we have just been discussing can be formulated as follows :

Problem (L). Given a homogeneous linear functional equation in a domain and a closed set E in the domain, find a formula for the solutions of the equation in a neighbourhood of E .

We have proved the following :

Lemma 6.2. *If problem (A) is solvable for a certain homogeneous linear functional equation, so is the corresponding problem (L).*

3°. Application to problem (C₁).

Problem (C₁) is related to finding a formula for the solutions of a homogeneous linear functional equation by the following proposition.

Lemma 6.3. *Consider two closed cylindrical sets A' , A'' in the space (x) . Suppose that they intersect, and that all their components but one are the same, and set $A' \cup A'' = A$, $A' \cap A'' = A_0$. Consider a holomorphic function Φ and a finite collection $(F_1, F_2, \dots, F_p)$ of holomorphic functions on a neighbourhood of A . Suppose that we have relations*

$$\Phi = A'_1 F_1 + A'_2 F_2 + \dots + A'_p F_p$$

and

$$\Phi = A''_1 F_1 + A''_2 F_2 + \dots + A''_p F_p$$

where A'_i, A''_i ($i = 1, 2, \dots, p$) are functions holomorphic in a neighbourhood of A' , A'' respectively. Suppose also that we have a formula for the solutions of the functional equation

$$A_1 F_1 + A_2 F_2 + \dots + A_p F_p = 0 \tag{4}$$

in the neighbourhood of A_0 which has the following properties :

1. The kernel of this formula consists of functions holomorphic on A ;
2. Any solution of the equation in a neighbourhood of A_0 can be represented globally in a neighbourhood of A_0 in the form (3).

Under these conditions Φ can be represented in the form

$$\Phi = B_1 F_1 + B_2 F_2 + \cdots + B_p F_p,$$

where the B_i ($i = 1, 2, \dots, p$) are functions holomorphic in a neighbourhood of A .

In fact, let

$$A_i = \sum_j C_{ij} \pi_{ij} \quad (i = 1, 2, \dots, p; j = 1, 2, \dots, N) \quad (5)$$

with some identities among (C) be the formula in question, having the properties indicated. Let us consider the functions

$$\gamma_i = A'_i - A''_i \quad (i = 1, 2, \dots, p)$$

which are holomorphic in a neighbourhood of A_0 and satisfy there the equation (4). They can therefore be represented, globally in the neighbourhood of A_0 , in the form

$$\gamma_i = \sum_j d_{ij} \pi_{ij},$$

the d_{ij} being holomorphic in a neighbourhood of A_0 .

Since A_0 is cylindrical, we can find, using the Cousin integral, functions $a_{ij}(x)$, $b_{ij}(x)$ holomorphic in a neighbourhood of A' , A'' respectively, such that

$$a_{ij} - b_{ij} = d_{ij}$$

identically. Further, since the functions (d) satisfy the identities in the formula (5) for solutions, we can choose (a) and (b) so as to satisfy them also. Let us then set

$$\alpha_i = \sum_j a_{ij} \pi_{ij}, \quad \beta_i = \sum_j b_{ij} \pi_{ij};$$

we have

$$\alpha_i - \beta_i = \gamma_i;$$

moreover, we have

$$\sum \alpha_i = 0, \quad \sum \beta_i = 0$$

identically in a neighbourhood of A_0 by our requirements on the formula (5) for solutions of (4), and, since the functions (π) are holomorphic in a neighbourhood of A , the functions (α) are holomorphic in a neighbourhood of A' , and satisfy there the equation (4); the functions (β) have similar properties relative to A'' .

We set

$$B'_i = A'_i - \alpha_i, \quad B''_i = A''_i - \beta_i \quad (i = 1, 2, \dots, p).$$

The functions (B'), (B'') are holomorphic, respectively, in a neighbourhood of A' , A'' and we have

$$\Phi = B'_1 F_1 + B'_2 F_2 + \cdots + B'_p F_p$$

in a neighbourhood of A' ,

$$\Phi = B''_1 F_1 + B''_2 F_2 + \cdots + B''_p F_p,$$

in a neighbourhood of A'' . Now, since

$$B'_i - B''_i = (A'_i - \alpha_i) - (A''_i - \beta_i) = \gamma_i - \gamma_i = 0$$

for each i , the functions B'_i and B''_i are but parts of the same holomorphic function B_i in a neighbourhood of A . Thus

$$\Phi = B_1 F_1 + B_2 F_2 + \cdots + B_p F_p$$

in a neighbourhood of A . Q.E.D.

4. Reduction of Problem (C₁)

1°.

Let us consider, in a neighbourhood of a closed polycylinder Δ , a finite system of holomorphic functions

$F_i(x)$ ($i = 1, 2, \dots, p$) in a neighbourhood of Δ , not identically zero, and the corresponding problem (A). Suppose that problem (A) is solved in neighbourhoods of Δ' and of Δ'' . Then, problem (A) is solvable in a neighbourhood of Δ .

If the conclusion in this lemma is established, we can use Lemma 1 to find a formula for the solution of equation (1) with the properties required in Lemma 2.

We shall prove this proposition by induction on p . For $p = 1$, there is only the single ideal $(I_1) = 0$ in problem (A), and it is obviously solved, so that the proposition is true in this case. Let us therefore consider the case p , assuming the proposition true for $1, 2, \dots, p - 1$.

Let $(\varphi_1, \varphi_2, \dots, \varphi_q)$ be a pseudobasis for (I_1) in a neighbourhood of Δ' such that each φ_i belongs globally to (I_1) on this neighbourhood; and let $(\varphi'_1, \varphi'_2, \dots, \varphi'_{q'})$ be a similar pseudobasis for Δ'' . The systems (φ') , (φ'') of functions are equivalent at every point of Δ_0 . Since, by hypothesis, any problem (C₁) is solvable for Δ_0 , the equivalence is global. Since Δ_0 is simply connected, one finds, by Cartan's theorem, a system $(\varphi_1, \varphi_2, \dots, \varphi_s)$ of holomorphic functions in a neighbourhood of Δ such that $(\varphi) \sim (\varphi')$ globally for Δ' , $(\varphi) \sim (\varphi'')$ globally for Δ'' .

Since φ_i ($i = 1, 2, \dots, q$) belongs globally to (I_1) in a neighbourhood of Δ' , we have, globally on this neighbourhood,

$$\varphi_i F_1 \equiv 0 \pmod{(F_2, F_3, \dots, F_p)} \quad (i = 1, 2, \dots, q);$$

the same is therefore true for $\varphi_i F_1$ ($i = 1, 2, \dots, s$), and an analogous result holds for Δ'' . Now, since the functional equation

$$A_2 F_2 + A_3 F_3 + \dots + A_p F_p = 0$$

is in the case $(p - 1)$, we can find a formula for its solutions, having the properties stated in Lemma 2 on a neighbourhood of Δ_0 . By Lemma 2, we therefore have

$$\varphi_i F_1 \equiv 0 \pmod{(F_2, F_3, \dots, F_p)} \quad (i = 1, 2, \dots, s)$$

globally on a neighbourhood of Δ , that is to say, the functions φ_i belong globally to (I_1) on this neighbourhood.

The system (φ) of functions, which has the above properties, is a pseudobasis of (I_1) in a neighbourhood of Δ' since $(\varphi) \sim (\varphi')$; the same holds for Δ'' . (φ) is thus a pseudobasis for (I_1) in a neighbourhood of Δ . The part of problem (A) relating to the ideal (I_1) is thereby solved; the rest of problem (A) is solvable by induction hypothesis.

2°.

We now apply the preceding results to closed polycylinders. In the space (x) , let us consider a closed polycylinder

$$E_{2n} : |x_i - x_i^0| \leq r_i \quad (i = 1, 2, \dots, n),$$

where the x_i^0 being complex numbers, y^0 , real. Let E_1 be the part of this line lying in the closed polycylinder (C) ; in general, E_1 is a segment; sometimes it reduces to a point.

We introduce $E_2, E_3, \dots$ similarly by successively raising the real dimension, and terminating with E_{2n} . E_2 , for example, is a closed cylindrical set with a point of (C_i) as component in the x_i -plane for $i = 1, 2, \dots, n - 1$, and (C_n) as component in the x_n -plane; E_{2n} stands for the polycylinder (C) .

We shall apply the lemmas of No. 3 to this geometric configuration and find that problem (C₁) and (A) can be solved successively for $E_0, E_1, \dots, E_{2n}$, so long as problem (K) is always solvable.

Case of E_0 . Problem (C₁) is trivially solvable in the neighbourhood of a point; by our hypothesis about problem (K), problem (I) is also solvable.

Case of E_1 . We consider any one of the sets E_1 and denote it by the same symbol E_1 ; we can suppose that it is a real segment.

1°. Consider holomorphic functions $F_1(x), F_2(x), \dots, F_p(x)$, not identically zero, in a neighbourhood of E_1 and for which problem (A) is solvable at every point of E_1 .

The set E_1 is a cylindrical set whose component in $(x_1, x_2, \dots, x_{n-1})$ -space is a point which we denote by Q . The component in x_n -space is a segment which we denote by I ; I is horizontal, and its left extremity will be denoted m_0 , its right extremity by m_q ; we introduce $(q-1)$ points $m_1, m_2, \dots, m_{q-1}$ on this segment, from left to right, and denote the closed segment (m_{i-1}, m_i) by I_i ($i = 1, 2, \dots, q$). Let M_i be the cylindrical set (Q, m_i) in the space (x) , L_i the set (Q, I_i) . We set

$$L_1 \cup L_3 \cup L_5 \cup \dots = L'_1, \quad L_2 \cup L_4 \cup L_6 \cup \dots = L'_2;$$

we then have

$$L = L'_1 \cup L'_2 = E_1, \quad L'_0 = L'_1 \cap L'_2 = M_1 \cup M_2 \cup \dots \cup M_{q-1}.$$

Now, problem (A) is solvable for the system (F) at every point of E_1 ; hence, if the segments L_i are sufficiently small, the problem is solvable in a neighbourhood of each of them (as is obvious by the Borel-Lebesgue lemma). It is therefore solvable for L'_1 and for L'_2 . Since problem (C₁) is solvable for L'_0 , it follows from Lemma 3 that problem (I) is always then solvable for E_1 . (F) and (E_1) being arbitrary, problem (A) is always solvable in a neighbourhood of any segment E_1 .

2°. Let us now consider problem (C₁) for a general segment E_1 . Problem (A) is always solvable in the neighbourhood of E_1 , hence, by Lemma 1, we can find a formula for the solutions of the linear equation corresponding to the functions (F) having the properties indicated in Lemma 2. By Lemma 2, problem (C₁) is solvable for the set E_1 . Since (F) and E_1 are arbitrary, any problem (C₁) is solvable in the neighbourhood of any segment E_1 .

Case of E_2 . We repeat the above argument.

1°. Problem (A). Consider a set E_2 and denote by Q its component in $(x_1, x_2, \dots, x_{n-1})$ -space; its component in the x_n -plane will be (C_n) . Let us also consider holomorphic functions $F_1(x), F_2(x), \dots, F_p(x)$, not identically zero, and the corresponding problem (A) in the neighbourhood of E_2 . This problem has already been solved for any E_1 contained in E_2 .

Between the lower and the upper extremities of the closed circle (C_n) , we consider $(q-1)$ horizontal lines which partition (C_n) into q closed sets; we denote these sets by $\alpha_1, \alpha_2, \dots, \alpha_q$ (from bottom to top). Denote the closed polycylinder (Q, α_i) ($i = 1, 2, \dots, q$) in the space (x) by A_i and set

$$A = A_1 \cup A_3 \cup A_5 \cup \dots = \tilde{A}_1, \quad A_2 \cup A_4 \cup A_6 \cup \dots = \tilde{A}_2;$$

we then find that

$$A = \tilde{A}_1 \cup \tilde{A}_2 = E_2$$

and that the intersection $A_0 = \tilde{A}_1 \cap \tilde{A}_2$ consists of a finite number of sets E_1 . Since problem (A) is solvable in a neighbourhood of an arbitrary E_1 -subset of E_2 , the problem will be solvable in the neighbourhood of each $\tilde{A}_i$ if the partition of E_2 is sufficiently fine (once again by the Borel-Lebesgue lemma), and consequently in neighbourhoods of $\tilde{A}_1$ and $\tilde{A}_2$. Since any problem (C₁) is solvable for any set E_1 , and therefore for A_0 , problem (A) is solvable for the set E_2 by Lemma 3; again, since (F) and E_2 are arbitrary, problem (A) is always solvable in the neighbourhood of any set E_2 .

2°. Problem (C₁). We fix a set E_2 and a problem (C₁) for this set E_2 . We can then realise the geometric configuration above in such a way that problem (C₁) is already solved in the

neighbourhood of $\tilde{A}_1$ and of $\tilde{A}_2$. Since the corresponding problem (A) is solvable for A and any problem (C_1) is solvable for A_0 , problem (C_1) is solvable for the set E_2 by Lemmas 1 and 2. The problem (C_1) and the set E_2 being arbitrary, problem (C_1) is always solvable in the neighbourhood of any set E_2 .

By continuing in this way, we find that : Problems (C_1) and (A) are always solvable in the neighbourhood of a closed polycylinder E_{2n} , so long as problem (K) is always solvable. This is the desired result.

5. The Remainder Theorem

The following theorem will furnish us with a useful tool in what follows.

Theorem 6.4 (The Remainder Theorem). *Consider a domain of the form $[D, (C)]$ in the space $(x_1, x_2, \dots, x_n, y)$ where D is a (univalent, finite) domain in the space (x) and (C) is a circle in the y -plane. Let us consider a holomorphic function $F(x, y)$ in $[D, (C)]$ such that, for any point (x^0) of D , the equation $F(x^0, y) = 0$ has Λ roots in (C) , where Λ is a finite number, independent of (x^0) . Then, any holomorphic function $f(x, y)$ in $[D, (C)]$ can be represented in the form*

$$f(x, y) = f_0(x, y) + \varphi(x, y)F(x, y),$$

where f_0 and φ are holomorphic functions on $[D, (C)]$; f_0 is a polynomial with respect to y which is identically zero if $\Lambda = 0$, and is of degree at most $\Lambda - 1$ in y if $\Lambda > 0$. Furthermore, the representation is unique.

We shall start by proving the uniqueness. If $\Lambda = 0$, we have $f_0 = 0$; suppose therefore that $\Lambda > 0$. In this case, if we had two representations with (f_0, φ) and (f'_0, φ') , we would have

$$(f_0 - f'_0) + (\varphi - \varphi')F = 0$$

identically. Now, for any point (x^0) of D , the equation $F(x^0, y) = 0$ has Λ roots in (C) ; hence, if we set $f_0 - f'_0 = \psi$, the equation $\psi(x^0, y) = 0$ would have at least Λ roots in (C) ; but, since ψ is a polynomial in y of degree at most $(\Lambda - 1)$, it must necessarily be identically zero, that is $f_0 = f'_0$.

Let us look now at the form of $F(x, y)$. Let us fix a point (ξ) in D arbitrarily. Thanks to the theorem of Weierstrass, when (x) is near (ξ) and y is in (C) , we have

$$F(x, y) = \omega(x, y)F_0(x, y),$$

where ω and F_0 are holomorphic functions, ω is never zero and

$$F_0(x, y) = y^\Lambda + a_1(x)y^{\Lambda-1} + \dots + a_\Lambda(x) \quad (\Lambda > 0),$$

the $a_i(x)$ ($i = 1, 2, \dots, \Lambda$) being holomorphic functions in a neighbourhood. Now, since such an expression for F is obviously unique, it holds globally on $[D, (C)]$.

Let us examine the special condition we imposed, that the equation $F(x^0, y) = 0$ have always the same number Λ of roots in (C) . In the space of two complex variables x, y , let $F = y - x$, and $f(x, y) = f(y)$ where $f(y)$ is a holomorphic function of the single variable y having $|y| < 1$ for its exact domain of existence. If we take the unit circle for (C) , the condition is only valid for $|x| < 1$. If now D is a (connected) domain in the x -plane which extends both outside and inside the unit circle, the condition on F above is not satisfied.

The condition imposed on F is thus inevitable. This is one of the curious phenomena that one meets in the dealing with analytic functions when one leaves the field of functions of a single variable.

Let us consider the case of a single variable; this is when F and f are functions of a single variable Y and the condition on F above is always satisfied. The theorem remains true in this case (if it is true for several variables).

We shall now prove the proposition. Suppose that

$$F(x, y) = y^\Lambda + A_1(x)y^{\Lambda-1} + \cdots + A_\Lambda(x), \quad \Lambda > 0,$$

where the A_i are holomorphic in D .

Suppose now that $F(x, y)$ has no multiple factors. Then, the analytic variety

$$F(x, y) = 0, \quad \frac{\partial F}{\partial y} = 0$$

has complex dimension $(n - 1)$; moreover, because of the form of F , the projection S of the variety on the space (x) [i.e. the set of (x') for which (x', y') is on the variety for a suitable value y'] is a characteristic surface. For any point (x) of D outside S we have $\Delta = F_0$ because the roots y_i ($i = 1, 2, \dots, \Lambda$) of the equation $F(x, y) = 0$ are all distinct; consequently, we have

$$y_i^\mu = \Delta_i / \Delta \quad (i = 1, 2, \dots, \Lambda), \quad (2)$$

where Δ is the Vandermonde determinant. We shall estimate the absolute value of y_i .

Let (x') be any point of $D' = D - S$; let y_i ($i = 1, 2, \dots, \Lambda$) be the roots of $F(x', y) = 0$ in (C) and let ρ be a number less than the radius of (C) . Let R be the closed set of points (x') of D' for which at least one of the $|y_i|$ is $\geq \rho$. If R is not empty, the function $1/(y - y_i)$ is holomorphic for $|y| < \rho$ and $(x') \in R$. By the maximum principle,

$$\sup_{|y| < \rho} \frac{1}{|y - y_i|} = \frac{1}{||y_i| - \rho|}.$$

We can therefore apply the theorem of holomorphic functions to D' and obtain a representation.

In the general case, where F may have multiple factors, we adjoin a new variable t and set

$$F^*(x, y, t) = F(x, y) + tG(x, y),$$

where G is a polynomial in y of degree less than Λ . We can choose G so that F^* has no multiple factors in the domain $[(x) \in D', |t| < \rho, |y| < 1]$ and, in addition, has no multiple factors. By what we have just seen, we have

$$f(x, y) = G(x, y, t) + H(x, y, t)\Phi(x, y, t)$$

identically in this domain; here G and H are holomorphic, and G is a polynomial with respect to y of degree at most $\Lambda - 1$. Setting $t = 0$, we obtain

$$f(x, y) = G(x, y, 0) + H(x, y, 0)F(x, y);$$

this is the desired relation for $[D', (C)]$; since D' is arbitrary and such a representation is unique, the relation holds for $[D, (C)]$. Q.E.D.

6. Solution of the Local Problem (K)

We now devote ourselves to problem (K).

1°.

We suppose that we are in the space of n complex variables $(x_1, x_2, \dots, x_n)$. We write $x = (x_2, \dots, x_n)$ so that $(x_1, x_2, \dots, x_n) = (x_1, x)$. Let (C) , (C') be circles in the x_1 -plane and x -space respectively and set $C = [(C), (C')]$. Let Δ be the closed polycylinder C .

We propose to solve problem (K) at the origin for equation (1) relative to a system of functions

$$F_i(x) \quad (i = 1, 2, \dots, p)$$

holomorphic on Δ , not identically zero.

2°.

We can suppose that $F_p(x)$ is not divisible by x_1 at the origin; for if all the $F_i(x)$ were divisible by x_1 , we would divide equation (1) by x_1 , which does not change the ideal (I_1) .

Let us therefore suppose that $F_1(0) = F_2(0) = \dots = F_p(0) = 0$. In this case, it is clear that there exists a number m such that all the functions $F_i(x)$ are divisible by x_1^m at the origin, and among these functions one at least, say $F_p(x)$, is not divisible by x_1^{m+1} . The above formula for the solutions is then still valid.

Problem (K) is therefore solved for a space (E) of one complex dimension. It is therefore sufficient to solve it for a space (E) of dimension $(n + 1)$, assuming it to be solvable for spaces of smaller dimension.

3°.

Let us represent the space (E) by $(x_1, x_2, \dots, x_n, y_1, \dots, y_m)$ and the functions $F_1, F_2, \dots, F_p$ by $F_1(x, y), F_2(x, y), \dots, F_p(x, y)$. Let (C) , (C') be circles in x -space and y -space respectively, and set $\Delta = [(C), (C')]$. We suppose that the functions $F_i(x, y)$ are holomorphic on Δ , not identically zero. We suppose also that, if $F_1(x, y)$ is expressed as a polynomial in y ,

$$F_1(x, y) = y^\Lambda + A_1(x)y^{\Lambda-1} + \dots + A_\Lambda(x),$$

the equation $F_1(x, y) = 0$ has no multiple roots for any $(x) \in (C)$. Finally we suppose that, for the functions $F_2, F_3, \dots, F_p$, the degrees in y are less than Λ .

Let $\varphi_1(x, y), \varphi_2(x, y), \dots, \varphi_q(x, y)$ be the pseudobasis of (I_1) in the neighbourhood of $\Delta' = [(0), (C')]$. By hypothesis, there exist formulae for the solutions of the equation

$$A_2F_2 + A_3F_3 + \dots + A_pF_p = 0$$

in the neighbourhood of Δ' ; by the preceding section, we can find a pseudobasis (φ) having the properties stated. Let us set

$$\varphi_iF_1 = \psi_{i2}F_2 + \psi_{i3}F_3 + \dots + \psi_{ip}F_p \quad (i = 1, 2, \dots, q)$$

in the neighbourhood of Δ ; the ψ_{ij} are holomorphic functions having degrees in y at most $(\Lambda - 1)$. Let (π) be the system of functions consisting of (φ) and (ψ) ; we have seen that the formula

$$A_i = C_{i1}\pi_{i1} + C_{i2}\pi_{i2} + \dots + C_{iN}\pi_{iN} \quad (i = 1, 2, \dots, p) \quad (2)$$

with some identities among (C) represents the solutions of equation (1) in a neighbourhood of the origin.

Let us now set

$$\begin{cases} B_i = B_i^0 + P_iF_p & (i = 1, 2, \dots, p-1) \\ B_p = B - (\beta_1F_1 + \beta_2F_2 + \dots + \beta_{p-1}F_{p-1}), \end{cases} \quad (C)$$

where the B_i^0 and P_i ($i = 1, 2, \dots, p-1$) are polynomials in y , holomorphic when $(x) \in (Y)$ and the B_i^0 have degree at most $\Lambda - 1$ in y . The function B defined by (C) is therefore a polynomial in y , holomorphic for $(x) \in (Y)$ and the system of functions (B) satisfies equation (1) for $(x) \in (Y)$. B therefore satisfies

$$B \cdot F_p = \Phi, \quad \Phi = -(B_1^0 F_1 + B_2^0 F_2 + \dots + B_{p-1}^0 F_{p-1});$$

here Φ has degree at most $2\Lambda - 1$ in y since $\Lambda_i < \Lambda$ ($i = 1, 2, \dots, p-1$). Thus B is of degree at most $\Lambda - 1$ in y .

4°.

We therefore suppose that, in the functional equation

$$A_1 F_1 + A_2 F_2 + \dots + A_p F_p = 0, \quad (1)$$

the functions $F_1, F_2, \dots, F_p$ have the same properties relative to the polycylinder $[(C), (C')]$ as the functions $\varphi_1, \varphi_2, \dots, \varphi_p$.

Our aim is to find a formula

$$A_i = C_{i1}\pi_{i1} + C_{i2}\pi_{i2} + \dots + C_{iN}\pi_{iN} \quad (i = 1, 2, \dots, p) \quad (2)$$

with some identities among (C) for the solutions of this equation (1) in a neighbourhood of the origin.

Let us consider the system of equations

$$\alpha_{ij} = \sum \beta_{ijk} P_{ijk} \quad (i = 1, \dots, p; j = 0, \dots, \Lambda - 1; k = 1, \dots, \nu) \quad (B)$$

with some identities among (β) , the P_{ijk} being well determined functions on a polycylinder (C_0) contained in (C) , which has the following properties :

1°. Any solution (α) of equation (B) at any point (x^0) of (C_0) can be represented in the form (C) at (x^0) .

2°. For any collection (β) of holomorphic functions at an arbitrary point (x^0) of (C_0) , the corresponding system of functions (α) defined by (C) is a solution of (B) at (x^0) .

We shall show that such an expression (C) always exists. The condition that the system (B) be independent of P_{ijk} is equivalent to saying that the P_{ijk} are holomorphic functions on (C_0) .

5°.

Let us consider a domain D in the space (x) and holomorphic functions $A_{ij}(x)$ ($i = 1, 2, \dots, q; j = 1, 2, \dots, p$) on this domain; (of these functions, some can be identically zero). Consider now the system of simultaneous (homogeneous linear) functional equations

$$\begin{aligned} A_{11}F_1 + A_{12}F_2 + \dots + A_{1p}F_p &= 0 \\ A_{21}F_1 + A_{22}F_2 + \dots + A_{2p}F_p &= 0 \\ &\dots \\ A_{q1}F_1 + A_{q2}F_2 + \dots + A_{qp}F_p &= 0 \end{aligned} \quad (1)$$

with certain identities among the (A) . The (A) are the unknown functions; between them, there are in general, certain identities (without this, the system would reduce to q separate equations; and would be precisely what we have considered above).

Let (E_r) be the system consisting of the first r equations of (1). We have seen that we can find a formula for the solutions of (E_r) in a neighbourhood of Δ_r ; and this formula has the required properties. We want to find a similar formula for (E_{r+1}) in a neighbourhood of Δ_{r+1} .

We replace the letters $A_{r+1,i}$ and F_i ($i = 1, 2, \dots, p$) by new letters B_i, ψ_i in the last equation of (E_{r+1}) ; we then have

$$\begin{aligned} A_{11}F_1 + A_{12}F_2 + \dots + A_{1p}F_p &= 0 \\ &\dots \\ A_{r1}F_1 + A_{r2}F_2 + \dots + A_{rp}F_p &= 0 \\ B_1\psi_1 + B_2\psi_2 + \dots + B_p\psi_p &= 0 \end{aligned} \tag{3}$$

with certain identities among (A, B) .

By hypothesis, we can find a formula for solutions of (E_r)

$$A_{ij} = \sum C_{ijk} \pi_{jk} \quad (k = 1, 2, \dots, \mu) \tag{4}$$

with certain identities among the (C) , (this in a neighbourhood of Δ_r). We are going to substitute this in the equation

$$B_1\psi_1 + B_2\psi_2 + \dots + B_p\psi_p = 0. \tag{5}$$

The new expressions so obtained can be written

$$\begin{cases} A_{ij} = \sum \varepsilon_{ijk} P'_{ijk} \\ B_i = \sum \varepsilon_{ik} P_{ik} \end{cases} \quad (i = 1, \dots, p; j = 1, \dots, r; k = 1, \dots, t), \tag{9}$$

with certain identities among $(\varepsilon, \varepsilon')$; the (P, P') stand for the kernel, and $(\varepsilon, \varepsilon')$ are the unknown functions. Among the identities indicated, some come from (4), some from (8), and others are new.

We shall study the nature of this expression. Let us consider an arbitrary solution (A, B) of the equation (3) at an arbitrary point P of Δ . Since (A) satisfies (E_r) , it can be represented in the form (4) at P . Since (B) satisfies the last equation of (3), we have the representation (8) at P . The identities among (C) and (γ) imply the identities among $(\varepsilon, \varepsilon')$ indicated in (9).

6°. Existence theorem.

Suppose that we can make correspond to the system (B) of simultaneous equations an expression (C) with the properties indicated. Then, the union of (3) and (D) gives a formula for the solutions of equation (1) in the polycylinder $[(C_0), (C')]$, if we regard the δ_i ($i = 1, 2, \dots, p-1$) and $\varepsilon_{ij} = (\varphi \cdot C_i)_j$ ($i = 1, 2, \dots, p; j = 1, 2, \dots, N'$) as the independent unknown functions.

In fact, the union of these two relations takes the same form as (2), whose kernel consists of well determined functions (F) and (θ) holomorphic on $[(C_0), (C')]$. Let us consider a solution (A) of equation (1) at an arbitrary point (x_0, y_0) of $[(C_0), (C')]$. Then, by what we have seen above, the solution is represented by the union of (3) and (D). Conversely, let us consider arbitrary holomorphic functions δ_i and ε_{ij} at (x_0, y_0) . The corresponding functions (A) form a solution of equation (1) at (x_0, y_0) . This establishes the existence theorem.

7. Formulae for the Solutions of Simultaneous Equations

1°.

Suppose that we can make correspond to the system (B) of simultaneous equations an expression

$$\alpha_{ij} = \sum \beta_{ijk} P_{ijk}$$

with some identities among (β) , the P_{ijk} being well determined functions on a polycylinder (C_0) contained in (C) , which has the following properties :

1°. Any solution (α) of equation (B) at any point (x^0) of (C_0) can be represented in the form (C) at (x^0) .

2°. For any collection (β) of holomorphic functions at an arbitrary point (x^0) of (C_0) , the corresponding system of functions (α) defined by (C) is a solution of (B) at (x^0) .

We shall show that such an expression (C) always exists. The condition that the system (B) be independent of P_{ijk} is equivalent to saying that the P_{ijk} are holomorphic functions on (C_0) .

2°.

The existence of such a formula (C) is established by induction on the number p of equations in the system (B). For $p = 1$, the result follows from the preceding section. Suppose the result established for systems of $(p - 1)$ equations; we shall prove it for p equations.

3°. Formulation of the result.

We can now state the fundamental theorem :

Theorem 6.5. *Given a system of simultaneous homogeneous linear functional equations with holomorphic coefficients in a domain D , there exists, in the neighbourhood of any closed polycylinder contained in D , a formula for the solutions having the properties indicated.*

8. Solution of Problem (J)

1°. Statement of the result.

We can now solve problem (J) for a certain class of ideals, namely those which we have called “ideals with indeterminate domains”.

Theorem 6.6. *Let Δ be a closed polycylinder in the space (x) and (I) an ideal with indeterminate domains in a neighbourhood of Δ . Suppose that problem (K) is always solvable for the ideals (I) at any point of Δ . Then (I) has a finite pseudobasis on Δ .*

2°. Geometric ideals.

We shall now consider a particular class of ideals, which we call “geometric ideals”. Let V be a characteristic variety in a neighbourhood of a closed polycylinder Δ and let (I) be the set of pairs (f, δ) such that f vanishes on $V \cap \delta$. Then (I) is an ideal with indeterminate domains, which we call the geometric ideal associated to V .

3°. Pseudobases for geometric ideals.

We shall prove that geometric ideals have finite pseudobases on closed polycylinders. The proof is based on the remainder theorem and on the results concerning problem (K).

Theorem 6.7. *Let Δ be a closed polycylinder in the space (x) and V a characteristic variety in a neighbourhood of Δ . Then the geometric ideal associated to V has a finite pseudobasis on Δ .*

The proof proceeds by induction on the dimension of the variety V . If V is of dimension zero, the result is immediate. Suppose the result established for varieties of dimension less than n ; we shall prove it for dimension n .

By the Weierstrass preparation theorem, we can choose coordinates so that V is defined, in a neighbourhood of any point, by a system of equations of the form

$$F_i(x, y) = 0 \quad (i = 1, 2, \dots, m),$$

where the F_i are polynomials in y with holomorphic coefficients in x .

4°. Application to the local pseudobasis problem.

We can now solve problem (J) for geometric ideals on closed polycylinders. The proof uses the results of the preceding sections and proceeds by an induction on the number of equations defining the variety.

Theorem 6.8. *Let Δ be a closed polycylinder and (I) a geometric ideal in a neighbourhood of Δ . Then (I) has a finite pseudobasis on Δ .*

9. The Cousin Problems

Let us take for A a closed circle; let A_1 be the part of A for which $X < \delta$, where X is the real part of x and δ is a sufficiently small positive number; and let A_2 be the part of A for which $X > -\delta$. We have $A = A_1 \cup A_2$ and $A_1 \cap A_2$ consists of two closed segments.

We consider the Cousin problem for the domain A . Let f_1 be a holomorphic function on A_1 and f_2 a holomorphic function on A_2 such that $f_1 - f_2$ is holomorphic on $A_1 \cap A_2$. We want to find a holomorphic function f on A such that $f - f_1$ is holomorphic on A_1 and $f - f_2$ is holomorphic on A_2 .

This problem can be solved by the methods developed in the preceding sections. The solution uses the formula for the solutions of homogeneous linear functional equations and the pseudobasis theorem for geometric ideals.

1°. The first Cousin problem.

We consider a domain D in the space (x) and a covering of D by a finite number of domains D_i . Suppose given, on each intersection $D_i \cap D_j$, a holomorphic function f_{ij} such that $f_{ij} + f_{jk} + f_{ki} = 0$ on $D_i \cap D_j \cap D_k$. We want to find holomorphic functions f_i on D_i such that $f_i - f_j = f_{ij}$ on $D_i \cap D_j$.

2°. The second Cousin problem.

We consider a domain D in the space (x) and a covering of D by a finite number of domains D_i . Suppose given, on each D_i , a meromorphic function f_i such that f_i/f_j is holomorphic and non-zero on $D_i \cap D_j$. We want to find a meromorphic function f on D such that f/f_i is holomorphic and non-zero on D_i .

10. Summary of Results

We have established the following fundamental results :

1. Problem (C_1) is always solvable for closed polycylinders, provided problem (K) is always solvable.
2. Problem (K) is solvable for homogeneous linear functional equations with holomorphic coefficients.
3. Geometric ideals have finite pseudobases on closed polycylinders.
4. The Cousin problems are solvable for domains satisfying suitable convexity conditions.

These results constitute the foundation of the theory of functions of several complex variables as developed by the author in the preceding memoirs.

Comments on Paper VII

Note du commentateur. Les notions et resultats de ce Memoire VII sont a la base de la theorie des fonctions analytiques de plusieurs variables complexes. OKA y a developpe systematiquement la theorie des “ideaux de domaines indetermines”, qui est en substance equivalente a celle de faisceaux d’ideaux introduite par CARTAN en 1949.

OKA pose ici une serie de problemes fondamentaux. Le probleme (J), en termes de faisceaux, est le suivant : un faisceau analytique d’ideaux est-il coherent ? OKA donne lui-meme un contre-exemple. Il semble qu’a cette epoque il savait que le faisceau d’ideaux defini par un sous-ensemble analytique est coherent (cf. les 5 dernieres lignes du Memoire) ; mais il n’a pas publie de demonstration.

Au point de vue technique, il est bon d’attirer l’attention du lecteur sur l’important “theoreme du reste” (n°5). Il ressemble au lemme donne par H. CARTAN (Ann. E.N.S. 1944, p. 170) auquel on pourrait d’ailleurs le ramener, en observant que $F(x, y)$ est equivalente, dans le domaine considere par OKA, a un polynome en y de degre Λ .

Le theoreme I du Memoire VII dit que si (I) est un ideal de domaines indetermines dans un espace $\mathbb{C}^n$, et si Δ est un polycylindre ferme dans cet espace, alors (I) admet une pseudobase finie sur Δ , pourvu que le probleme (K) soit toujours resoluble.

Le theoreme II dit que le probleme (K) est toujours resoluble pour les equations fonctionnelles lineaires homogenes a coefficients holomorphes.

Le theoreme III affirme que si l’on se donne sur chaque voisinage (γ) un systeme fini (f) de fonctions holomorphes, de telle sorte que pour chaque paire $(\gamma'), (\gamma'')$, les systemes correspondants $(f'), (f'')$ soient equivalents en chaque point de l’intersection, alors on peut trouver un systeme fini (F) de fonctions holomorphes dans un voisinage de Δ tel que $(F) \sim (f)$ en tout point de Δ .

Le theoreme IV dit que si l’on se donne des F_i holomorphes en nombre fini sur Δ , le faisceau des relations lineaires (a coefficients holomorphes) entre les F_i est engendre par un nombre fini de sections de ce faisceau (cela resulte du theoreme A parce que OKA a prouve que ce faisceau est coherent).

Enfin le theoreme IV dit que si l’on se donne des F_i holomorphes en nombre fini sur Δ , le faisceau des relations lineaires (a coefficients holomorphes) entre les F_i est engendre par un nombre fini de sections de ce faisceau (cela resulte du theoreme A parce que OKA a prouve que ce faisceau est coherent).

Chapter 7

VIII. Fundamental Lemma

Journal of the Mathematical Society of Japan 3 (1951), pp. 204–214

Introduction

The principal problems we have dealt with since Memoir I are the following: Cousin problems, the problem of expansions and the problem of (different types of) convexity¹. In Memoirs I–VI², we have seen, to put it in one word, that these problems can be solved affirmatively for univalent domains without points at infinity³. Furthermore, the author has verified, albeit without publishing this, that these results remain valid at least as far as domains without points at infinity and without interior ramification points⁴.

We must therefore either introduce suitable points at infinity or allow points of ramification. Now, one will find that almost nothing is known about domains with interior ramification. For example, what happens to the local expansion of a function? We shall therefore deal first with the second problem.

Now, the fundamental idea of our researches is symbolised by Theorem II⁵ of Memoir I. We have actually used it in the form of Theorem II of Memoir V. In the present Memoir VIII, we shall use it essentially in the form of Theorem IV of Memoir VII, which asserts that problem (K) is always solvable⁶. This theorem is a fundamental tool in the study of domains ramified in the interior; this is the fundamental lemma of the title of the present Memoir VIII, which was written just to establish this lemma.

To establish this lemma for (finite) domains without points of ramification, it is obviously sufficient to solve problems (C₂) and (E) and to find, locally, a finite pseudobasis for a geometric ideal with indeterminate domains. Of these, we solved problems (C₂) and (E) in Memoir VII, and, more recently H. Cartan has solved the last problem, based on Theorem IV of Memoir VII asserting that problem (K) is always solvable⁷. However, if we allow points of ramification, we meet the new difficulty that a holomorphic function on a characteristic variety is not necessarily the restriction of a holomorphic function in (ambient) space. As a consequence, this generates a class of problems (J) which contains, in a certain sense, geometric ideals and is more extensive.

In the present memoir, we shall solve this problem, again using Theorem IV of Memoir VII (see Theorem 2 below). We shall also establish the fundamental lemma and apply it to the study

1. See Memoir VI, pp. 107–110.

2. K. OKA, Sur les fonctions analytiques de plusieurs variables, I–VI.

3. A point of the space (x) is called a point at infinity of a domain over the space (x) if it is not the base-point of any point of the domain.

4. See No. 5 below.

5. K. OKA, Sur les fonctions analytiques de plusieurs variables, I, Theorem 2, p. 9.

6. See Memoir VII, Theorem IV, p. 107.

7. H. Cartan, Varietes analytiques reelles et varietes analytiques complexes, Bull. Soc. Math. France 85 (1957), pp. 77–99.

of domains with interior ramification.

1. Preliminaries

We shall recall some notions from Memoir VII and introduce some new ones.

1°. Ideals with indeterminate domains.

We consider, in the space of n complex variables $(x_1, x_2, \dots, x_n)$, sets of pairs (f, δ) where δ is a domain (connected or not) and f is a holomorphic function on δ . Such a set (I) is called an ideal with indeterminate domains if it satisfies the following conditions:

1°. If $(f, \delta) \in (I)$ and a is a holomorphic function on a (not necessarily connected) domain δ' , then $af \in (I)$ on $\delta \cap \delta'$;

2°. If $(f, \delta) \in (I)$ and $(f', \delta') \in (I)$, then $f + f' \in (I)$ on $\delta \cap \delta'$.

In the present memoir, we shall, in general, simply call this an ideal. The following topological property follows immediately from the definition: if (I) is an ideal and if $(f, \delta) \in (I)$ and $\delta' \subset \delta$, then $(f, \delta') \in (I)$. We can therefore tell whether or not $f \in (I)$ at a point P .

A point P of the space (x) will be called a lacunary point of the ideal (I) if no holomorphic function belongs to (I) at P . The set of lacunary points of (I) is clearly closed. A point P of the space (x) will be called a zero of (I) if any holomorphic function belonging to (I) at P vanishes at P . The set of zeros of (I) is clearly closed in the complement of the set of lacunary points of (I) .

Conversely, every closed set of points in the space (x) is obviously the set of zeros of a suitable ideal in this space.

A finite system of holomorphic functions F_i ($i = 1, 2, \dots, p$) on a domain $\mathcal{D}$ of the space (x) will be called a (finite) pseudobasis of the ideal (I) if it has the following properties:

1°. Each F_i belongs to (I) at every point of $\mathcal{D}$;

2°. Any element of (I) at a point of $\mathcal{D}$ belongs to (I) globally in a neighbourhood of this point, by means of the functions F_i .

2°. Theorem IV of Memoir VII.

The fundamental tool for our researches is the following theorem, which we have established in Memoir VII:

Theorem 7.1. *Given a homogeneous linear functional equation with holomorphic coefficients in a domain D , there exists, in the neighbourhood of any closed polycylinder contained in D , a formula for the solutions with the required properties.*

Let us begin by analysing some general principles which follow immediately from this theorem.

1°. **H. Cartan's Corollary.** Let $(I_1), (I_2)$ be two holomorphic ideals with indeterminate domains in the space (x) and let $(I) = (I_1) \cap (I_2)$; (I) is clearly also such an ideal. For (I) to have a pseudobasis at a point (x^0) , it is sufficient that (I_1) and (I_2) do. (The proof is immediate.)

2°. **Corollary 1.** Suppose given a holomorphic ideal with indeterminate domains $(I) = \{(f, \delta)\}$ in the space (x) and a point (x^0) . We transform the ideal (I) with the help of functions $F(x), \Phi_1(x), \dots, \Phi_p(x)$ holomorphic on a neighbourhood V of (x^0) and form the set $(J) = \{(\varphi, \delta')\}$ as follows:

$$\varphi = fF + A_1\Phi_1 + \dots + A_p\Phi_p, \quad \delta' = V \cap \delta \cap \mu_1 \cap \dots \cap \mu_p$$

where A_i ($i = 1, 2, \dots, p$) is an arbitrary holomorphic function on μ_i (and $(f, \delta) \in (I)$). (J) is clearly an ideal. Suppose that (J) has a pseudobasis at the point (x^0) and that the functional

equation

$$A_0F + A_1\Phi_1 + \cdots + A_p\Phi_p = 0$$

has the following property: If $(A_0, A_1, \dots, A_p)$ is a solution of this equation at a point of μ ; then A_0 belongs necessarily to (I) at this point.

Then (I) has a pseudobasis at (x^0) .

3°. Corollary 2. Suppose given a holomorphic ideal with indeterminate domain, (I) , in the space (x) and a function $\Phi(x)$ holomorphic in a neighbourhood of a point (x^0) . If both the adjoint and the quotient of (I) with respect to (Φ, V) have pseudobases at (x^0) , so then does the original ideal (I) .

In fact, let $(\varphi_1, \varphi_2, \dots, \varphi_p)$ be a pseudobasis of the adjoint on the neighbourhood V' of the point (x^0) ($V' \subset V$). If we have chosen V' sufficiently small, we have, identically $C_i\varphi_i$.

Example 7.2. In the space of two complex variables (x, y) , consider the ideal generated by the 2 elements (xy, Δ) , $(1, \Delta')$, where Δ is the whole space (x, y) and Δ' is given by $|y| > 0$. (I) does not have a pseudobasis at the origin. However, the adjoint of (I) with respect to (y, Δ) has the pseudobasis (y) on Δ , which is also a pseudobasis for the quotient of (I) with respect to (x, Δ) . Thus the converse statement, either just for the adjoint, or just for the quotient, is not true. But we see that for (y, Δ) , the quotient does not have a pseudobasis at the origin, nor does the adjoint with respect to (x, Δ) . This leads us to the following fact:

3°. Geometric ideals.

Let V be a characteristic variety in a neighbourhood of a point (x^0) and let (I) be the set of pairs (f, δ) such that f vanishes on $V \cap \delta$. Then (I) is an ideal with indeterminate domains, which we call the geometric ideal associated to V .

We shall prove that geometric ideals have finite pseudobases. More precisely:

Theorem 7.3. *Let (x^0) be a point in the space (x) and (I) a geometric ideal associated to a characteristic variety V passing through (x^0) . Then (I) has a finite pseudobasis at (x^0) .*

The proof proceeds by induction on the dimension of the variety V . Let V be of dimension n in a space of $n + m$ variables. If $m = 0$, the result is trivial. Suppose the result established for varieties of dimension n in spaces of $n + m'$ variables with $m' < m$; we shall prove it for m .

We can suppose, thanks to the Weierstrass preparation theorem, that V is defined, in a neighbourhood of (x^0, y^0) , by equations of the form

$$F_i(x, y) = 0 \quad (i = 1, 2, \dots, m),$$

where F_1 is a distinguished polynomial in y_1 and F_i ($i = 2, \dots, m$) are polynomials in y_i with coefficients holomorphic in $(x, y_1, \dots, y_{i-1})$.

2. Characteristic Varieties and Ramification

1°. Definitions.

We recall that a characteristic variety in a domain of the space of n complex variables is a set defined locally by the vanishing of a finite number of holomorphic functions. A point of a characteristic variety is called regular if, in a neighbourhood of this point, the variety is a complex manifold; otherwise it is called singular.

2°. Functions on ramified domains.

Let $\mathcal{D}$ be a domain with interior ramification over the space (x) and L the corresponding characteristic variety. A function f on $\mathcal{D}$ will be called holomorphic if, for each point P of $\mathcal{D}$, there exists a neighbourhood V of P such that f is the restriction of a holomorphic function on V .

3°. The Cousin problems on ramified domains.

We can now formulate the Cousin problems for domains with interior ramification. The first Cousin problem is formulated as follows: given a domain $\mathcal{D}$ with interior ramification and a covering of $\mathcal{D}$ by domains $\mathcal{D}_i$, with meromorphic functions f_i on $\mathcal{D}_i$ such that f_i/f_j is holomorphic and non-zero on $\mathcal{D}_i \cap \mathcal{D}_j$, find a meromorphic function f on $\mathcal{D}$ such that f/f_i is holomorphic and non-zero on $\mathcal{D}_i$.

4. Pseudobases on Ramified Domains

1°. The projection method.

We return to the domain $\mathcal{D}$ over the space (x) and the corresponding variety L as at the beginning of No. 7. If we attempt to define a distribution of zeros on $\mathcal{D}$, extending the procedure of the second Cousin problem on univalent domains, we immediately run into a new phenomenon, namely:

In general, a characteristic surface on a domain with interior ramification cannot be represented, even locally, as the set of zeros of a single holomorphic function.

2°. (W)-functions.

Let $\mathcal{D}$ be a domain over the space (x) and L a characteristic variety in the space (x, y) . A holomorphic function $F(x, y)$ on a domain Δ containing points of L will be called a (W)-function on Δ relative to L if F vanishes identically on $L \cap \Delta$.

3°. (Z)-ideals.

We return to the domain $\mathcal{D}$ over the space (x) and the corresponding variety L as at the beginning of No. 7. We consider, in the space (x, y) , the set (I) of pairs (f, δ) such that δ is the domain $(x) \in \mathcal{D}$ and such that, if (f', δ') is the image (on $\mathcal{D}$) of the restriction of (f, δ) to L , then f' has at least the zero distribution of the given system. This set (I) is an ideal with indeterminate domains, which we call a (Z)-ideal.

4°. The fundamental lemma.

We can now state the fundamental lemma:

Theorem 7.4 (Fundamental Lemma). *Let $\mathcal{D}$ be a finite domain without points at infinity, over the space of n complex variables, with interior ramification. Let L be the characteristic variety defining the ramification. Then any holomorphic function on $\mathcal{D}$ can be extended to a holomorphic function in a neighbourhood of L .*

The proof of this lemma uses the results of the preceding sections and proceeds by induction on the dimension of the domain.

5. Properties of Ramification Surfaces

1°. First and second kind.

Let $\mathcal{D}$ be a domain over the space (x) and η a holomorphic function on $\mathcal{D}$. We consider the characteristic variety S_0 defined by the equation $\eta(P) = \eta(P')$ for pairs of points P, P' of $\mathcal{D}$. Let $\mathcal{U}$ be the surface of ambiguity defined by this equation.

A point M of S_0 will be said to have property (H) with respect to $\mathcal{U}$ if there exists a neighbourhood of M on which any holomorphic function can be extended to the ambient space. Any ramification surface having this property will be called a ramification surface of the first kind with respect to $\mathcal{U}$.

Case 2, when $a_1 = 0$ on a_0 . Let T_0 be the set of points of S_0 corresponding to points of a_0 . In this case, the function $(x_1)^{1/l}$ shows that no point of T_0 has property (H) with respect to S_0 . We say that a ramification surface with this property is of the second kind with respect to $\mathcal{U}$.

Let P be a point of $\mathcal{D}$; if there exists another point P' of $\mathcal{D}$ having the same coordinates as P and such that $\eta(P) = \eta(P')$, we shall call P a point of ambiguity of $\mathcal{D}$ with respect to η .

2°. The surface of ambiguity.

We consider a non-singular transformation L of the space (x, y) of the form

$$\begin{aligned} x'_i &= \sum A_{ik} x_k + \sum A_{i,n+l} y_l, \\ y'_j &= \sum A_{n+j,k} x_k + \sum A_{n+j,n+l} y_l \end{aligned}$$

$(i, k = 1, 2, \dots, n; j, l = 1, 2, \dots, m),$

$$|A_{pp} - 1| < \varepsilon, \quad |A_{pq}| < \varepsilon \quad (p \neq q; p, q = 1, 2, \dots, n + m),$$

ε being a positive number which we shall choose later. The transformation L is close to the identity and preserves the structure of the equations defining L .

3°. Pseudobases on the surface of ambiguity.

The functions ψ_i ($i = 1, 2, \dots, p$) have a pseudobasis in a neighbourhood of Δ by Cartan's theorem, consequently a pseudobasis $(\psi_1, \psi_2, \dots, \psi_p)$ on Δ by Theorem III of Memoir VII. The set of common zeros of the functions ψ_i ($i = 1, 2, \dots, p$) is E in Δ . Now, each ψ_i is a (W) -function on Δ because it is identically zero on E . Hence $T \subset E$.

6. The Nullstellensatz

We have the following lemma:

Lemma 7.5 (Lemma of Hilbert-Rückert). *Let $F_i(x)$ ($i = 1, 2, \dots, p$), $f(x)$ be holomorphic functions at a point (x^0) . If $f(x)$ vanishes identically on the set of common zeros of the F_i , we can find a positive integer Λ such that $f^\Lambda \equiv 0 \pmod{(F)}$.*¹⁰

If F_i ($i = 1, 2, \dots, p$) and f are holomorphic on a polycylinder (C) around (x^0) , the result holds globally on (C) .

10. See Rückert, Math. Ann. 107 (1933), pp. 259–273.

7. (Z)-Ideals and Their Properties

We return to the domain $\mathcal{D}$ over the space (x) and the corresponding variety L as at the beginning of No. 7. If we attempt to define a distribution of zeros on $\mathcal{D}$, extending the procedure of the second Cousin problem on univalent domains, we immediately run into a new phenomenon, namely:

In general, a characteristic surface on a domain with interior ramification cannot be represented, even locally, as the set of zeros of a single holomorphic function.

1°. Property (H) for varieties.

Let P_0 be a point of L and $U(x, y)$ a holomorphic function at P_0 in the space (x, y) , such that the image (on $\mathcal{D}$) of the restriction of U to L is u , i.e. if $U(x, \eta) = u$, we shall say that u has property (H) at P_0 with respect to L .

Let us now consider a distribution of zeros on $\mathcal{D}$; suppose that it is everywhere locally defined by a finite system of holomorphic functions on $\mathcal{D}$ which have property (H) with respect to L . We consider, in the space (x, y) , the set (I) of pairs (f, δ) such that δ is the domain $(x) \in \mathcal{D}$ and such that, if (f', δ') is the image (on $\mathcal{D}$) of the restriction of (f, δ) to L , then f' has at least the zero distribution of the given system.

2°. The main theorem on (Z)-ideals.

We can now state the main theorem:

Theorem 7.6. *Let $\mathcal{D}$ be a domain over the space (x) and L a characteristic variety in the space (x, y) . Let (I) be a (Z)-ideal on $\mathcal{D}$ relative to L . Then (I) has a finite pseudobasis at every point of $\mathcal{D}$.*

The proof proceeds in several steps. First, we establish the result for the case where the singular points of L are of codimension at least 2 in L . Then we reduce the general case to this special case by considering a suitable ramified covering.

3°. Reduction to the special case.

We suppose that the set S of singular points of L is of codimension > 2 in L at the point considered. Let E be the set of zeros of (I) at (x^0, y^0) . We suppose that (x', y') is a point of E and set $\Psi_i = 0$ ($i = 1, 2, \dots, r$); we then have the identity

$$B' \Phi^{W'} + A \cdot F' = 0.$$

Now, the order of vanishing of the function $\Phi^{W'}$ on L is zero. Hence B' vanishes on L at least to the order of vanishing of F' , and consequently, $B \in (I)$ at (x', y') by definition. The proposition is thus true in the special case.

The general case can be reduced to the above special case by a suitable ramified covering. Let $\pi: \tilde{L} \rightarrow L$ be a ramified covering such that the singular points of $\tilde{L}$ are of codimension > 2 . We apply the special case to $\tilde{L}$ and obtain a pseudobasis on $\tilde{L}$. By descending to L , we obtain the desired result.

4°. The projection method.

Let L' be the projection of L on the space (x, y_j) ; then, for $(x) \in \delta$, L' is a characteristic surface. Now, by Theorem 1, (J) has a pseudobasis at every point of $[(x) \in \delta, |y_j - y_j^0| < \rho]$; L' is necessarily the set of zeros of (J) . Since Φ_j is identically zero on L' , we can find an integer

ν' such that $\Phi_j^{\nu'} \in (J)$ at (x^0, y_j) . Then $\Phi_j^{\nu'} \in (I)$ at (x^0, y^0) . Let ν be the largest of the ν' for $j = 1, 2, \dots, m$; we have

$$\Phi_j^\nu \equiv 0 \pmod{(F)} \quad \text{at } (x^0, y^0).$$

Let us consider the characteristic variety T given by $\Phi_j = 0$ ($j = 1, 2, \dots, m$) in Δ . L consists of irreducible components of T .

8. The Cousin Problems on Ramified Domains

1°. Reduction to univalent domains.

Let us take for A a closed circle; let A_1 be the part of A for which $X < \delta$, where X is the real part of x and δ is a sufficiently small positive number; and let A_2 be the part of A for which $X > -\delta$. We have $A = A_1 \cup A_2$ and $A_1 \cap A_2$ consists of two closed segments.

We consider the Cousin problem for the domain A . The problem can be solved by reducing it to the corresponding problem on univalent domains, using the results of the preceding sections.

2°. Application of the fundamental lemma.

The fundamental lemma allows us to extend holomorphic functions from a ramified domain to a neighbourhood of the characteristic variety defining the ramification. This extension preserves the properties of the functions and allows us to apply the results established for univalent domains.

3°. Holomorphic functions on the covering.

We consider a domain $\mathcal{D}$ over the space (x) and a ramified covering $\pi : \tilde{\mathcal{D}} \rightarrow \mathcal{D}$. A function f on $\tilde{\mathcal{D}}$ will be called holomorphic if it is locally the restriction of a holomorphic function on the ambient space. The fundamental lemma ensures that such functions can be extended to a neighbourhood of the characteristic variety.

4°. Extension of the results.

We can now extend the results of Memoirs I–VII to domains with interior ramification. The key tool is the fundamental lemma, which allows us to reduce problems on ramified domains to problems on univalent domains.

9. Conclusion

We have established the fundamental lemma and applied it to the study of domains with interior ramification. The main results are:

1. The fundamental lemma: any holomorphic function on a finite domain without points at infinity, with interior ramification, can be extended to a neighbourhood of the characteristic variety defining the ramification.
2. (Z) -ideals have finite pseudobases at every point.
3. The Cousin problems are solvable for domains with interior ramification, under suitable conditions.

These results complete the theory developed in the preceding memoirs and open the way to the study of more general domains.

Comments on Paper VIII

La lecture de ce Memoire VIII est encore compliquee par le fait que OKA melange l'etude de ces problemes de nature locale a des considerations globales, qui faisaient deja l'objet du Memoire VII. C'est la raison pour laquelle la partie I du present Memoire est consacree a l'approfondissement technique de notions relatives aux faisceaux coherents d'ideaux, notamment le theoreme 1 qui a servi d'une maniere essentielle dans la preuve du theoreme 2 de la Partie II du Memoire.

OKA donne aussi une demonstration originale de la coherence du faisceau d'ideaux attache a un sous-ensemble analytique de $\mathbb{C}^n$ (qu'il appelle le "theoreme de H. CARTAN"), et donne divers criteres de coherence, notamment le "corollaire 2". Il donnera aussi un critere de coherence dans l'Appendice.

Mais le but essentiel d'OKA est l'etude des "domaines interieurement ramifies". Un tel domaine D est, par definition, un revetement ramifie a un nombre fini de feuilletts d'un domaine $\tilde{D}$ de $\mathbb{C}^n$. On peut le considerer comme l'image d'un sous-ensemble analytique L de $\mathbb{C}^{n+m}$ par la projection $p : \mathbb{C}^{n+m} \rightarrow \mathbb{C}^n$, p definissant une bijection de l'ensemble des points reguliers de L sur un ouvert dense de D . OKA definit alors ce qu'il entend par fonction holomorphe sur D (c'est une fonction continue qui est holomorphe aux points reguliers de D); en transportant cette definition a L , on obtient les fonctions holomorphes dans l'ouvert des points reguliers de L et qui ont une limite en chaque point singulier lorsqu'on reste dans une composante irreductible de L au point considere.

Les germes de fonctions holomorphes en $a \in L$ ne sont autres que les elements de la cloture integrale de l'anneau local de L en a dans son corps des fractions. OKA montre que cette cloture integrale est un anneau noetherien (theoreme du denominateur universel, lemme 2); de plus, etant donne un point $a \in E$, il existe un systeme fini de fonctions U_i holomorphes dans l'espace ambiant au voisinage de a , telles que chaque U_i soit un denominateur universel en tout point de E assez voisin de a , et telles que l'annulation de toutes les U_i definisse exactement l'ensemble des points singuliers de E voisins de a .

Nous esperons que les indications qui precedent aideront le lecteur dans l'etude de ce Memoire VIII. Le coeur du Memoire est constitue par le theoreme 2, qui concerne ce que OKA appelle les "ideaux (Z)". Le commentateur avoue que cette notion d'ideal (Z) lui demeure un peu obscure, mais il croit que le theoreme 2 fournit essentiellement le resultat annonce au debut de ce commentaire, a savoir que le faisceau des clotures integrales $f_a(E)$ est un faisceau coherent sur l'espace analytique E (muni de la structure analytique induite par l'espace ambiant).

La demonstration se fait en deux temps : on examine d'abord le "cas special" ou, au voisinage de $a \in E$, le sous-ensemble S des points singuliers de E est de codimension > 2 dans E . On ramene ensuite le cas general a ce cas special en considerant un revetement ramifie $E' \rightarrow D$ qui se factorise par $E' \rightarrow E \rightarrow D$; E' est un sous-ensemble analytique d'un espace $\mathbb{C}^{n+m+r}$, tel que l'ensemble des points singuliers de E' soit de codimension > 2 dans E' au point considere. On applique alors le cas special a E' et on obtient un faisceau coherent sur E' . Il reste a en deduire la coherence du faisceau etudie sur E , et pour cela on est ramene a utiliser le theoreme 1 de la Partie I du present Memoire, qui a ete etabli justement pour cela.

Chapter 8

IX. Unramified Domains Without Points at Infinity

Japanese Journal of Mathematics 27 (1953), pp. 97–155

Introduction

This is the ninth in a series of memoirs, the first of which was published in 1936. We shall first cast an eye over the terrain in which we find ourselves.

The general theory of analytic continuation in a single variable is like an open field; despite many efforts one has not found any facts which could not have been predicted by formal logic. The case of several variables, on the other hand, seems to us like mountainous country, very precipitous.

In 1902, Fabry remarked that the radii of convergence of a double series are not arbitrary; from this, we were led by Hartogs in 1906 to the very fundamental and really curious fact that every domain of holomorphy is pseudoconvex.

From then on until 1932, each new problem in the field gave rise to other problems; the lines of this accumulation of difficulties, as well as the flow of ideas, were drawn with great emphasis in the following work:

BEHNKE–THULLEN, *Theorie der Funktionen mehrerer komplexer Veränderlichen*, 1934.

The main problems dealt with in this book are: the inverse Hartogs problem, first and second Cousin problems, problem of expansion of functions.

In 1935, Weil took the first step in the opposite direction, that is, in the direction of solving these problems; thanks to this, the last three of the above problems could be solved for rationally convex domains.

It was just at this time, and to study these problems, that we began our researches. The following memoirs of Behnke–Stein, Caratheodory, and others opened up new perspectives on these classical problems.

We shall devote the present memoir to the study of unramified domains without points at infinity, and to the solution of the Cousin problems for such domains.

Continued in the next volume of the collected papers.

End of Part III

Pages 101–150 of the Collected Papers

Typeset with LuaLaTeX

Continuation of Memoir VIII and Memoir IX

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Chapter I

Complement to the Fundamental Lemma

0.1 The Problem

We want to solve the inverse Hartogs problem without invoking the Weil integral. This gives rise to the following problem concerning the lemma established in the preceding memoir.

In the lemma of Memoir VIII, we found that to each holomorphic function u on a neighbourhood of Δ_0 on $\Re$, there corresponds a holomorphic function $F(x, y)$ on a neighbourhood of (A_0, B_0) such that $F = u$ on L and $F = 0 \pmod{\Phi}$ globally. Let now

$$F = A_1\Phi_1 + A_2\Phi_2 + \cdots + A_\mu\Phi_\mu \quad (0.1)$$

identically, the A_i ($i = 1, 2, \dots, \mu$) being holomorphic functions in a neighbourhood of (A_0, B_0) ; suppose that

$$|u| < M \quad \text{on } V_0,$$

where V_0 is a domain (not necessarily connected) such that $\Delta_0 \subset V_0 \subset \Re$, and M is a positive number. We ask if it is possible to choose the A_i ($i = 1, 2, \dots, \mu$) with the bound

$$|A_i| < KM,$$

K being a positive constant independent of u .

We shall solve this by inspecting successively the quantitative relations in the theorems which went into the lemma.

0.2 The Remainder Theorem

Let us start with the remainder theorem formulated in No. 5 of Memoir VII (in what follows, we shall omit the word “Memoir”) as follows:

Consider a domain of the form $[\mathfrak{D}, (C)]$ in the space $(x_1, x_2, \dots, x_n, y)$, where $\mathfrak{D}$ is a univalent (finite) domain in the space (x) and (C) is a circle in the y -plane. Let us consider a holomorphic function $F(x, y)$ in $[\mathfrak{D}, (C)]$ such that, for any point (x^0) of $\mathfrak{D}$, the equation $F(x^0, y) = 0$ has λ roots in (C) , λ being a finite integer independent of (x^0) . Then, any holomorphic function

$f(x, y)$ on $[\mathfrak{D}, (C)]$ can be put in the form

$$f(x, y) = f_0(x, y) + \varphi(x, y)F(x, y), \quad (0.2)$$

where f_0 and φ are holomorphic on $[\mathfrak{D}, (C)]$; f_0 is a polynomial in y of degree at most $\lambda - 1$ (identically zero if $\lambda = 0$). Further, such a decomposition is unique.

This theorem, due to W. Rückert ^{extsuperscript11}), has been examined quantitatively by H. Cartan ^{extsuperscript12}); we shall simply formulate his result:

Let (C_0) be a circle concentric with (C) and contained in (C) ; suppose that all λ roots of $F(x^0, y)$ remain in (C_0) for an arbitrary point (x^0) of $\mathfrak{D}$. Then, if $|f|$ has an upper bound M on $[\mathfrak{D}, (C)]$, we have

$$|f_0| < KM, \quad |\varphi| < KM$$

on $[\mathfrak{D}, (C_0)]$, K being a positive constant independent of f .

0.3 The Local Problem (C_1)

We now inspect problem (C_1) quantitatively. Let us first consider the problem locally; we shall prove the following:

Proposition 0.1. *Given holomorphic functions $F_1, F_2, \dots, F_p$ in a polycylinder (C) , $|x_i| < R$ ($i = 1, 2, \dots, n$) in the space (x) , there correspond a positive constant r smaller than R and a positive constant K which have the following property: Let $f(x)$ be a holomorphic function on (C) such that $f = 0 \pmod{F}$ at every point of (C) , and such that $|f| < M$ on (C) (M is a positive number), one can then find holomorphic functions $A_j(x)$ ($j = 1, 2, \dots, p$) on the polycylinder (γ) , $|x_i| < r$ ($i = 1, 2, \dots, n$) with the bounds*

$$|A_j(x)| < KM \quad \text{on } (\gamma)$$

in such a way that we have $f = A_1F_1 + A_2F_2 + \dots + A_pF_p$ identically.

We shall generalise problem (C_1) as follows:

Suppose given a system of simultaneous functional equations

$$f_i = A_{i1}F_{i1} + A_{i2}F_{i2} + \dots + A_{ip}F_{ip} \quad (i = 1, 2, \dots, q), \quad (0.3)$$

where f_i, F_{ij} ($j = 1, 2, \dots, p$) stand for given holomorphic functions on a domain $\mathfrak{D}$ over the space (x) , and the A_{ij} stand for the unknown functions, together with a certain number of identities of the form

$$A_{ij} = A_{kl} \quad (i, k = 1, \dots, q; j, l = 1, \dots, p).$$

If this system of equations has a solution (A) at every point of $\mathfrak{D}$, find a solution on $\mathfrak{D}$.

Let us rearrange the form of the functional equation. Suppose given, for example,

$$f_1 = A_{11}F_{11} + A_{12}F_{12}, \quad f_2 = A_{21}F_{21} + A_{22}F_{22} \quad \text{with } A_{12} = A_{21}.$$

Let us set

$$\begin{aligned}\Phi_{11} &= F_{11}, & \Phi_{12} &= F_{12}, & \Phi_{13} &= 0, \\ \Phi_{21} &= 0, & \Phi_{22} &= F_{21}, & \Phi_{23} &= F_{22};\end{aligned}$$

the given functional equation then takes the form

$$f_i = B_1\Phi_{i1} + B_2\Phi_{i2} + B_3\Phi_{i3} \quad (i = 1, 2).$$

By applying this method of arranging the functional equation (together with the identities indicated), we can always put the system in the form

$$f_i = B_1\Phi_{i1} + B_2\Phi_{i2} + \cdots + B_r\Phi_{ir} \quad (i = 1, 2, \dots, q), \quad (0.4)$$

where $(\Phi_{i1}, \Phi_{i2}, \dots, \Phi_{ir})$ is equal, up to order, to $(F_{i1}, F_{i2}, \dots, F_{ip}, 0, \dots, 0)$.

We shall justify the proposition for the problem so generalised. Since it is obviously true for $(0, q)$, it is sufficient to justify it for $(n, q+1)$ and $(n+1, 1)$ assuming it to be true for all (n', q') such that $n > n' > 0$, $q > q' > 0$. We begin with the case of $(n, q+1)$.

Consider the simultaneous functional equations

$$f_i = A_1F_{i1} + A_2F_{i2} + \cdots + A_pF_{ip} \quad (i = 1, 2, \dots, q+1) \quad (0.5)$$

in the space $(x_1, x_2, \dots, x_n)$, where f_i, F_{ij} ($j = 1, \dots, p$) are given holomorphic functions on the domain $\mathfrak{D}$.

0.4 The Global Problem (C_1)

We now inspect problem (C_1) quantitatively and globally. For this, it is necessary to continue to treat it in generalised form.

Let us recall the geometric configuration of No. 4 in VII.

Let us take a closed circle (C_i) in each x_i -plane ($i = 1, \dots, n$) and let (C) be the closed polycylinder having the (C_i) as components. Denote by E_0 an arbitrary point of (C).

We separate the variable x_n into its real and imaginary parts, $x_n = X + iY$; and consider a line of the form $X_i = X_i^0$ ($i = 1, \dots, n-1$), $Y = Y^0$, the X_i^0 being complex numbers, Y^0 real. Let E_1 be the intersection of this line with the closed polycylinder (C); E_1 is a segment which may reduce to a point.

We consider similarly $E_2, E_3, \dots$ by raising each time the real dimension by 1, terminating with E_{2n} ; E_2 , for example, is a closed cylindrical set with a point of (C_i) as component in the x_i -plane for $i = 1, 2, \dots, n-1$, and (C_n) as component in the x_n -plane; E_{2n} stands for the polycylinder (C).

We have just seen that the problem (C_1) (generalised and quantitative) is solvable in the neighbourhood of E_0 ; let us therefore pass to the case of E_1 .

We consider an arbitrary one of the sets E_1 and denote it by the same letter; we may suppose that it is a real segment.

The component of E_1 in $(x_1, \dots, x_{n-1})$ space is a point which we denote by Q and its component in the x_n -plane is a segment which we denote by I ; I is horizontal, its left extremity will be denoted by m_0 , its right extremity by m_q ; we introduce $q - 1$ points $m_1, \dots, m_{q-1}$ on the segment from left to right. Let I_i be the closed segment $[m_{i-1}, m_i]$ ($i = 1, \dots, q$). Let M_i be the cylindrical set (Q, m_i) in the space (x) and L_i the cylindrical set (Q, I_i) . We set

$$L_1 \cup L_3 \cup L_5 \cup \dots = L^{(1)}, \quad L_2 \cup L_4 \cup L_6 \cup \dots = L^{(2)}.$$

We then have

$$L^{(1)} = L^{(1)} \cup L^{(2)} = E_1, \quad L^{(1)} \cap L^{(2)} = M_1 \cup M_2 \cup \dots \cup M_{q-1}.$$

In a polycylinder (C') concentric with (C) and containing (C) , we now consider the functional equation

$$f_i = A_1 F_{i1} + A_2 F_{i2} + \dots + A_p F_{ip} \quad (i = 1, 2, \dots, p'), \quad (0.6)$$

assuming that it has a solution at every point of (C') . Since this problem (C_1) is solvable (quantitatively) at every point of E_1 , we can solve it in the neighbourhood of each I_h if we choose the I_h ($h = 1, 2, \dots, q$) sufficiently small; more precisely, we can find a circle (χ_h) in the x_n -plane around the midpoint of I_h which contains I_h and a polycylinder (β) in the space $(x_1, \dots, x_{n-1})$ with centre Q which are independent of the I_h , in such a way that there are holomorphic functions A_j ($j = 1, 2, \dots, p$) in (β, χ_h) satisfying the identities and the condition

$$|A_j| < K_1 M,$$

where K_1 is a positive constant independent of the h and M is an upper bound for the $|f_i|$ on (C') .

0.5 Problem (C_2)

Between problems (C_1) and (C_2) , there is the relation which we have given in No. 1 of VII; hence the following result is obtained immediately from the above result:

Consider two concentric polycylinders (C) , (C_0) in the space (x) having radii R_i , R_i^0 ($i = 1, 2, \dots, n$) respectively, where $R_i > R_i^0$. Let r be a positive number with $r < R_i - R_i^0$, and let (x^0) be an arbitrary point of (C_0) . We describe the polycylinder (γ) with centre (x^0) and radius r . Let $F_j(x)$ ($j = 1, 2, \dots, p$) be a system of holomorphic functions on (C) . Under these circumstances, there exists a positive constant K with the following property: Given a holomorphic function $\varphi(x)$ on (γ) such that $|\varphi| < M$, M being a positive number independent of (γ) , and such that for any pair $((\gamma_1), (\gamma_2))$ of these (γ) , the corresponding functions φ_1, φ_2 satisfy $\varphi_1 = \varphi_2 \pmod{F}$ at every point of $(\gamma_1) \cap (\gamma_2)$, we can find a holomorphic function $\Phi(x)$ on (C_0) such that $|\Phi| < KM$ and $\Phi = \varphi \pmod{F}$ at every point of (γ) .

We have treated problems (C_1) , (C_2) on polycylinders, but the results hold on univalent cylindrical domains; to verify this, it is enough to apply the usual procedure of passing to higher-dimensional spaces.

0.6 The Nullstellensatz

We have to examine the quantitative aspect of the theorem which we have called the Nullstellensatz (No. 8 of VIII) in the following form:

Let us consider, in the space $(x_1, x_2, \dots, x_n, y_1, \dots, y_\mu)$, a holomorphic function $W(x, y)$; let L be the characteristic variety $W(x, y) = 0$ and S_0 the set of its singular points. Let $U(x, y)$ be a holomorphic function which vanishes identically on S_0 . Then $U = 0 \pmod{W}$ in a neighbourhood of any point of S_0 . More precisely, if we denote by L_0 the intersection of L with a closed polycylinder (C) having (x^0, y^0) as centre, where (x^0, y^0) is a point of S_0 , we have

$$U = W^\lambda \Omega \quad (\lambda \text{ being a positive integer}), \quad (0.7)$$

Ω being a holomorphic function on (C) .

We want to prove that if $|U| < M$ on (C) , we can choose Ω such that $|\Omega| < KM$ on (C_0) , (C_0) being a concentric polycylinder, smaller than (C) , and K being a positive constant independent of U .

Let us recall the reasoning in No. 8 of VIII. We first consider a point (x^0, y^0) of S_0 and denote by L_1 the intersection of L with a sufficiently small neighbourhood of (x^0, y^0) .

1°. If (x^0, y^0) is not in S_0 , we have $\partial W / \partial y_1 \neq 0$, for example; hence, by the implicit function theorem, we have, in a neighbourhood of (x^0, y^0) , $y_1 = f(x, y_2, \dots, y_\mu)$, where f is holomorphic. We set

$$y'_1 = y_1 - f(x, y_2, \dots, y_\mu), \quad y'_i = y_i \quad (i = 2, \dots, \mu).$$

In the new variables, W takes the form $W = y'_1 \cdot W_1(x, y')$, where W_1 does not vanish in the neighbourhood. U vanishes on $y'_1 = 0$; hence $U = y'_1 \cdot U_1 = W \cdot (U_1/W_1)$.

2°. If (x^0, y^0) belongs to S_0 , we can find $\mu + 1$ functions W_i ($i = 1, 2, \dots, \mu + 1$) holomorphic in a neighbourhood of (x^0, y^0) such that the set of common zeros of the W_i is contained in S_0 and such that $U = 0 \pmod{W_1, W_2, \dots, W_{\mu+1}}$.

3°. Suppose that (x^0, y^0) belongs to S_0 . Let $U(x, y)$ be a holomorphic function in a neighbourhood of (x^0, y^0) which vanishes identically on S_0 . By the Nullstellensatz, we find that $U = 0 \pmod{W_1, W_2, \dots, W_{\mu+1}}$ in a neighbourhood of (x^0, y^0) , λ being a positive integer.

4°. Let us now consider the problem posed in No. 1 above (see No. 10 of VIII). Because of the results of No. 4 onwards, we find easily (using the Borel lemma) that the answer is affirmative, as long as the integer λ is chosen sufficiently large.

Chapter II

Pseudoconvex Domains; a Second Lemma

0.7 Generalities

Let us recall what we said in the introduction to Memoir VI. In 1906, F. Hartogs made what I think is an epoch-making discovery in the theory of analytic functions of several variables; namely that every domain of holomorphy is subject to a very curious restriction which we call being pseudo-convex.

The same restriction was found successively in different parts of the theory; we shall rapidly recall them. In 1910, E. E. Levi showed that domains of meromorphy are also pseudoconvex; G. Julia indicated in 1926 that so also are domains of normality for holomorphic functions. In 1931, W. Saxer added the statement that the same is true (in a certain sense) for families of meromorphic functions. And finally, in 1934, the author indicated the corresponding result for families of characteristic surfaces.

These theorems were studied, and diverse consequences deduced, by the discoverers themselves, and other mathematicians, especially by Hartogs and E. E. Levi.

The present chapter is divided into three parts A, B, C. In A, we shall expound, abstractly, what we owe to Hartogs and to E. E. Levi.

0.8 Domains

The domains which we shall consider from here on in the present memoir contain neither points at infinity nor interior points of ramification, with a single exception in No. 24. We shall not, therefore, indicate these conditions. Domains of exactly this kind are explained in the book of Behnke-Thullen; what we shall do is simply to introduce and add some terminology and notation which will be useful in what follows.

We consider the space of n complex variables $x_1, x_2, \dots, x_n$; and over this space (x) , we consider a point P ; a point is a thing (Ding) which has well-defined coordinates (x) . The point (x) in the space (x) is called the projection of the point P (on the space (x)); and is denoted by $\overline{P}$. We shall sometimes say that P lies over $\overline{P}$. If E is a set of points, the set of projections $\overline{E}$ is called the projection of E and is denoted by $\overline{E}$.

In general, we shall use, for sets of points over the space (x) , the terms and the notation of the theory of abstract sets.

We shall call a set of points over the space (x) a domain if it possesses a system $(\mathfrak{S})$ of subsets, called neighbourhoods (Umgebungen) which satisfies “neighbourhood postulates” (“Umgebungspostulate”) and, in addition, is connected in a sense which respects $(\mathfrak{S})$.

Consider a domain $\mathfrak{D}$ over the space (x) . $\mathfrak{D}$ has, by definition, a definite system $(\mathfrak{S})$ of neighbourhoods. We can therefore draw continuous curves on $\mathfrak{D}$. Let ξ, η be an arbitrary pair of points of $\mathfrak{D}$; we denote by $d(\xi, \eta)$ the lower bound of the lengths of rectifiable curves on $\mathfrak{D}$ joining ξ and η ; $d(\xi, \eta)$ satisfies the distance postulates (“Entfernungspostulate”). We shall define the distance (Entfernung) of points ξ, η on $\mathfrak{D}$ as this $d(\xi, \eta)$. By introducing this, we can use the terminology of “metric spaces” for domains.

0.9 Pseudoconvex Domains

Definition 0.1. Let $\mathfrak{D}$ be a domain over the space (x) . We say that $\mathfrak{D}$ is pseudoconvex if for every point P_0 of $\mathfrak{D}$, there exists a neighbourhood $\mathfrak{U}$ of P_0 in $\mathfrak{D}$ such that $\mathfrak{U}$ is a domain of holomorphy.

This definition is equivalent to the following: Let $\varphi(P)$ be a real-valued continuous function on $\mathfrak{D}$. We say that φ is pseudoconvex if for every one-dimensional complex analytic curve C in $\mathfrak{D}$, the restriction of φ to C is subharmonic.

0.10 Pseudoconvex Functions

Definition 0.2. A real-valued continuous function $\varphi(P)$ on a domain $\mathfrak{D}$ is called pseudoconvex if for every one-dimensional complex analytic variety A passing through any point P_0 of $\mathfrak{D}$, the restriction of φ to A is subharmonic in a neighbourhood of P_0 .

More precisely, let P_0 be a point of $\mathfrak{D}$ having coordinates (x^0) . We consider a one-dimensional complex analytic variety A passing through P_0 and having in a neighbourhood of this point the form

$$x_j = f_j(t) \quad (j = 1, 2, \dots, n), \quad (0.8)$$

where the $f_j(t)$ are holomorphic functions of t in a neighbourhood of the origin (and such that $f_j(0) = x_j^0$). We separate t, x_j into real and imaginary parts and set

$$t = u + iv, \quad x_j = u_j + iv_j$$

(i being the imaginary unit). We shall denote $\varphi(x)$ by

$$\varphi(u_1, u_2, \dots, u_n; v_1, v_2, \dots, v_n) = \varphi(u_j, v_j).$$

0.11 The Main Property

Let $\varphi(x_1, x_2, \dots, x_n, y)$ be a real continuous function in the neighbourhood of a point (x^0, y^0) of the space (x, y) . We separate x_j, y ($j = 1, 2, \dots, n$) into their real and imaginary parts and

set

$$x_j = u_j + iv_j, \quad y = y_1 + iy_2,$$

and denote $\varphi(x, y)$ by $\varphi(u, v, y_1, y_2)$. Let us suppose that $\varphi(u, v, y_1, y_2)$ has continuous partial derivatives up to second order, and that $W(\varphi; \alpha, \beta)$ is everywhere positive for all systems of real values of (α, β) except for $(0, 0)$; we shall, in what follows, denote this simply by

$$W(\varphi; \alpha, \beta) > 0$$

(if there is no confusion).

Suppose, further, that

$$\frac{\partial^2 \varphi}{(\partial y_1)^2} + \frac{\partial^2 \varphi}{(\partial y_2)^2} > 0 \quad \text{at } (x^0, y^0).$$

Under these circumstances, we propose to find a characteristic surface σ passing through (x^0, y^0) in such a way that it does not leave the portion of the space where

$$\varphi(x, y) > \varphi(x^0, y^0)$$

in the neighbourhood of (x^0, y^0) , except at this point itself.

Theorem 0.1. *Suppose given a real continuous function $\varphi(x_1, x_2, \dots, x_n)$ in a neighbourhood of a point (x_0) in the space (x) such that, if u_j, v_j are the real and imaginary parts of x_j ($j = 1, 2, \dots, n$), the function $\varphi(x) = \varphi(u, v)$ has continuous partial derivatives up to second order, we have $W(\varphi; \alpha, \beta) > 0$ and, furthermore, at least one of the derivatives $\partial\varphi/\partial u_j$ ($j = 1, 2, \dots, n$) is non-zero at (x_0) . We can then find a characteristic surface σ passing through (x_0) and remaining in the part of the space where $\varphi(x) > \varphi(x_0)$ except at (x_0) itself. Moreover, if one of the derivatives $\partial\varphi/\partial u_n, \partial\varphi/\partial v_n$ is not zero at (x_0) , we can make σ take the form $x_n = f(x_1, x_2, \dots, x_{n-1})$, f being a polynomial in $x_1, x_2, \dots, x_{n-1}$ of degree at most 2.*

0.12 Modification of Pseudoconvex Functions

We are now concerned with modifying pseudoconvex functions to make them satisfy the conditions of the above theorem.

Let us begin with the problem of raising the degree of continuity. This problem has already been solved for subharmonic functions by T. Radó and F. Riesz, whose methods can be applied to our case; we now explain this.

1°. Let $\mathfrak{D}$ be a domain over the space $(x_1, x_2, \dots, x_n)$. Following the book of Behnke-Thullen, we consider the boundary distance (Randdistanz) as follows. Let P be an arbitrary point of $\mathfrak{D}$ and (x_0) its coordinates. Consider a polycylinder γ with centre (x_0) and radius r such that there is a domain γ^* satisfying the conditions $\gamma^* \cong \gamma$, $\gamma^0 \subset \mathfrak{D}$ and $P \in \gamma^*$; γ^* will be called a polycylindrical neighbourhood on $\mathfrak{D}$ with centre P and radius r . Let $\bar{r}$ be the upper bound of these r . We call $\bar{r}$ the boundary distance of P with respect to $\mathfrak{D}$.

Let now $\varphi(P)$ be a single valued real upper semicontinuous bounded function on $\mathfrak{D}$. Let P be an arbitrary point of $\mathfrak{D}$, let P' be a definite point in $\mathfrak{D}_1 = \mathfrak{D}(r)$ where r is a fixed positive

number. We consider a polycylindrical neighbourhood γ with centre P' and radius r , and the arithmetic mean of $\varphi(P)$ on γ using the Lebesgue integral

$$\varphi_1(P') = \frac{1}{(\pi r^2)^n} \int_{\gamma} \varphi(P) dV.$$

The function $\varphi_1(P)$ is defined on $\mathfrak{D}_1$ and takes a well-defined finite value at every point of $\mathfrak{D}_1$. It is obviously continuous. We shall denote, following F. Riesz, the operation of constructing $\varphi_1(P)$ from $\varphi(P)$ by

$$\varphi_1(P) = A_r[\varphi(P)].$$

We suppose, in what follows, that $\varphi(P)$ is pseudoconvex and bounded on $\mathfrak{D}$. We shall first show that $\varphi_1(P)$ is also pseudoconvex on $\mathfrak{D}_1$. Suppose in fact that $\mathfrak{D}$ is a univalent domain in the space (x) ; there is no loss of generality in this.

0.13 A Second Modification

Let $\varphi(P)$ be a pseudoconvex function on a domain $\mathfrak{D}$ over the space $(x_1, x_2, \dots, x_n)$. In view of the preceding result, we can suppose that $\varphi(P)$ is continuous and has continuous partial derivatives up to second order with respect to u_j, v_j ($j = 1, 2, \dots, n$), the u_j, v_j being, respectively, the real and imaginary parts of x_j . To make $\varphi(P)$ have the main property (of No. 16), it is sufficient to make it satisfy the following conditions:

$$\begin{aligned} W(\varphi; \alpha, \beta) &> 0, \\ \sum_{j=1}^n \left[\left(\frac{\partial \varphi}{\partial u_j} \right)^2 + \left(\frac{\partial \varphi}{\partial v_j} \right)^2 \right] &> 0, \quad (j = 1, 2, \dots, n). \end{aligned}$$

Recall that the second condition did not appear in the case of two complex variables (No. 13, Memoir VI).

0.14 A Boundary Problem

Problem 0.1 (Boundary Problem). Given a pseudoconvex domain $\mathfrak{D}$ over the space $(x_1, x_2, \dots, x_n)$, find a pseudoconvex function $\psi(P)$ on $\mathfrak{D}$ having the following properties: 1°. $\psi(P)$ possesses property (P_1) . 2°. For any real number α , the set $\mathfrak{D}_\alpha$ of points of $\mathfrak{D}$ such that $\psi(P) < \alpha$ satisfies the condition $\mathfrak{D}_\alpha \subseteq \mathfrak{D}$.

The significance of this problem will be understood if one recalls the results expounded in “III. A Complementary Problem” in Memoir VI.

Now, in the present case, since $\mathfrak{D}$ is not univalent, we have to search for a new kind of pseudoconvex function on $\mathfrak{D}$, apart from the Hartogs radii, in order to make $\mathfrak{D}_\alpha$ always compact on $\mathfrak{D}$. (For the same reason, one sees that the theorem of Cartan-Thullen is not sufficient to treat the given problems on multivalent domains of holomorphy.)

Problem 0.2 (Auxiliary Problem). Let $\mathfrak{D}$ be a domain over the space (x) and $\mathfrak{D}_0$ a subdomain whose polycylindrical boundary distance with respect to $\mathfrak{D}$ is not zero. Find a pseudoconvex

function $\lambda_0(P)$ on $\mathfrak{D}_0$ such that for any real number α , the set Δ_α of points of $\mathfrak{D}_0$ such that $\lambda_0(P) < \alpha$ satisfies the condition $\Delta_\alpha \Subset \mathfrak{D}$.

0.15 Normal Families of Continuous Curves

We shall apply, for this purpose, Montel's concept to continuous curves. This is not indispensable, but is useful in handling the auxiliary problem.

Given a family $\mathfrak{F}$ of continuous curves in the space $(x_1, x_2, \dots, x_n)$, we shall say that $\mathfrak{F}$ is a normal family if, from any infinite sequence C_j ($j = 1, 2, \dots$) in $\mathfrak{F}$, we can extract an infinite subsequence C_{p_j} ($j = 1, 2, \dots$) in such a way that for each curve C_{p_j} , we can choose an equation of the form

$$x_i = \varphi_i^{(p_j)}(t), \quad 0 < t < 1, \quad (i = 1, 2, \dots, n)$$

representing C_{p_j} (the $\varphi_i^{(p_j)}$ being continuous functions) such that the sequences $\varphi_i^{(p_j)}(t)$ ($j = 1, 2, \dots$) are uniformly convergent on $0 < t < 1$.

0.16 Solution of the Auxiliary Problem

In the auxiliary problem, let us for a moment ignore the condition of being pseudoconvex. We then find diverse functions of the kind wanted right at hand; of these we shall choose the following:

Let O be a fixed point of $\mathfrak{D}_0$ and P an arbitrary point of $\mathfrak{D}_0$. Let $d(P_1, P_2)$ be the distance between P_1 and P_2 on the domain $\mathfrak{D}_0$ (more precisely, the lower bound of the lengths of curves on $\mathfrak{D}_0$ joining P_1 and P_2) and let us set

$$\lambda(P) = d(O, P).$$

$\lambda(P)$ is a real continuous function on $\mathfrak{D}_0$. Let α be an arbitrary real number, and Δ_α the set of points of $\mathfrak{D}_0$ satisfying $\lambda(P) < \alpha$. Since the polycylindrical boundary distance of $\mathfrak{D}_0$ with respect to $\mathfrak{D}$ is not zero, it is clear that $\Delta_\alpha \Subset \mathfrak{D}$ because of the criterion we just saw. $\lambda(P)$ is thus one of the functions with the required property.

0.17 Solution of the Boundary Problem; a Second Lemma

We consider the following problem to complete the problem of No. 18.

Problem 0.3. Let $\mathfrak{D}$ be a pseudoconvex domain over the space (x) , let $\varphi(P)$ be a continuous pseudoconvex function on $\mathfrak{D}$ having the following property (P_0) (on $\mathfrak{D}$): for any real number α , the set of points of $\mathfrak{D}$ given by $\varphi(P) < \alpha$ is completely interior to $\mathfrak{D}$. Modify the function $\varphi(P)$ in such a way that it has property (P_1) , preserving the other properties on $\mathfrak{D}$.

To solve this problem, let α_p ($p = 1, 2, \dots$) be a sequence of positive numbers such that $\alpha_p < \alpha_{p+1}$, and having the limit $+\infty$. Let us denote the set of points of $\mathfrak{D}$ defined by $\varphi(P) < \alpha_p$ by $\mathfrak{D}_p$; we then have $\mathfrak{D}_p \Subset \mathfrak{D}$. Since the problem of No. 18 is solvable, given a positive number

ε , we can find a pseudoconvex function $\psi_p(P)$ on $\mathfrak{D}_p$, having property (P_1) and such that

$$|\varphi(P) - \psi_p(P)| < \varepsilon.$$

Lemma 0.1 (Lemma II). *Given a pseudoconvex domain over the space (x) without points at infinity or interior points of ramification, we can always find a pseudoconvex function $\psi(P)$ on $\mathfrak{D}$ having the following properties: 1°. $\psi(P)$ possesses property (P_1) . 2°. For any real number α , the set of points of $\mathfrak{D}$ given by $\psi(P) < \alpha$ remains completely interior to $\mathfrak{D}$.*

0.18 Subsequent Problems

We shall inspect rapidly the nature of the boundary problem when one allows either points at infinity (in a suitable sense) or non-transcendental points of ramification as interior points.

In the first case, we cannot use the auxiliary function $\mu(x) = \sum |x_i|^2$ ($i = 1, 2, \dots, n$); this gives rise to a strange problem.

In the second case, the Hartogs radii no longer play a role; this presents a difficulty which seems to me to be truly great.

As for details concerning the form of these difficulties, we shall content ourselves with presenting them in later memoirs.

Chapter III

The Main Problems

0.19 General Notions

Let us consider a domain $\mathfrak{D}$ over the space $(x_1, x_2, \dots, x_n)$ (without points at infinity or interior points of ramification); let P be an arbitrary point of $\mathfrak{D}$ and (x) its coordinates. A function $f(P)$ of P is called holomorphic on $\mathfrak{D}$ if it is single valued on $\mathfrak{D}$ and, further, is a holomorphic function of (x) at every point of $\mathfrak{D}$. Let $f(P)$ be a holomorphic function (of P) on $\mathfrak{D}$. We say that $\mathfrak{D}$ is the domain of existence (or domain of holomorphy) of $f(P)$ if this function always has different Taylor expansions at any two different points of $\mathfrak{D}$ and in addition, if this function cannot be continued analytically across any boundary point of $\mathfrak{D}$. $\mathfrak{D}$ is called a domain of holomorphy if there exists at least one function for which $\mathfrak{D}$ is the domain of existence.

A domain of holomorphy is pseudoconvex by Hartogs' theorem (1906). Here the notion of being pseudoconvex is a kind of local convexity (one can define it locally). Now, in 1926, Julia proposed the following global problem: Is every domain of normality a domain of holomorphy? And in 1931, in order to handle Julia's problem, H. Cartan introduced, for the first time, a kind of global convexity (which cannot, a priori, be expressed locally); his idea is essentially as follows:

Let $\mathfrak{D}$ be a domain over the space (x) (not necessarily connected), and let (Φ) be a family of holomorphic functions on $\mathfrak{D}$. We say that $\mathfrak{D}$ is convex with respect to (Φ) , or simply (Φ) -convex, if it satisfies the following conditions:

1°. The family (Φ) contains at least one function $f_0(P)$ having different Taylor expansions at any pair of distinct points on $\mathfrak{D}$.

2°. For any domain Δ (connected or not) such that $\Delta \subseteq \mathfrak{D}$, there exist two domains Δ' , Δ'' (connected or not) such that $\Delta \subseteq \Delta' \subseteq \Delta'' \subseteq \mathfrak{D}$ and such that to any point P_0 of $\Delta'' - \Delta'$ corresponds a function $f(P)$ of (Φ) for which $|f(P_0)| > \max |f(\Delta)|$ (where the term on the right denotes the upper bound of $|f(P)|$ on Δ).

As a special case, if $\mathfrak{D}$ is convex with respect to the family of holomorphic functions on $\mathfrak{D}$, we say, with Behnke, that $\mathfrak{D}$ is holomorph-convex (regular-konvex). Concerning this notion, we shall see later that the first condition is necessarily a consequence of the second.

More generally, if $\mathfrak{D}$ is convex with respect to the family consisting of functions holomorphic on a domain $\mathfrak{D}'$ such that $\mathfrak{D}' \supset \mathfrak{D}$, we shall say that $\mathfrak{D}$ is holomorph-convex with respect to $\mathfrak{D}'$.

In 1932, H. Cartan and Thullen found that any univalent domain of holomorphy without

points at infinity is holomorph-convex. (This remains true for the case of a bounded number of sheets.) This is a truly fundamental fact for univalent domains, and gives the reason to distinguish domains of holomorphy from other kinds of pseudoconvex domains, domains of meromorphy, domains of normality, etc. We have, in fact, solved the Cousin problems and the problem of expansions for univalent (finite) domains of holomorphy with the help of this theorem.

However, in the present case, the (really simple) argument of Cartan-Thullen is no longer valid as we have already remarked. Consequently, one has to take the notion of holomorph-convex domains as the basic notion. This leads, first, to the following problem:

Problem 0.4 (Inverse Hartogs Problem). Is every pseudoconvex domain over the space (x) (without points at infinity or interior points of ramification) holomorph-convex?

We shall, from now on, call this the inverse Hartogs problem.

0.20 Expansion of Holomorphic Functions

Using the idea of Memoir I and Lemma I of No. 24, we obtain the following theorem.

Theorem 0.2. *Let $\mathfrak{D}$, $\mathfrak{D}'$ be two domains over the space (x) , without points at infinity or interior ramification points such that $\mathfrak{D} \Subset \mathfrak{D}'$; $\mathfrak{D}$ is not necessarily connected. If $\mathfrak{D}$ is holomorph-convex with respect to $\mathfrak{D}'$, every function holomorphic on $\mathfrak{D}$ can be expanded in a series of functions holomorphic on $\mathfrak{D}'$ converging uniformly on any region completely interior to $\mathfrak{D}$.*

Proof. In fact, let $f(P)$ be a function holomorphic on $\mathfrak{D}$. Since $\mathfrak{D}$ is holomorph-convex with respect to $\mathfrak{D}'$, it can be expressed as the limit of an increasing sequence of analytic polyhedra Δ_p of the form

$$|x_i| < r_i, \quad |\varphi_j(P)| < 1, \quad P \in \mathfrak{G} \quad (i = 1, 2, \dots, n; j = 1, 2, \dots, \nu),$$

where $\mathfrak{G}$ stands for a domain (connected or not) such that $\Delta_p \Subset \mathfrak{G} \subset \mathfrak{D}$, the r_i are positive constants and the $\varphi_j(P)$ are holomorphic functions on $\mathfrak{D}'$ such that the ordered systems of values of $(\varphi_1, \varphi_2, \dots, \varphi_\nu)$ are different at any pair of distinct points in a neighbourhood of Δ_p which have the same coordinates.

Let us consider the closed polycylinder C ,

$$|x_i| < r_i, \quad |y_k| < 1 \quad (i = 1, 2, \dots, n; k = 1, 2, \dots, \nu)$$

in the space $(x_1, \dots, x_n, y_1, \dots, y_\nu)$ and the characteristic variety L given by $y_j = \varphi_j(P)$ ($j = 1, 2, \dots, \nu$), $P \in \mathfrak{G}$ (the coordinates of P being (x)). By Lemma I, we can find a holomorphic function $F(x, y)$ in a neighbourhood of C such that $F = f$ on L . If we expand $F(x, y)$ in a Taylor series about the origin and substitute $y_j = \varphi_j(P)$ ($j = 1, 2, \dots, \nu$), we obtain an expansion of $f(P)$ in a neighbourhood of Δ_p consisting of terms holomorphic on $\mathfrak{D}'$ and converging uniformly in a neighbourhood of Δ_p . The theorem follows immediately from this. $\square$

0.21 The First Cousin Problem

The same applies to the first Cousin problem as we shall verify. We start by verifying the following lemma.

Lemma 0.2. *Let us consider the analytic polyhedron Δ over the space (x) as in Lemma I, and a hyperplane L meeting Δ . L partitions Δ into two closed sets Δ_1, Δ_2 ; let J_0 be the part of J on L . With this geometric configuration, given a holomorphic function $\varphi(P)$ in a neighbourhood of J_0 , we can find two holomorphic functions $\varphi_1(P), \varphi_2(P)$ in the neighbourhood of Δ_1 and of Δ_2 respectively such that we have $\varphi_1(P) - \varphi_2(P) = \varphi(P)$ identically.*

Proof. In fact, suppose that L is defined by $u = 0$, where, to simplify the notation, we assume that u is the real part of x_1 ; let Δ_1 be the part of Δ over $u < 0$; Δ_2 is then that over $u > 0$. Δ is given in the form $x_i \in A_i, f_j(P) \in B_j$ ($i = 1, 2, \dots, n; j = 1, 2, \dots, \nu$) (with the same notation as indicated in Lemma I). Let us draw a circle of the form $|x_1| < R$ in the x_1 -plane which contains A_1 . Let L be the vertical diameter. By Lemma I, we have then a function $\psi(x, y)$, holomorphic in a neighbourhood of the intersection $(x_1 \in L) \cap (A, B)$ such that $\psi = \varphi$ on Γ ($y_j = f_j(P), j = 1, 2, \dots, \nu$). Consider the Cousin integral

$$\frac{1}{2\pi i} \int_L \frac{\psi(t, x_2, \dots, x_n, (y))}{t - x_1} dt,$$

where i is the imaginary unit and the integral is taken from bottom to top. Thanks to Cousin, we know well the properties of this integral (as well as the role it plays). Substituting $y_j = f_j(P)$ ($j = 1, 2, \dots, \nu$), we obviously obtain the required pair of functions. $\square$

Theorem 0.3. *The first Cousin problem is always solvable on a holomorph-convex domain over the space (x) without points at infinity or interior points of ramification.*

Proof. Let $\mathfrak{D}$ be a domain over the finite portion of the space (x) , without interior ramification points. We can give ourselves a Cousin problem on $\mathfrak{D}$ just as in the previous cases; we shall not repeat this.

Suppose that $\mathfrak{D}$ is holomorph-convex. $\mathfrak{D}$ is then the limit of an increasing sequence of analytic polyhedra Δ_p of the form $|x_i| < r_i, |f_j(P)| < 1, P \in \mathfrak{G}$ ($i = 1, 2, \dots, n; j = 1, 2, \dots, \nu$), where $\Delta_p \Subset \mathfrak{G} \subset \mathfrak{D}$, the r_i are positive constants and the $f_j(P)$ are holomorphic functions on $\mathfrak{D}$ having the properties indicated in Lemma I. Let (p) be the given poles on $\mathfrak{D}$. By the lemma which we have just established, we can find a meromorphic function $G_p(P)$ in a neighbourhood of Δ_p having the poles (p) .

Let $H_p(P) = G_{p+1}(P) - G_p(P)$; H_p is a function holomorphic in a neighbourhood of Δ_p . Let ε_p be an arbitrary positive number. By Theorem I, we obtain a function $K_p(P)$ holomorphic on $\mathfrak{D}$ such that $|K_p - H_p| < \varepsilon_p$ on Δ_p . From this, we obtain by the usual argument the theorem. $\square$

0.22 The Inverse Hartogs Problem: The Point of Departure

Problem 0.5 (An Auxiliary Problem). We shall deal with the inverse Hartogs problem using Lemmas I, II and Theorems I, II. Following the idea of Memoir VI, we shall start by looking for something which plays the role of the Cousin integral.

Let $\mathfrak{D}$ be a bounded domain over the space (x) (without interior ramification points). Let u be the real part of x_1 , let a_1, a_2 be real numbers such that $a_1 < 0 < a_2$. We denote the part of $\mathfrak{D}$ over $u < a_2$ by $\mathfrak{D}_1$, that over $u > a_1$ by $\mathfrak{D}_2$, and the part over $a_1 < u < a_2$ by $\mathfrak{D}_3$. We suppose that $\mathfrak{D}_1$ and $\mathfrak{D}_2$ are non-empty and are holomorphy-convex. The same is then true of $\mathfrak{D}_3$.

Let $f_j(P)$ ($j = 1, 2, \dots, \nu$) be holomorphic functions on $\mathfrak{D}_3$. Consider the following subset E of $\mathfrak{D}$: Every point which does not belong to $\mathfrak{D}_3$ belongs to E . As for a point P of $\mathfrak{D}_3$, it belongs to E if and only if it satisfies simultaneously the conditions $|f_j(P)| < 1$ ($j = 1, 2, \dots, \nu$).

We suppose that E has connected components extending over both the portion $u < a_1$ and the portion $a_2 < u$ at once; one of these components will be denoted by Δ . We suppose, relative to this Δ , that the functions (f) satisfy the following conditions (which we call conditions (C)):

1°. There exists a positive number δ_1 such that if $\mathfrak{U}$ is the part of Δ over $|u| < \delta_1$, we have $\mathfrak{U} \subseteq \mathfrak{D}$.

2°. There exists a positive number δ_2 and a positive number ε_0 smaller than 1 such that, if p is any one of $1, 2, \dots, \nu$, the set of points of $\mathfrak{D}_3$ satisfying $|f_p(P)| > 1 - \varepsilon_0$ remains outside $|u - a_1| < \delta_2$ and $|u - a_2| < \delta_2$.

3°. The ordered system of values of $(f_1, f_2, \dots, f_\nu)$ do not coincide for any pair of distinct points of $\mathfrak{U}$ which have the same coordinates.

Given a function $\varphi(P)$ holomorphic on $\mathfrak{D}_3$, find a function $\varphi_1(P)$ holomorphic on Δ_1 and a function $\varphi_2(P)$ holomorphic on Δ_2 so that they remain holomorphic on the part of Δ_0 over $u = 0$ and we have $\varphi_1(P) - \varphi_2(P) = \varphi(P)$ identically.

0.23 The Case of Holomorph-Convex Domains

We shall first consider the problem for $\mathfrak{D}_3$ instead of $\mathfrak{D}$. We shall obtain a solution (ψ_1, ψ_2) by the method expounded in No. 26; we are concerned with finding a suitable expression for the solution.

We consider the characteristic variety Γ in the space $(x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_\nu)$ given by $y_j = f_j(P)$ ($j = 1, 2, \dots, \nu$), P being a general point of $\mathfrak{D}_3$ having coordinates (x) . $\mathfrak{D}_3$ being bounded, we take a positive number r_0 so large that $\mathfrak{D}_3$ is contained in the polycylinder in the space (x) with centre the origin and radius r_0 ; and we consider a polycylinder C , $|x_i| < r$, $|y_k| < \rho$ ($i = 1, 2, \dots, n$; $k = 1, 2, \dots, \nu$) in the space (x, y) , where $r > r_0$, $\rho_0 < \rho < 1$.

Consider the subset $\mathfrak{U}'$ of $\mathfrak{U}$ given by $|u| < \delta$, $|f_k(P)| < \rho$ ($k = 1, 2, \dots, \nu$), where δ is a positive number $< \delta_1$. Since $\mathfrak{U} \subseteq \mathfrak{D}$, $\mathfrak{U}'$ is an open set such that $\mathfrak{U}' \subseteq \mathfrak{U}$.

Since $\varphi(P)$ is a holomorphic function on $\mathfrak{D}_3$, we can construct, by Lemma I, a holomorphic function $\Phi(x, y)$ on the intersection of C and $|u| < \delta$ so that $\Phi = \varphi$ on Γ . Let L be a segment of the imaginary axis in the x_1 -plane which contains the origin and has its extremities in the annulus $r_0 < |x_1| < r$.

Consider the Cousin integral

$$\psi(x, y) = \frac{1}{2\pi i} \int_L \frac{\Phi(x'_1, x_2, \dots, x_n, (y'))}{x'_1 - x_1} dx'_1,$$

where i is the imaginary unit and the integral is taken from bottom to top. If we substitute $y_k = f_k(P)$ in the expression for $\psi(x, y)$, we have

$$\frac{1}{2\pi i} \int_L \frac{\varphi(x'_1, x_2, \dots, x_n, f_1(P), \dots, f_\nu(P))}{x'_1 - x_1} dx'_1.$$

The holomorphic function on the intersection $\Delta_1 \cap \mathfrak{D}_3$ defined by $\psi(P)$ will be denoted by $\psi_1(P)$, that on $\Delta_2 \cap \mathfrak{D}_3$ by $\psi_2(P)$; these functions $\psi_1(P)$ and $\psi_2(P)$ remain holomorphic on the part of Δ_0 over $u = 0$ and satisfy the required relation $\psi_1(P) - \psi_2(P) = \varphi(P)$.

0.24 The General Case

In the expression (1), the cylindrical set of points (L, Γ) in the space (x, y') is the decreasing limit of a sequence of holomorph-convex domains which contain the set; let V be a domain of this sequence.

We construct a meromorphic function $X_1(x, y', P)$ on $(\mathfrak{D}_3 \times V)$ which has the same poles as $X(x, y', P)$ in $(\mathfrak{D}_3)$, but is otherwise holomorphic; this is possible if V is chosen sufficiently close to (L, Γ) because this is a first Cousin problem in view of the second of the conditions (C) and because $(\mathfrak{D}_3 \times V)$ is holomorph-convex (Theorem II).

Since the function $(X - X_1)$ is holomorphic on $(\mathfrak{D}_3 \times V)$ and $(\mathfrak{D}_3 \times V)$ is holomorph-convex with respect to $(\mathfrak{D}_3 \times V)$, we can expand it in a series of functions holomorphic on $(\mathfrak{D}_3 \times V)$ converging uniformly on any region completely interior to $(\mathfrak{D}_3)$ (Theorem I). For a given positive number ε , we can therefore construct a function $F_1(x, y', P)$ holomorphic on $(\mathfrak{D}_3 \times V)$ such that $|(X - X_1) - F_1| < \varepsilon$ on $(\mathfrak{D}_3 \times V)$, where V now stands for a new neighbourhood of (L, Γ) completely interior to the old one. Set

$$K_1(x, y', P) = X - X_1 - F_1;$$

K_1 is a holomorphic function on $(\mathfrak{D}_3)$ such that $|K_1| < \varepsilon$ on $(\mathfrak{D}_3 \times V)$.

We construct, entirely analogously, a function $K_2(x, y', P)$ corresponding to $\mathfrak{D}_2$. We modify the integral (1), using these functions, as follows:

$$I_1(P) = \int [X(x, y', P) - K_1(x, y', P)] \Phi(x, x_2, \dots, x_n, y') (dx, dy'), \quad (0.9)$$

$$I_2(P) = \int [X(x, y', P) - K_2(x, y', P)] \Phi(x, x_2, \dots, x_n, y') (dx, dy'). \quad (0.10)$$

Let us look at $I_1(P)$. Since $X - K_1 = X_1 + F_1$, the function $(X - K_1)$ is meromorphic on V for any fixed point (x, y') on (L, Γ) and is holomorphic for $P \in \Delta_1$. Hence $I_1(P)$ is a holomorphic function on Δ_1 . Similarly, $I_2(P)$ is holomorphic on Δ_2 .

The functions $I_1(P)$, $I_2(P)$ remain holomorphic at every point of Δ_0 over $u = 0$ because the function $\psi(P)$ given by (1) has this property and K_1 , K_2 are holomorphic. From the properties of $\psi(P)$, it follows that

$$I_1(P) - I_2(P) = \varphi(P) - \int [K_1(x, y', P) - K_2(x, y', P)] \Phi(x, x_2, \dots, x_n, y') (dx, dy'). \quad (0.11)$$

Let us set $K(x, y', P) = K_1(x, y', P) - K_2(x, y', P)$, and consider the identity. $\varphi(P)$ is holomorphic on $\mathfrak{D}$, K is holomorphic on $(\mathfrak{D}_3)$ and $\Phi(x, y)$ is holomorphic on $C \cap (|u| < \delta)$. The right-hand side is therefore a holomorphic function on $\mathfrak{D}$, and the same is therefore true of left-hand side. Let us set

$$\varphi_0(P) = I_1(P) - I_2(P);$$

this is the first approximation to $\varphi(P)$. We have thus seen that the given function $\varphi(P)$ and its first approximation $\varphi_0(P)$ are holomorphic on one and the same domain (connected or not), namely $\mathfrak{D}$. This is the fundamental idea.

Let us estimate the difference $\varphi' = \varphi - \varphi_0$. We have

$$\varphi' = \int K \cdot \Phi(dx, dy').$$

Suppose that φ is bounded on $\mathfrak{D}_3$; there is no loss of generality in this assumption. Let M be the upper bound of $|\varphi|$ on $\mathfrak{D}_3$. By Lemma I, we can choose the function $\Phi(x, y)$ holomorphic on $C \cap (|u| < \delta)$ such that $|\Phi| < N \cdot M$, N being a constant independent of φ . As for K , we have $K = K_1 - K_2$, where $|K_1| < \varepsilon$, $|K_2| < \varepsilon$ on (L, Γ, V) . We therefore have

$$|\varphi'(P)| < 2\varepsilon N \cdot N_1 \cdot M$$

on $\mathfrak{D}_3$, where $N_1 = 2r(2\pi\rho_0)^\nu$.

Let us choose ε so that $2\varepsilon N \cdot N_1 < \lambda$, λ being a positive number smaller than 1. We then have $|\varphi'(P)| < \lambda M$.

If we start again with $\varphi'(P)$ instead of $\varphi(P)$, we obtain similarly a first approximation to $\varphi'(P)$, say $\varphi_1(P)$, which is holomorphic on $\mathfrak{D}_3$ and such that, if $\varphi'' = \varphi' - \varphi_1$, then $|\varphi''| < \lambda^2 M$ on $\mathfrak{D}_3$. Repeating this process an infinity of times, we obtain the required solution.

Theorem 0.4. *The auxiliary problem (of No. 27) is always solvable.*

0.25 The Idea of Memoir II and Lemma II

Let us recall the idea in Memoir II. Let $\mathfrak{D}$ be a domain over the space (x) , let E be a set of points on $\mathfrak{D}$. We shall say that E is holomorph-convex (from the outside) with respect to $\mathfrak{D}$ if $E \subseteq \mathfrak{D}$ and the closure of E in $\mathfrak{D}$ is the limit of a decreasing sequence of subdomains (not necessarily connected) of $\mathfrak{D}$ which are holomorph-convex with respect to $\mathfrak{D}$. If Δ_1, Δ_2 are two subdomains (not necessarily connected) of $\mathfrak{D}$ which are holomorph-convex with respect to $\mathfrak{D}$, the intersection $\Delta_1 \cap \Delta_2$ has the same property. Let E be a set of points such that $E \subseteq \mathfrak{D}$ and F_1, F_2 two sets of points of $\mathfrak{D}$ containing E which are closed, compact, and holomorph-convex from the outside with respect to $\mathfrak{D}$. Then, by the preceding remark, the same is true of $F_1 \cap F_2$. Let F_p ($p = 1, 2, \dots$) be a decreasing sequence of sets of points of $\mathfrak{D}$ with this property; the same is then true of the limit.

Let $\mathfrak{D}$ be a holomorph-convex domain over the space (x) and let E be a set of points such that $E \subseteq \mathfrak{D}$. There then exists a set F of points of $\mathfrak{D}$ containing E which is closed, compact and holomorph-convex from the outside with respect to $\mathfrak{D}$. Let H be the intersection of all

these F ; F is clearly a set of points of $\mathfrak{D}$ containing E , closed and compact in $\mathfrak{D}$. We shall call H the (H) -closure of E in $\mathfrak{D}$. We claim that H is the intersection of a countable number of suitable sets F as above.

Theorem 0.5. *Let $\mathfrak{D}$ be a holomorph-convex domain over the space (x) , let E be a set of points such that $E \subseteq \mathfrak{D}$ and let H be the (H) -closure of E with respect to $\mathfrak{D}$. Let σ_t be a piece of a characteristic surface depending on the parameter t in the interval $0 < t < 1$, given in the form $f(P, t) = 0$, $P \in V$ where V is a domain such that $V \subseteq \mathfrak{D}$ and $f(P, t)$ is a holomorphic function of (P, t) in a neighbourhood of $(V, [0, 1])$ which is not identically zero. Then, H and the family $\{\sigma_t\}$ cannot be related to each other in the following way: 1°. The distance on $\mathfrak{D}$ between H and the boundary of any σ_t is larger than a positive constant. 2°. The distance between E and any σ_t is larger than a positive constant. And 3°. σ_0 passes through a point of H and σ_1 is situated at a positive distance from H .*

0.26 Solution of the Problem

Let us return to the auxiliary problem. We remark that, for the Δ of this problem, the first Cousin problem is always solvable on Δ .

In fact, since $\mathfrak{D}_3$ is holomorph-convex with respect to $\mathfrak{D}_1$ and the $f_j(P)$ ($j = 1, 2, \dots, \nu$) are holomorphic functions on $\mathfrak{D}_3$, we can, by Theorem I, expand each $f_j(P)$ in a series of functions holomorphic on $\mathfrak{D}_1$, converging uniformly on any domain completely interior to $\mathfrak{D}_3$. The part of Δ over $u < a_2$ is therefore holomorph-convex, as is the part over $a_1 < u$. Suppose given a first Cousin problem on Δ . By Theorem II, this problem is solvable on the part of Δ over $u < a_2$ and for the part over $a_1 < u$. Since the auxiliary problem is always solvable, the given first Cousin problem on Δ is solvable.

Let us consider how we might construct the domain Δ_0 . Suppose given a domain $\mathfrak{D}$ over the space (x) such that the domains $\mathfrak{D}_1, \mathfrak{D}_2$ are holomorph-convex. Since any holomorph-convex domain is pseudoconvex, $\mathfrak{D}_1$ and $\mathfrak{D}_2$, and consequently $\mathfrak{D}$, are pseudoconvex. We can therefore apply Lemma II to $\mathfrak{D}$, namely: there exists a pseudoconvex function $\varphi(P)$ on $\mathfrak{D}$ having property (P_1) and such that, for any real number α , the set $\mathfrak{D}_\alpha$ of points of $\mathfrak{D}$ given by $\varphi(P) < \alpha$ satisfies $\mathfrak{D}_\alpha \subseteq \mathfrak{D}$.

Let us denote the part of $\mathfrak{D}_\alpha$ over $u < a_2$ by $\mathfrak{D}_\alpha^{(1)}$, that over $a_1 < u$ by $\mathfrak{D}_\alpha^{(2)}$. We then have: $\mathfrak{D}_\alpha^{(1)}$ is holomorph-convex with respect to $\mathfrak{D}_1$; similarly $\mathfrak{D}_\alpha^{(2)}$ with respect to $\mathfrak{D}_2$.

Theorem 0.6. *Let $\mathfrak{D}$ be a domain such that $\mathfrak{D}_1$ and $\mathfrak{D}_2$ are holomorph-convex. Then $\mathfrak{D}$ is also holomorph-convex.*

Theorem 0.7. *Let $\mathfrak{D}$ be a pseudoconvex domain over the space (x) . Let P_0 be an arbitrary point on the boundary of the set $\mathfrak{D}_\alpha$. From the reasoning given in proving the theorem in No. 16 that every function having property (P_0) has also the main property, we deduce immediately that the intersection of $\mathfrak{D}_\alpha$ and a sufficiently small polycylindrical neighbourhood with centre P_0 is holomorph-convex. Consequently, by the theorem above, $\mathfrak{D}_\alpha$ is holomorph-convex; hence, by the above, so is $\mathfrak{D}$.*

0.27 Pseudoconvex Domains and Domains of Holomorphy

Theorem 0.8. *Every pseudoconvex domain over the space (x) without points at infinity or interior points of ramification is holomorph-convex.*

From this, it follows at once that in the definition of holomorph-convex domains, or of analytic polyhedra, the first condition is but a consequence of the second.

Corollary 0.1. *Every pseudoconvex domain over the space (x) without points at infinity or interior points of ramification is a domain of holomorphy.*

Proof. Let, in fact, $\mathfrak{D}$ be a pseudoconvex domain over the space (x) . By Theorem III, $\mathfrak{D}$ is holomorph-convex. There therefore exists a holomorphic function $F_1(P)$ on $\mathfrak{D}$ having distinct Taylor expansions at any pair of distinct points of $\mathfrak{D}$. Suppose that there is a holomorphic function $F_2(P)$ on $\mathfrak{D}$ such that it cannot be analytically continued across any boundary point of $\mathfrak{D}$, and consider $F(P) = F_1(P) + cF_2(P)$, c being a constant. We find that if we choose c suitably, F conserves both the character of F_1 and that of F_2 (for this, it suffices to avoid a countable number of values). It is therefore sufficient to construct a function $F_2(P)$.

Since the domain $\mathfrak{D}$ is holomorph-convex, it is the limit of a sequence of analytic polyhedra Δ_p ($p = 1, 2, \dots$) such that $\Delta_p \subseteq \Delta_{p+1}$. Each Δ_p is given in the form $|f_j(P)| < 1$, $P \in \mathfrak{G}_p$ ($j = 1, 2, \dots, \nu$), where the $f_j(P)$ are holomorphic functions on $\mathfrak{D}$ (depending on p), $\mathfrak{G}_p$ is a subdomain (connected or not) of $\mathfrak{D}$ such that $\mathfrak{G}_p \supseteq \Delta_p$ (and ν also depends on p).

As for boundary points of $\mathfrak{D}$, we can choose a sequence of points Q_p ($p = 1, 2, \dots$) of $\mathfrak{D}$ with the following properties: 1°. $Q_p \in (\mathfrak{G}_p - \Delta_p)$. 2°. Every boundary point of $\mathfrak{D}$ can be expressed by a suitable subsequence of Q_p ($p = 1, 2, \dots$).

Let G_p ($p = 1, 2, \dots$) be a sequence of positive numbers such that the sum $\sum G_p$ is convergent. To each Q_p corresponds one of the functions $f_j(P)$ such that, if we denote it by $\varphi_p(P)$, we have $\varphi_p(Q_p) = a_p$ where $|a_p| > 1$ and $|\varphi_p(P)| < 1$ on Δ_p .

Let us set $\psi_p(P) = [\varphi_p(P)/a_p]^p$. Let σ_p be the characteristic surface on $\mathfrak{D}$ given by $\varphi_p(P) = a_p$; σ_p passes through Q_p . ψ_p is then a holomorphic function on $\mathfrak{D}$ having zeros of order p on σ_p ; it does not have zeros on Δ_p .

By Theorem I, we can find a holomorphic function $\chi_p(P)$ on $\mathfrak{D}$ such that we have

$$1 - G_p < |\chi_p(P)\psi_p(P)| < 1 + G_p$$

on Δ_p . Let us set

$$F_2(P) = \prod_{p=1}^{\infty} [\psi_p(P) \cdot \chi_p(P)].$$

This product converges absolutely and uniformly on any domain completely interior to $\mathfrak{D}$ since $|\psi_p(P)\chi_p(P) - 1| < G_p$ on Δ_p . □

Memoir IX. Finite Domains Without Interior Critical Points

C. The Other Problems

33. The Second Cousin Problem

The limit $F_2(P)$ is therefore a holomorphic function on $\mathfrak{D}$ whose zeros are given as the union of the zeros of the factors (φ_p/λ_p) . Hence $F_2(P)$ has zeros of order at least p on the characteristic surface σ_p which passes through Q_p (p is arbitrary). Hence we cannot continue $F_2(P)$ analytically across any boundary point of $\mathfrak{D}$. $\mathfrak{D}$ is therefore a domain of holomorphy.

The case considered in the present memoir is the last for which the second COUSIN problem holds in its original form and generality. If we introduce points at infinity, it is a priori obvious that holomorphic solutions do not exist in general. If we permit non-transcendental interior points of ramification, then, as we remarked in the preceding memoir, zeros distributed on a characteristic surface are not even locally the set of zeros of a single holomorphic function in general. In the present case, the inverse HARTOGS problem having been solved, it is sufficient to treat the problem for holomorph-convex domains.

We dealt with this problem in Memoir III on univalent

domains without points at infinity; in a word, we extended the problem itself to the domain of continuous functions to acquire an auxiliary notion with the help of which we expressed our conclusion, namely : if there exists a non-analytic solution, analytic solutions also exist. It is this result which we shall extend.

Let us recall briefly the general notions in the present case. Let $\mathfrak{D}$ be a domain over the space $(x_1, x_2, \dots, x_n)$ (without points at infinity or interior ramification points) and let P be an arbitrary point of $\mathfrak{D}$. Let $f_1(P), f_2(P)$ be two continuous functions on $\mathfrak{D}$; ¹ we shall say that f_1 and f_2 are equivalent ($f_1 \sim f_2$) on $\mathfrak{D}$ if there exists a continuous function $\lambda(P)$, nowhere zero, such that we have $f_1(P) = \lambda(P)f_2(P)$ identically on $\mathfrak{D}$. If $f_1 \sim f_2$ on $\mathfrak{D}$ and the set of zeros has no interior point, $\lambda(P)$ is unique; otherwise $\lambda(P)$ may not be determined uniquely, which gives rise to strange phenomena. We shall therefore exclude this case in what follows (when a continuous function vanishes identically in the neighbourhood of a point).

Suppose given for any point P_0 of $\mathfrak{D}$ a polycylindrical neighbourhood γ around P_0 and a continuous function $f(P)$ defined on γ in such a way that for any pair of contiguous γ , the corresponding functions are equivalent on the intersection (equivalence condition). Find a continuous function $F(P)$ on $\mathfrak{D}$ such that we always have $F(P) \sim f(P)$ on γ . This is the generalized second Cousin problem. If the function $F(P)$ exists, we shall say that $F(P)$ has the given zeros $\mathfrak{z}$ in $\mathfrak{D}$, where $\mathfrak{z}$ is expressed by the system $\{(\mathfrak{z}, \gamma)\}$ satisfying the equivalence condition.

To distinguish the COUSIN problem itself from the generalized problem, we use the terms analytic zeros, analytic solutions; sometimes, if our aim is the opposite of this, we use the terms continuous zeros, non-analytic solutions. If we are given analytic zeros $\mathfrak{z}$ on a domain $\mathfrak{D}$, the two problems arise simultaneously.

1. It is always understood that continuous functions are single-valued.

Given zeros $\mathfrak{z}$ on a domain $\mathfrak{D}$, we shall say that $\mathfrak{z}$ can be *swept out* if to each point P_0 of $\mathfrak{D}$, there exists a polycylindrical neighbourhood γ with centre P_0 and a continuous function $f(P, t)$ defined for $P \in \gamma$, $0 < t < 1$ satisfying the following three conditions :

- 1°. $f(P, 0)$ has the zeros $\mathfrak{z}$, $f(P, 1)$ does not vanish anywhere.
- 2°. $f(P, t)$ does not vanish identically on any portion having real dimension $(2n + 1)$.
- 3°. Let $(f_1, \gamma_1), (f_2, \gamma_2)$ be any two couples in the system $\{(f, \gamma)\}$ and let $\delta = \gamma_1 \cap \gamma_2$; if δ is not empty, we have $f_1(P, t) \sim f_2(P, t)$ on δ ($P \in \delta$, $0 < t < 1$).

Given zeros $\mathfrak{z}$ on a domain $\mathfrak{D}$, if there exists a continuous function $F(P)$ on $\mathfrak{D}$ having the zeros $\mathfrak{z}$, then $\mathfrak{z}$ can, necessarily, be swept out on $\mathfrak{D}$.

To see this, we have only to consider

$$f(P, t) = (1 - t)F(P) + t.$$

If the zeros $\mathfrak{z}$ can be swept out, we can choose a system $\{(f(P, t), \gamma)\}$ so that we always have $f(P, 1) = 1$ identically; further, if the $\mathfrak{z}$ are also analytic, we can choose the system so that, in addition, $f(P, 0)$ is always holomorphic.

Relative to this notion, let $\lambda(P)$ be a continuous function nowhere zero on a domain $\mathfrak{D}$. We say that $\lambda(P)$ satisfies condition (α) on $\mathfrak{D}$ if there exists a continuous function $\lambda(P, t)$ without zeros on $(\mathfrak{D}, [0, 1])$ such that $\lambda(P, 0) = \lambda(P)$, $\lambda(P, 1) = 1$ for any P in $\mathfrak{D}$. We have the following lemma : In order that every determination of $\log \lambda(P)$ be single valued on $\mathfrak{D}$, it is necessary and sufficient that $\lambda(P)$ satisfy condition (α) on $\mathfrak{D}$. As for the proof, since there is nothing to modify, we shall not repeat it.

We are extending the conclusion of Memoir III to the present case. Now, for this, the most favourable method to use is that of Memoir II. For this reason, let us extend Theorem I of Memoir II.

Let us consider a closed polycylinder $C : |x_i| < r_i$ ($i = 1, 2, \dots, n$) in the space (x) and a characteristic variety L defined in a neighbourhood of C . Let L_0 be the part of L on C . Then L_0 is holomorph-convex from the outside with respect to the space (x) (in other words, with respect to polynomials).

In fact, let us consider the geometric ideal $\mathfrak{z}$ with indeterminate domain attached to L , which is defined in a certain neighbourhood of C . By the theorem of H. Cartan, $\mathfrak{z}$ has local pseudobases (see Memoir VIII). Consequently, by Theorem III of Memoir VII, $\mathfrak{z}$ possesses a pseudobasis on a neighbourhood of C . The above lemma follows from this.

Theorem 8.1. *Given analytic zeros $\mathfrak{z}$ on a pseudoconvex domain over the space (x) (without points at infinity or interior ramification points), if there exists a non-analytic solution, there also exists an analytic solution.*

In fact, the zeros $\mathfrak{z}$ can be swept out, so that there exists a system $\{(f(P, t), \gamma)\}$ satisfying the conditions indicated relative to $\mathfrak{z}$; we suppose also that each $f(P, 0)$ is holomorphic and that each $f(P, 1)$ is the constant 1.

The domain $\mathfrak{D}$ can be expressed as the limit of a sequence of analytic polyhedra Δ_p ($p = 1, 2, \dots$) such that $\Delta_p \subset \Delta_{p+1}$. Each Δ_p is represented, in the usual sense, in the form

$$|x_i| < r_i, \quad |g_j(P)| < 1, \quad P \in \mathfrak{D} \quad (i = 1, \dots, n; j = 1, \dots, \nu),$$

($g_j(P)$ being holomorphic on $\mathfrak{D}$). Let us consider the configuration of Lemma I relative to this polyhedron Δ_p ; we have C , Σ , V in the space $(x_1, \dots, x_n, y_1, \dots, y_\nu)$ (No. 24). We have seen that if we are given a holomorphic function $\varphi(P)$ on Δ_p , we can regard it as being defined on Σ . By the same method, we can regard the zeros $\mathfrak{z}$ (which can be swept out) as well as the corresponding system $\{(f(P, t), \gamma)\}$ as being given on V ; here V is holomorph-convex with respect to the space (x, y) because of the above lemma if it is suitably chosen sufficiently close to Σ . Hence, by what we have seen in Memoir III (No. 8), we obtain a special solution, having the property indicated there. Substituting $y_j = g_j(P)$ ($j = 1, 2, \dots, \nu$) in this solution, we obtain a special solution on Δ_{p-1} ($p > 1$).

Thus, we have a function $F_p(P, t)$ on Δ_p ; it is continuous on $[P \in \Delta_p, 0 < t < 1]$ and such that $F_p(P, 0)$ is a holomorphic function on Δ_p and $F_p(P, 1) = 1$ identically; furthermore

$$F_p(P, t) \sim f(P, t)$$

on $[P \in \gamma, 0 < t < 1]$. The theorem results from this; the argument being word for word the same as in Memoir III (No. 9), we omit it.

Applying the same method to the generalized problem, we obtain the following criterion more easily than the above.

Theorem 8.2. *In order that the generalized second Cousin problem be solvable, it is necessary and sufficient that the given zeros be such that they can be swept out.*

34. The Problems (C_1) , (C_2) , (E) ; Incomplete Solutions

We shall follow the plan of Memoir VII and apply the method of Memoir II to problems (C_1) , (C_2) and (E) .

Let us begin with the generalized problem (C_1) as presented in Chapter I.

(C_1) Suppose given a functional equation

$$\varphi_i = \sum_j \lambda_j F_{ij} \quad (i = 1, 2, \dots, p; \quad j = 1, 2, \dots, q) \quad (8.1)$$

in the neighbourhood of an analytic polyhedron Δ , where the φ_i , F_{ij} stand for given holomorphic functions and the λ_j are unknown functions. If this equation has a local solution everywhere in a neighbourhood of Δ , it has a (global) solution on a neighbourhood of Δ .

Suppose, in fact, that Δ is given, in the usual sense, in the form $|x_i| < r_i$, $|g_j(P)| < 1$, $P \in \mathfrak{D}$ ($i = 1, 2, \dots, n$; $j = 1, 2, \dots, \nu$). In the space (x, y) , consider the closed polycylinder C , $|x_i| < r_i$, $|y_j| < 1$ and the characteristic variety Σ given by $y_j = g_j(P)$, $P \in \mathfrak{D}$. With this configuration, we can, by Lemma I, find holomorphic functions $G_{kl}(x, y)$, $\Phi_k(x, y)$

($k = 1, 2, \dots, p; l = 1, 2, \dots, q$) in a neighbourhood of C such that we have $G_{kl} = F_{kl}$, $\Phi_k = \varphi_k$ on Σ . Let $\mathfrak{z}_0$ be the geometric ideal (with indeterminate domains) attached to Σ and defined in a neighbourhood of C , and let $(H_1, H_2, \dots, H_\mu)$ be a pseudobasis of $\mathfrak{z}_0$ on a neighbourhood of C .

Consider the functional equation

$$\Phi_i = \sum_j \lambda_j G_{ij} + \sum_k B_{ik} H_k \quad (i = 1, \dots, p) \quad (8.2)$$

($j = 1, \dots, q; k = 1, \dots, \mu$) in a neighbourhood of C ; the λ_j , B_{ik} are the unknown functions. Since the equation (8.1) has a local solution at every point of a neighbourhood of Δ , the same is true of the equation (8.2) in a neighbourhood of C . Hence, by what we have seen in No. 4, we can find holomorphic functions λ_j , B_{ik} in a neighbourhood of C satisfying the equation (8.2). Substituting $y_l = g_l(P)$ ($l = 1, 2, \dots, \nu$), the desired result follows from this.

Let us consider problem (C_2) :

(C_2) Let Δ be an analytic polyhedron over the space (x) , let $F_1(P), F_2(P), \dots, F_\omega(P)$ be a system of functions holomorphic on a neighbourhood of Δ . Suppose that for any point P_0 in a neighbourhood of Δ , we are given a polycylindrical neighbourhood γ with centre P_0 and a holomorphic function $\psi(P_0)$ on γ in such a way that for any pair (γ_1, γ_2) of these polycylindrical neighbourhoods, the corresponding functions ψ_1, ψ_2 satisfy $\psi_1 = \psi_2 \bmod (F)$ at every point of $\gamma_1 \cap \gamma_2$.

We can then find a holomorphic function $\Psi(P)$ such that $\Psi(P) = \psi(P)$ at any point of γ , for an arbitrary γ .

In fact, with the same geometric configuration as above, let $G_i(x, y)$ ($i = 1, 2, \dots, \omega$) be functions holomorphic in a neighbourhood of C such that $G_i = F_i$ on Σ . We consider a problem (C_2) in a neighbourhood of C in the following way : To each point (x_0, y_0) in a neighbourhood of C we attach a function $\psi(x, y)$; if (x_0, y_0) is a point of Σ , we take for ψ a function which takes the value ψ on Σ ; if not, we take ψ at random.

Under these conditions, find a function $\Psi(x, y)$, holomorphic in a neighbourhood of C , such that $\Psi = \psi \bmod (G, H)$ at every point (x_0, y_0) , (H) being the same pseudobasis of $\mathfrak{z}_0$ as above. This is obviously a problem (C_2) , hence $\Psi(x, y)$ exists by Theorem II of VII. If we set $\Psi(P) = \Psi(x, g(P))$, $\Psi(P)$ is the required function.

Next, let $\mathfrak{D}$ be a domain over the space (x) . We consider a pair $(f, \mathfrak{E})$ where $\mathfrak{E}$ is a subdomain of $\mathfrak{D}$ (not necessarily connected) and f is a holomorphic function defined on $\mathfrak{E}$. With these $(f, \mathfrak{E})$, we can define, exactly as in Memoir VII, (holomorphic) ideals $(\mathfrak{J})$ in $\mathfrak{D}$; the $(\mathfrak{J}, \mathfrak{E})$'s such that one has $\mathfrak{E} \subset \mathfrak{D}$ and, if $[x, g(P)]$ is an arbitrary point of Σ in $\mathfrak{E}$, the function $\zeta[x, g(P)] = f(P)$ belongs locally to $\mathfrak{J}$. Clearly $\mathfrak{J}$ is an ideal without lacunary points in $\mathfrak{D}$.

Let P_0 be an arbitrary point in the neighbourhood of Δ and $[f_1(P), f_2(P), \dots, f_\mu(P)]$ a pseudobasis of $\mathfrak{J}$ at P_0 . Let (x_0) be the coordinates of P_0 and let M_0 be the point on Σ having the coordinates $[x_0, g(P_0)]$. Let $\zeta_i(x, y)$, $(i = 1, 2, \dots, \mu)$ be a holomorphic function at M_0 such that $\zeta_i = f_i$ on Σ . Then $(\zeta_1, \dots, \zeta_\mu, H_1, \dots, H_\lambda)$ is a pseudobasis of $\mathfrak{J}$ at M_0 ((H) is the pseudobasis indicated above). At any point of V which does not belong to Σ , $\mathfrak{J}$ has (1) as pseudobasis. Hence, by Theorem III of VII, $\mathfrak{J}$ has a (finite) pseudobasis on a neighbourhood of C . The theorem follows from this on substituting $y_j = g_j(P)$ ($j = 1, 2, \dots, \nu$).

35. Complete Solution of the Congruence Problems

We are concerned with solving these problems on pseudoconvex domains themselves. This is possible for problems (C_1) , (C_2) , as we shall now show.

Theorem 8.3. *Suppose given on a pseudoconvex domain $\mathfrak{D}$ over the space (x) without points at infinity or interior points of ramification, a functional equation*

$$\varphi_i = \sum_j \lambda_j F_{ij} \quad (i = 1, 2, \dots, \mu; \quad j = 1, 2, \dots, \mu'), \quad (8.3)$$

where φ_i, F_{ij} represent given holomorphic functions, and the λ_j stand for unknown functions. If this equation has a local solution everywhere on $\mathfrak{D}$, it has a solution on $\mathfrak{D}$.

In fact, $\mathfrak{D}$ can be expressed as the limit of a sequence of analytic polyhedra Δ_p ($p = 1, 2, \dots$) such that $\Delta_p \subset \Delta_{p+1}$, each Δ_p being defined by functions holomorphic on $\mathfrak{D}$. By the intermediate result (α), there exists a solution $(\lambda_{p1}, \lambda_{p2}, \dots, \lambda_{p\mu'})$ on a neighbourhood of Δ_p . We shall construct another solution from this as follows :

(A_1) : For (A_1), we take $\lambda_i = \lambda_{1i}$ ($i = 1, 2, \dots, \mu'$).

To construct (A_2), set $B_i = \lambda_{2i} - \lambda_i$ ($i = 1, 2, \dots, \mu'$). The system (B) satisfies the linear functional equation

$$B_1 F_{i1} + B_2 F_{i2} + \dots + B_{\mu'} F_{i\mu'} = 0 \quad (i = 1, 2, \dots, \mu) \quad (8.4)$$

on a neighbourhood of Δ_1 .

Theorem 8.4. *Let $\mathfrak{D}$ be a pseudoconvex domain over the space (x) without points at infinity or interior ramification points. Let $F_1(P), F_2(P), \dots, F_\mu(P)$ be a system of functions holomorphic on $\mathfrak{D}$. Suppose that we are given, for every point P_0 of $\mathfrak{D}$, a polycylindrical neighbourhood γ with centre P_0 and a holomorphic function $\varphi(P)$ defined on γ such that for any pair (γ_1, γ_2) of these polycylindrical neighbourhoods, the corresponding functions φ_1, φ_2 satisfy $\varphi_1 = \varphi_2 \bmod (F)$ at every point of the intersection $\gamma_1 \cap \gamma_2$. We can then find a holomorphic function $\Phi(P)$ on $\mathfrak{D}$ such that we have $\Phi(P) = \varphi(P) \bmod (F)$ for every γ .*

In fact, continuing to regard $\mathfrak{D}$ as the limit of the sequence Δ_p ($p = 1, 2, \dots$), we have, by the intermediate result (β), a solution $\Phi_p(P)$ on a certain neighbourhood of Δ_p (because, by Theorem VI, the following two terms mean the same thing : $\Phi = \varphi$ at every point of γ ; and : $\Phi = \varphi$ on γ). Let ε_p ($p = 1, 2, \dots$) be a sequence of positive numbers such that the sum $\sum \varepsilon_p$ is convergent. We shall, recursively, construct a new sequence of solutions.

Theorem 8.5. *Problem (E) is always solvable on a pseudoconvex domain $\mathfrak{D}$ over the space (x) without points at infinity or interior ramification points, provided that the set of zeros of the coefficients has no interior point.*

This can already be found among geometric ideals with indeterminate domains.

Example 8.6. We take a positive integer ν such that $\nu > 3$ and consider the following four functions in (x_1, x_2, y_1, y_2) -space :

$$\begin{aligned} F_1 &= y_1^\nu - x_1^{\nu-1}, & F_2 &= y_2^\nu - x_1^\nu x_2, \\ F_3 &= y_1 y_2 - x_1 x_2, & F_4 &= x_1^{\nu-2} + x_2^\nu; \end{aligned}$$

we denote the characteristic variety defined by $F_1 = F_2 = F_3 = F_4 = 0$ by L and the geometric ideal with indeterminate domain attached to L by $\mathfrak{z}_0$; this is defined on the space (x_1, x_2, y_1, y_2) . We shall prove that the number of elements in any pseudobasis of the ideal $\mathfrak{z}_0$ at the origin is $> \nu - 1$.

Suppose that the ideal $\mathfrak{z}_0$ has a pseudobasis $[\varphi_1(x, y), \dots, \varphi_p(x, y)]$ at the origin, where $p = \nu - 2$.

Consider the RIEMANN surface (of complex dimension 2) of the function $t = x_1^{1/\nu}$ over (x_1, x_2) -space; we denote it by $\mathfrak{R}$. Let P be an arbitrary point of $\mathfrak{R}$, let (x_1, x_2) be the coordinates of P , and let t be the value of $x_1^{1/\nu}$ at P . Let S be the characteristic variety defined by $F_1 = F_2 = F_3 = 0$, and let M be an arbitrary point of S ; what are the coordinates of M ?

Since $F_1 = 0$, we have, first $y_1^\nu = x_1^{\nu-1}$, so $y_1 = t^{\nu-1}$. From $F_2 = 0$, we next obtain $y_2^\nu = x_2 x_1^\nu = \varepsilon x_2 t^\nu$, where ε is a number such that $\varepsilon^\nu = 1$, but is otherwise arbitrary. Now, since $F_3 = 0$, we must have $x_1 x_2 = y_1 y_2 = \varepsilon x_2 t^\nu = \varepsilon x_1 x_2$, that is, $\varepsilon = 1$. Hence M is given by $(x_1, x_2, t^{\nu-1}, x_2 t)$. Let us make the point $P(x_1, x_2, t)$ of $\mathfrak{R}$ and the above point M correspond to each other; this correspondence is biunique.

Let $\mathfrak{z}_0$ be the image on $\mathfrak{R}$ of the restriction of the ideal $\mathfrak{z}_0$ to S ; more precisely, let $\mathfrak{z}_0 = \{(f, \delta)\}$, let δ' be the image on $\mathfrak{R}$ of the part of δ on S , and let $f'(P)$ be the value at P

of the function induced by f on δ ; we denote the family of these (f', δ') by $\mathfrak{J}_0$. Then $\mathfrak{J}_0$ is a geometric ideal on $\mathfrak{R}$.

Each h_i has property (H) with respect to S ; in fact, we have on S

$$t \frac{\partial F_4}{\partial x_i} = \frac{\partial Y}{\partial x_i} - t \frac{\partial x_1^{\nu-2}}{\partial x_i} + Y \frac{\partial x_1^{\nu-1}}{\partial x_i} \quad (i = 1, 2, \dots, p).$$

Moreover, the function $h(t, x_2)$ vanish identically on $F_4 = 0$; they therefore belong to $\mathfrak{J}_0$ on $\mathfrak{R}$. Consequently, we have, in the neighbourhood of P_0 ,

$$h_i = c_{i1}\varphi_1 + c_{i2}\varphi_2 + \dots + c_{ip}\varphi_p \quad (i = 1, 2, \dots, p+1),$$

where the c_{ij} ($j = 1, 2, \dots, p$) represent holomorphic functions of (t, x_2) in the neighbourhood of $(0, 0)$ which have property (H) with respect to S .

Consider a holomorphic function $f(P)$ in the neighbourhood of P_0 having property (H) with respect to S ; say $f(P) = f(t, x_2)$. There then exists a holomorphic function $F(x_1, x_2, y_1, y_2)$ near the origin of (x_1, x_2, y_1, y_2) -space such that, identically,

$$F(x_1, x_2, t^{\nu-1}, x_2 t) = f(t, x_2).$$

If we expand $f(t, x_2)$ in a TAYLOR series about the origin and set $x_2 = 0$, we therefore have, necessarily,

$$f(t, 0) = a_0 + a_{\nu-1}t^{\nu-1} + a_{\nu}t^{\nu} + \dots,$$

($a_0, a_{\nu-1}, a_{\nu}, \dots$ being constants).

The functions $\varphi_i(t, x_2)$ must vanish on $F_4 = 0$. Since $F_4 = 0$ does not have multiple factors, they must be of the form

$$\varphi_i(x_1, x_2) = (x_1^{\nu-2} + x_2^{\nu})h_i(t, x_2),$$

the $h_i(t, x_2)$ being holomorphic. We therefore have, in the neighbourhood of $t = 0$,

$$\varphi_i(t, 0) = x_1^{\nu-2}(A_{i1} + A_{i2}t + A_{i3}t^2 + \dots + A_{i,p+1}t^p) + Bx_1^{\nu-2}t^{p+1} + \dots$$

$$= \chi_i(t) + Bx_1^{\nu-2}t^{p+1} + \dots$$

(where A_{ij} ($j = 1, 2, \dots, p+1$) and B are constants).

Let us set

$$c_{ij}(t, 0) = \gamma_{ij} + \delta_{ij}' t^{\nu-1} + \delta_{ij}'' t^\nu + \dots \quad (i = 1, \dots, p+1; j = 1, \dots, p).$$

Since $\nu - 1 = p + 1$ and

$$h_i(t, 0) = x_1^{\nu-2} t^{i-1} = c_{i1}(t, 0) \varphi_1(t, 0) + \dots + c_{ip}(t, 0) \varphi_p(t, 0)$$

($i = 1, 2, \dots, p+1$), we necessarily have

$$x_1^{\nu-2} t^{i-1} = \sum_j \gamma_{ij} \chi_j(t),$$

identically, where one at least of the $\gamma_{p+1,k}$ ($k = 1, 2, \dots, p$) is not zero because $\gamma_{p+1,1} A_{1,p+1} + \dots + \gamma_{p+1,p} A_{p,p+1} = 1$; thus $L_1, L_2, \dots, L_p$ are linearly dependent. On the other hand, since

$$\gamma_{i1} L_1 + \gamma_{i2} L_2 + \dots + \gamma_{ip} L_p = Z_i \quad (i = 1, 2, \dots, p),$$

$L_1, L_2, \dots, L_p$ are necessarily linearly independent. These conclusions are contradictory. Hence, the number of elements in any pseudobasis of $\mathfrak{z}_0$ at the origin must be $> \nu - 1$. $\square$

Let us now consider, in the space $(x_1, x_2, y_1, y_2, y_3)$ the characteristic variety given by

$$y_3 = b_\nu, \quad F_i(x_1, x_2, y_1, y_2) = 0 \quad (i = 1, 2, 3, 4),$$

where ν is a positive integer with $\nu > 3$, b_ν is a constant, and (x_1, x_2, y_1, y_2) stand for the functions we have just considered. For the geometric ideal attached to Γ_ν and defined in a domain containing the point $(0, \dots, 0, b_\nu)$, the number of elements in a local pseudobasis at this point is always $> \nu - 1$.

Let us consider the polycylinder C , $|x_i| < 1$, $|y_j| < 1$ ($i = 1, 2$; $j = 1, 2, 3$) in the space $(x_1, x_2, y_1, y_2, y_3)$; let us take the sequence $b_\nu = 1 - 2^{-\nu}$ ($\nu = 3, 4, \dots$) and consider the characteristic variety $\Gamma = \Gamma_3 + \Gamma_4 + \dots$ in C . Let $\mathfrak{z}_1$ be the geometric ideal with indeterminate domains attached to Γ . Because of what we have just seen, it is clear that $\mathfrak{z}_1$ does not possess any finite pseudobasis on C .

base $x_0 \in D$ et $x_1 \in D_1$ tels que $p(x_0) = p_1(x_1)$, on dit que D_1 est étalé sur D (et on écrit $D_1 \Subset D$) s'il existe une application continue $\varphi : D_1 \rightarrow D$ telle que $\varphi(x_1) = x_0$ et $p_1 = p \circ \varphi$; une telle φ est alors unique, et c'est une application étale de D_1 dans D . On définit alors facilement l'"intersection" de deux domaines étalés à point-base.

Rappelons que (D, p) est un *domaine d'holomorphie* s'il jouit de la propriété suivante : chaque fois que l'on a un domaine étalé $p_1 : D_1 \rightarrow \mathbb{C}^n$, deux points $x_0 \in D$ et $x_1 \in D_1$ tels que $p(x_0) = p_1(x_1)$, une fonction f holomorphe dans D et une fonction f_1 holomorphe dans D_1 qui "prolongent" l'une l'autre au voisinage des points-base, alors f_1 est le prolongement de f par l'application $\varphi : D_1 \rightarrow D$. Il revient au même de dire que D est holomorphiquement convexe.

OKA démontre d'abord que la propriété $H(D)$ (condition de HARTOGS globale) entraîne que D est holomorphiquement convexe. Il lui faut pour cela, comme dans le Mémoire VI, prouver l'existence d'une fonction psh dans D qui satisfasse aux propriétés (P) et (P') ; on prouve alors (fin du n°31) que si D est en outre holomorphiquement convexe, tout D_ρ (défini par $\varphi(x) < \rho$) est holomorphiquement convexe, et même holomorphiquement convexe relativement à D (c'est-à-dire en utilisant seulement des fonctions holomorphes dans D). Il s'ensuit que toute fonction holomorphe dans D_ρ est limite, uniformément sur tout compact, de fonctions holomorphes dans D (il s'agit là d'un théorème connu, que OKA rappelle sous l'énoncé du théorème I (n°25)). La suite des raisonnements utilise le fait que le premier problème de COUSIN est résoluble dans tout D_ρ holomorphiquement convexe (ce que OKA prouve en utilisant les polyèdres analytiques contenus dans D puis en passant à la limite : c'est l'objet du théorème II). Ensuite OKA prouve ce que nous avons appelé un "lemme de recollement" dans notre commentaire du Mémoire VI : si D est un domaine étalé dans $\mathbb{C}^n$ (coordonnées $x_1, \dots, x_n$), et si D_1 (défini comme l'ensemble des $x \in D$ dont l'image dans $\mathbb{C}^n$ satisfait $\operatorname{Re}(x_1) > -1$) et D_2 (défini comme l'ensemble des points de D dont l'image dans $\mathbb{C}^n$ satisfait

$\operatorname{Re}(x_1) < +1$) sont holomorphiquement convexes, alors D est holomorphiquement convexe. C'est là l'énoncé (13) du n°31.

La preuve de ce lemme de recollement est très technique. On observe d'abord que D_1 et D_2 sont des domaines d'holomorphie (plus exactement, leurs composantes connexes sont des domaines d'holomorphie), donc ils sont pseudo-convexes. Or la pseudo-convexité a un caractère local à la frontière; il s'ensuit que D lui-même est pseudo-convexe. Mais à ce stade du mémoire, OKA n'a pas encore prouvé que la pseudo-convexité entraîne la propriété de HARTOGS (ce qui sera l'objet du chapitre III); aussi doit-il montrer directement que D possède la propriété $H(D)$, en utilisant les polyèdres analytiques contenus dans D .

Le chapitre III est consacré aux problèmes généraux de COUSIN et aux problèmes (C_1) , (C_2) et (E) . OKA utilise la méthode de "l'immersion dans un espace plus grand" (déjà utilisée au Mémoire VII) : si Δ est un polyèdre analytique donné par $|x_i| < r_i$, $|g_j(P)| < 1$ ($i = 1, \dots, n$; $j = 1, \dots, \nu$), on plonge Δ dans le polycylindre C de l'espace (x, y) défini par $|x_i| < r_i$, $|y_j| < 1$, et on considère la variété caractéristique Σ définie par $y_j = g_j(P)$; alors les problèmes pour Δ se ramènent à des problèmes analogues pour C , ce qui permet de leur donner une solution complète.

Le dernier paragraphe (n°35) est particulièrement intéressant : OKA y montre que les idéaux géométriques cohérents n'ont pas nécessairement de pseudobase finie. Il donne à cet effet un exemple remarquable, où il considère dans $\mathbb{C}^4$ la variété caractéristique L définie par

$$y_1^\nu = x_1^{\nu-1}, \quad y_2^\nu = x_1^\nu x_2, \quad y_1 y_2 = x_1 x_2, \quad x_1^{\nu-2} + x_2^\nu = 0,$$

et il montre que l'idéal géométrique attaché à L admet, à l'origine, une pseudobase ayant un nombre d'éléments $> \nu - 1$. Comme OKA l'observe, cela donne un exemple où les sections d'un faisceau cohérent sur D ne sont pas combinaisons linéaires (à coefficients holomorphes dans D) d'un nombre fini d'entre elles.

Ce séminaire ayant été rédigé en octobre 1952, il semble

que OKA n'en ait pas eu connaissance lorsqu'il écrivit ce Mémoire IX, terminé en octobre 1953. Aussi é prouve-t-il le besoin, bien naturel, d'étendre les résultats globaux de son Mémoire VII (où ils étaient énoncés pour les polycylindres compacts) à tous les domaines d'holomorphie. C'est l'objet de la dernière partie du chapitre III. OKA y montre aussi sur un exemple que, en général, les sections d'un faisceau cohérent sur D ne sont pas combinaisons linéaires (à coefficients holomorphes dans D) d'un nombre fini d'entre elles.

Memoir X. A New Method of Generating Pseudoconvex Domains

UNE MODE NOUVELLE ENGENDRANT LES DOMAINES PSEUDOCONVEXES,

Japanese Journal of Mathematics **32** (1962), p. 1–12.

Introduction

In a series of earlier Memoirs², we treated a group of problems which were posed precisely in the beautiful book of H. BEHNKE–P. THULLEN. But, as we remarked in the preceding memoir, there still remain unsolved problems, some of which seem to us to be among the most difficult. It is our intention to treat these problems successively.

I shall not go into technical detail in this introduction. Rather, I should like to refer to the feeling for the seasons, which has been special to Japanese people since time immemorial, to explain what I feel in completing the present Memoir.

There is a tendency towards abstraction in the progress of the mathematical sciences of today. Even in the field of our own research, theorems have become more and more general, and some of them have even gone outside the space of

2. Memoirs I–IX of the present collection.

complex variables. I felt that this was winter. I have waited for a long time for the return of spring and have wanted to make some studies which would make this felt. The present memoir is but the first result. (See the theorem in No. 6.)

1. Preliminaries

For simplicity, we shall restrict ourselves to the space of two complex variables x, y . Let us consider a univalent domain D without points at infinity and a characteristic surface Σ in D . Let M_0 be a point of Σ . Suppose that, in the neighbourhood of M_0 , we can represent the point $M(x, y)$ of Σ by

$$x = \varphi(t), \quad y = \psi(t),$$

where φ, ψ are holomorphic functions of the complex variable t ; then, if t_0 corresponds to M_0 , we have

$$dx = \varphi'(t) dt, \quad dy = \psi'(t) dt,$$

and consequently, setting $t = t_1 + it_2$,

$$|dx|^2 + |dy|^2 = (|\varphi'(t)|^2 + |\psi'(t)|^2) (dt_1^2 + dt_2^2).$$

The area of the characteristic surface is therefore expressed in the form

$$da = (|\varphi'(t)|^2 + |\psi'(t)|^2) dt_1 dt_2.$$

We can also define the area in another way: the characteristic surface is given by an equation $f(x, y) = 0$; on this surface we have

$$\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = 0,$$

and consequently

$$da = \frac{|dx|^2 + |dy|^2}{\left| \frac{\partial f}{\partial x} \right|^2 + \left| \frac{\partial f}{\partial y} \right|^2} \cdot \left| \frac{\partial f}{\partial y} \right|^2.$$

In view of the CAUCHY-RIEMANN condition, this reduces to

$$da = \left| \frac{dx}{dt} \right|^2 dt_1 dt_2 + \left| \frac{dy}{dt} \right|^2 dt_1 dt_2.$$

We see thus that the area of the characteristic surface L is expressed as the sum of the areas of its projections on the coordinate planes.

2. The Cousin Representation

Let us consider again the characteristic surface L introduced above, and a dicylinder $[(C), (C')]$ in the interior of the domain D (that is to say, $D \supset [(C), (C')]$), and we shall find a representation of L in the dicylinder following COUSIN. The projection of the portion of L in the dicylinder onto the x -plane consists of pieces of RIEMANN surfaces (as in the Figure) which we denote by $\mathfrak{R}_x$. Let us suppose, for simplicity, that the characteristic surface L does not contain any plane of the form $x = \text{const.}$ or $y = \text{const.}$, and that the circles (C) , (C') are given by

$$|x| < R, \quad |y| < R'$$

respectively.

Let us look at the RIEMANN surface $\mathfrak{R}_x$ (not necessarily connected). Let x', x'' be two boundary points of $\mathfrak{R}_x$ such that

$$|x'| < R, \quad |x''| < R,$$

and let us suppose, to facilitate our explanations, that they cannot have the same coordinate except for a finite number of points.

Let us look at the figure, from which we can easily find the general case. The circle (C) is partitioned into 5 portions $\Pi_1, \Pi_2, \dots$ by 3 curves $\theta_1, \theta_2, \theta_3$ which, together with the circumference C constitute the projection on the x -plane of the boundary of $\mathfrak{R}_x$. The number of sheets of $\mathfrak{R}_x$ is 0 over Π_1 , 1

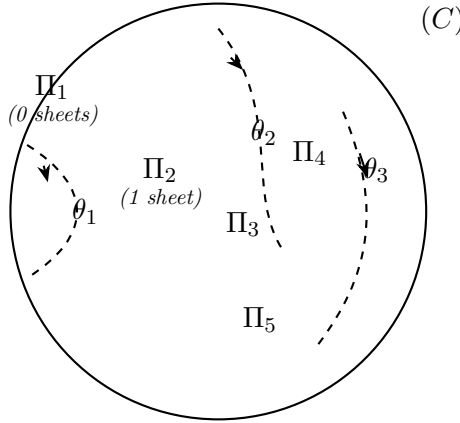


Figure 8.1 – Projection of $\mathfrak{R}_x$ onto the x -plane, showing the partition by curves $\theta_1, \theta_2, \theta_3$ and the number of sheets over each region Π_i .

over Π_2 , which one recognises from the direction of the curve θ_1 .

We are going to decompose the curves $\theta_1, \theta_2, \theta_3$ into pieces δ_i ($i = 1, 2, \dots, t$). Let θ denote any one of these curves. The positive sense of θ is that of the projection, C' , and that of β_j is the direction corresponding to this. Let $q + 1$ be the number of sheets of the RIEMANN surface over the point Π_0 . If we let Y trace out the curve β_j starting at Π_0 , Y will pass successively through q points (over Π_0) which we denote by $\eta_1, \eta_2, \dots, \eta_q$; let $\zeta_1, \zeta_2, \dots, \zeta_q$ be the corresponding points of θ_1 . The arc θ_{j1} is then divided into $q + 1$ arcs $(\zeta_0, \zeta_1), (\zeta_1, \zeta_2), \dots, (\zeta_{q-1}, \zeta_q)$ and what remains; we denote these by $\delta_1, \delta_2, \dots, \delta_{q+1}$. As for θ_{j2} , since it is a closed curve, there is just one difference: we take first a point η_0 over the point R' . Just as above, we obtain points of division $\eta_{q+1}, \eta_{q+2}, \dots, \eta_{q+r}$ on θ_{j2} . We mark points of division $\eta_{q+r+1}, \eta_{q+r+2}, \dots, \eta_{q+r+s}$ on θ_{j3} . We thus obtain $\delta_1, \delta_2, \dots, \delta_t$ where

$$t = q + r + s + 2.$$

To every x in Δ_j , there correspond j' points in (C') :

$$y_{j,1}, y_{j,2}, \dots, y_{j,\nu}, \dots, y_{j,j'}$$

such that the points $(x, y_{j,\nu})$ lie on Σ . If we regard x as variable in the neighbourhood of the initial position, these $y_{j,\nu}$ represent functions of x , holomorphic on the RIEMANN surface. To the domain Δ_j we make correspond the holomorphic function $\Phi_j(x, y)$ as follows:

$$\Phi_j(x, y) = (y - y_{j,1})(y - y_{j,2}) \cdots (y - y_{j,j'}) \quad \text{if } j' \neq 0,$$

$$\Phi_j(x, y) = 1 \quad \text{if } j' = 0.$$

Let x be an arbitrary point on the curve θ_j considered on the RIEMANN surface (obtained by continuation of $\Re_x$); to x , there corresponds a single point on β_j . This determines a function $Y_j(x)$ attached to θ_j . In the situation of the figure, we have

$$\Phi_2 = (y - Y_1)\Phi_1. \quad (8.5)$$

There is therefore at the point 1 on the curve θ_1 . We are concerned with $I_1(x, y) + I_2(x, y)$. Now since

$$\log[y - Y_2(x)] - \log[y - Y_1(x)] = -2\pi i, \quad (8.6)$$

$I_2(x, y) + I_1(x, y) - \log(\zeta_1 - x)$ is a holomorphic function in the neighbourhood of $x \in \Delta'$.

For each domain of the form $[\Delta_n \cap (C')]$, $n = 1, 2, \dots$, we now consider the function

$$F(x, y) = \prod_j \Phi_j \cdot e^{-I_j} \prod_k (\zeta_k - x),$$

where the summation $\sum$ extends over all the arcs δ_j and the product $\prod$ is taken over all the points of division. By what we have seen above, we find that $F(x, y)$ is holomorphic on a neighbourhood of the dicylinder $[(C), (C')]$ having the property that $F(x, y)/\Phi_j$ on Δ_n is holomorphic and non-zero in the neighbourhood of each Δ_n .

Let us consider the general case. One finds without essential difficulty that we can construct the function $F(x, y)$ by the same method and that this function has the same property.

We also remark that while we have chosen a dicylinder among all dicylindrical domains (A, A') , this is just for simplicity; the essential condition is that A' be simply connected.

3. The Main Estimate

We now assert the following: Consider circles (C) and (Γ) in the x -plane, (Γ) being interior to (C) and concentric circles (C') , (C'') in the y -plane, (C'') being interior to (C') . Let Σ be a characteristic surface defined in a neighbourhood of the dicylinder $[(C), (C')]$.

With this geometric configuration, suppose that the area of Σ is restricted as follows:

- 1°. If (C'', C') is the annulus in the y -plane between C' and C'' , the area of Σ in the dicylindrical domain $[(C), (C'', C')]$ is smaller than Ω .
- 2°. The same holds for the area in $[(\Gamma_0), (C')]$, where (Γ_0) is a circle interior to (Γ) .

Then we can find a circumference C_1 of the form $|y| = R_1$ ($R'' < R_1 < R'$) having the following property: Let $\Re_x$ be the projection on the x -plane of the portion of Σ in $[(C), (C_1)]$, and $\Re_y$ the corresponding projection on the y -plane. Let β_i ($i = 1, 2, \dots, p$) be the curves forming part of the boundary of $\Re_y$ for which $|y| = R_1$, and let γ_i be the corresponding curves in the x -plane (multiplicity being taken into account). Let L be the total length of the curves γ_i and ℓ be that of the β_i . We then have:

$$L + \ell < K_1 \Omega, \quad K_1 = \frac{2\pi}{R' - R''}.$$

The circle (C) is then partitioned by the curves γ_i into several portions Δ_n where $\Re_x$ has n sheets. Let X_0 be a point

of Δ_n ; we write

$$n = \psi(X_0).$$

In fact, suppose that the circles (C) , (C') are given by $|x| < R$, $|y| < R'$ respectively, where $R < \frac{1}{2}$, $R' < \frac{1}{2}$, which is permitted. Suppose, in addition, that the characteristic surface Σ does not contain any characteristic plane of the form $x = \text{const.}$ or $y = \text{const.}$; this is also permissible.

Let $\Re_x$, $\Re_y$ be the RIEMANN surfaces over the x -, y -planes respectively corresponding to the portion of Σ in $[(C), (C''), (C')]$. If we denote their areas by A_x , A_y , we have

$$A_y = \iint_{\Re_y} \left| \frac{dx}{dy} \right|^2 d\omega, \quad A_x = \iint_{\Re_x} \left| \frac{dy}{dx} \right|^2 d\omega, \quad (8.7)$$

where $d\omega$ is the area element on the RIEMANN surface $\Re_x$ and $x(y)$ is the function defined by Σ . Now, since

$$1 + X^2 > 2X$$

for any positive number X , we have

$$A_x + A_y < \Omega,$$

and therefore

$$\iint_{\Re_y} \left(1 + \left| \frac{dx}{dy} \right|^2 \right) d\omega < \Omega. \quad (8.8)$$

Let us draw on Σ the following (one or several) curves:

$$|x| < R, \quad |y| = R'_k, \quad R'_1 < R'_2 < R',$$

R being the radius of C . Let us denote by γ_i , β_i the projections of these curves on the x - and y -planes respectively and their lengths by ℓ_i , ℓ'_i . It then follows from the above inequality that there exists a radius R_1 in the interval (R'_1, R') satisfying

$$\sum_i (\ell_i + \ell'_i) < \frac{K_1 \Omega}{2\pi R_1}. \quad (8.9)$$

Let (C_1) be the circle $|y| < R_1$.

We apply the method of the preceding section to the polycylinder $[(C), (C_1)]$ thus obtained to construct the function $F(x, y)$ associated to Σ . We continue to use the same notations.

We look first at the upper bound. We set

$$\log |F| = u = u_1 + u_2 + u_3,$$

where

$$u_1 = \log |\Phi_j|, \quad u_2 = \sum \log |\zeta_k - x|, \quad u_3 = \Re \left[- \sum I_j \right],$$

where the logarithms are real and the operation $\Re$ means that we take the real part. Since $R < \frac{1}{2}$, $R' < \frac{1}{2}$, we find at once that

$$u_1 < 0, \quad u_2 < 0.$$

Let R_1 be the radius of C_1 .

4. The Converse Proposition

We are concerned now with a certain converse of the preceding proposition. We shall prove the following:

Theorem 8.7. *Let $F(x, y)$ be a holomorphic function on the polycylinder $[(C), (C')]$ and having the following bounds:*

$$1^\circ. \log |F| < M,$$

$$2^\circ. m(\log |F|) > -K_1 \cdot M$$

on $[(C), (C')]$, K_1, M being positive constants. Then, for any given polycylinder $[(C_1), (C'_1)]$ in the interior of $[(C), (C')]$, the area a_1 of the characteristic surface Γ given by $F(x, y) = 0$ is restricted by the inequality

$$a_1 < K_2 \cdot M,$$

K_2 being a positive constant independent of M and of F .

1°. To verify this, we equip ourselves with the following auxiliary proposition (which one could also reduce to the JENSEN formula).

Let (C) , (C_1) be the circles $|z| < R$, $|z| < R_1$ respectively where $R_1 < R$. Let $f(z)$ be a holomorphic function on (C) , bounded in such a way that $|f(z)| < e^M$, and having at least n zeros on the closed circle (C_1) ; we then necessarily have

$$|f| < M\lambda^n$$

on (C_1) , λ being a positive constant independent of M and of f such that $0 < \lambda < 1$.

To check this, we may suppose that $R = 1$. We denote R_1 by θ .

Suppose first that $f(z)$ is zero at the point a on (C_1) . If $a = 0$, we have, by SCHWARZ's lemma,

$$|f(z)| < e^M \theta$$

on (C_1) . Suppose, therefore, that $a \neq 0$. We can suppose that a is real and positive. Consider the well-known transformation

$$z' = \frac{z - a}{ab - z} \quad (ab = 1),$$

which leaves the circle (C) invariant. Let

$$\ell = \max |z'| \quad \text{for } |z| < \theta.$$

We see that

$$0 < \ell < 1, \quad \theta < 1.$$

By virtue of SCHWARZ, we have

$$|f| < M\ell$$

on (C_1) .

Suppose next that f is zero at a, b . We can suppose that $a \neq 0$ and that a is real and positive. Consider the function

$$\varphi = \frac{f}{z}.$$

By the preceding result, we have

$$|\varphi| < \sqrt{M} \ell$$

on (C_1) . We therefore have

$$|f| = |\varphi| \cdot |z| < \sqrt{M} \ell \cdot \theta = M\lambda^2.$$

Continuing in this way, we arrive at the result indicated.

2°. Under the circumstances explained at the beginning of this section, suppose that Γ does not contain any piece of characteristic surface of the form $x = \text{const.}$ or $y = \text{const.}$ Suppose that (C) , (C_1) , (C') , (C'_1) are given by $|x| < R$, $|x| < R_1$ ($R_1 < R$), $|y| < R'$, $|y| < R'_1$ ($R'_1 < R'$) respectively and that $R < \frac{1}{2}$, $R' < \frac{1}{2}$. Let $\mathfrak{R}_x$, $\mathfrak{R}_y$ be the projections of Γ on the x - and y -planes respectively.

Let us use the same notation as in No. 2, except that we replace $[(C), (C')]$ by $[(C_1), (C'_1)]$. We have

$$\sum_p \int_p j' du = \Omega_j.$$

Let X_0 be a point of Δ_p ; we have p' roots of $F(x_0, y) = 0$ in (C'_1) . By the preliminary proposition, we have

$$|F(x_0, y)| < e^M \lambda^{p'}$$

for $x_0 \in \Delta_p$, $|y| < R'_1$. Hence

$$\int_{(C_1)} \log |F(x, y)| du < M \cdot \pi R_1^2 - \log \frac{1}{\lambda} \sum \Omega_j,$$

$$m(\log |F(x, y)|) < M - \frac{\log \frac{1}{\lambda}}{\pi R_1^2} \sum \Omega_j.$$

Now $m(\log |F(x, y)|) > -K_1 M$, therefore

$$\log \frac{1}{\lambda} \cdot \sum \Omega_j \leq \pi R_1^2 (K_1 + 1) \cdot M = K'_2 M.$$

3°. Let us consider the general case in which Γ can be expressed in the form

$$\Gamma = \Gamma_1 + \Gamma_2 + \Gamma_3,$$

where Γ_2 consists of pieces of characteristic planes of the form $x = \text{const.}$, Γ_3 of pieces $y = \text{const.}$, and Γ_1 is the rest.

Let us estimate the area Ω_2 of the portion of Γ_2 in the dicylinder $[(C_1), (C'_1)]$. Let m' be the number of planes of the form $x = \text{const.}$, multiplicity being taken into account; the area Ω_2 is then

$$\Omega_2 = m' \pi R_1'^2.$$

Now, by the preliminary proposition,

$$|F| < e^M \lambda^{m'}$$

on $[(C_1), (C'_1)]$. Hence

$$-K_1 M < M - m' \log \frac{1}{\lambda}, \quad \text{therefore} \quad m' < \frac{(K_1 + 1)M}{\log \frac{1}{\lambda}}.$$

The area Ω_2 is therefore bounded in the way indicated. The same holds for Ω_3 .

For the area Ω_1 , consider the function

$$\Phi(x, y) = F(x, y) / \prod_j (x - a_j) \prod_k (y - b_k),$$

where $x = a_j$ ($j = 1, 2, \dots, m'$) and $y = b_k$ ($k = 1, 2, \dots, n$) represent roots of the equation $F(x, y) = 0$, multiplicity being taken into account. We then have, by the maximum modulus principle,

$$\log |\Phi| < M + m' \log \frac{R}{R_1} + n \log \frac{R'}{R'_1} = K' M,$$

K' being a positive constant.

We still have

$$m(\log |\Phi|) > -K_1 M.$$

By the preceding case, we therefore know that the area Ω is also bounded in the way indicated above, and consequently, we have

$$\Omega < K_2 \cdot M. \quad \text{Q.E.D.}$$

5. A Lemma on the Hartogs Figure

Let us draw two circles (C) and (Γ_0) of the form $|x| < R$, $|x| < R_0$, $R > R_0$, in the x -plane, and two circles (C') , (C'') of the form $|y| < R'$, $|y| < R''$, $R' > R''$ in the y -plane. Let us consider a characteristic surface Σ in the neighbourhood of $[(C), (C')]$ (not necessarily irreducible). We consider the case when Σ does not contain any characteristic plane of the form $x = \text{const.}$ or $y = \text{const.}$. We suppose that the area of Σ in $[(C), (C'', C')]$ is smaller than Ω , as is the area of Σ in $[(\Gamma_0), (C')]$.

Let C be a given circumference of the form $|y| = R$, $R'' < R < R'$. As we saw in No. 3, we find another circumference C_1 of the form $|y| = R_1$, $R < R_1 < R'$ having the following property: Let $\mathfrak{R}_x$ be the projection on the x -plane of the portion of Σ in $[(C), (C_1)]$, and $\mathfrak{R}_y$ the corresponding projection on the y -plane. Let β_i , $i = 1, 2, \dots, p$, be the curves forming part of the boundary of $\mathfrak{R}_y$ for which $|y| = R$, and let γ_i be the corresponding curves in the x -plane (multiplicity being taken into account). Let L be the total length of the curves γ_i and ℓ be that of the β_i . We then have:

$$L + \ell < K_1 \Omega, \quad K_1 = \frac{2\pi}{R' - R''}.$$

The circle (C) is then partitioned by the curves γ_i into several portions Δ_n where $\mathfrak{R}_x$ has n sheets. Let X_0 be a point of Δ_n ; we write

$$n = \psi(X_0).$$

Consider a circle (Γ) of the form $|x| < R_1$ ($R_0 > R_1$) and the curves γ_i . The curves γ_i are analytic, and its elements are regular except perhaps at finitely many points.

6. The Main Theorem

Let us consider a univalent domain D in the space (x, y) ; we shall suppose that D has no points at infinity to simplify the language. We consider a sequence

$$(S) \quad L_1, L_2, \dots, L_n, \dots$$

of characteristic surfaces in D and a sequence

$$(T) \quad V_1, V_2, \dots, V_n, \dots$$

of positive numbers. The sequence (T) is arbitrary, but we shall regard it as being unbounded, in general.

Using these sequences (S) and (T) , we distinguish between two kinds of points in D .

- 1°. If, for a point P_0 , there exists a dicylinder (γ) with centre P_0 such that the sequence

$$\frac{\Omega_1}{V_1}, \frac{\Omega_2}{V_2}, \dots, \frac{\Omega_n}{V_n}, \dots, \quad (8.10)$$

where Ω_n is the area of L_n in (γ) , is bounded, we shall call P_0 a point of the first kind.

- 2°. If there exists no dicylinder (γ) having this property, we shall call P_0 a point of the second kind.

We shall see that the following result holds:

Theorem 8.8. *The set of points of the first kind is pseudo-convex in the domain D .*

In fact, it is sufficient to verify the continuity theorem (C) . Let E be the set of points of the second kind in D . E is clearly closed in D . Let us consider a dicylinder $[(C), (C')]$ in D . Let (Γ) be a circle interior to (C) and let (C'') be a circle concentric with (C') and interior to it. Suppose that there are no points of E in $[(\Gamma), (C')]$, nor in $[(C), (C''), (C')]$. It is sufficient to affirm that there are no points of E in $[(C), (C')]$.

Suppose, by contradiction, that there exists a point (ξ, η) of E in $[(C), (C')]$. Let us consider a dicylinder $[(C_1), (C'_1)]$ with centre (ξ, η) and contained in $[(C), (C')]$. By the lemma, if Ω'_n is the area of L_n in $[(C_2), (C'_1)]$, we have

$$\Omega'_n < K \max(\Omega_n, \Omega''_n),$$

K being a positive constant independent of n . The sequence Ω'_n/V_n is therefore bounded. This contradicts (ξ, η) being a point of the second kind. $\square$

étant donné, on cherche, parmi les F associées à Σ , une F particulière qui satisfasse à certaines majorations. Cela donne lieu à une analyse assez subtile d'OKA (voir les n^{os} 2 et 3).

Venons-en au contenu du “lemme” du n^o5. Il concerne la “marmite de HARTOGS”, qui a joué un rôle essentiel dans l'étude de la “pseudo-convexité” des domaines (cf. Mémoires VI et IX). Rappelons que la “marmite” $M(r, r')$ du dicylindre A est la réunion des ouverts $|x| < r$, $|y| < 1$ et $|x| < 1$, $1 - r' < |y| < 1$ (avec $0 < r < 1$, $0 < r' < 1$). Le lemme en question affirme qu'il existe un nombre $K > 0$ jouissant de la propriété suivante : si Σ est un diviseur dans A , et si Ω désigne l'aire (supposée finie) de la portion de Σ située dans la marmite $M(r, r')$, alors l'aire totale de Σ dans A est finie et majorée par $K\Omega$.

Note on Families of Multivalued Analytic Functions etc.

NOTE SUR LES FAMILLES DE FONCTIONS ANALYTIQUES MULTIFORMES ETC.,

Journal of Science of the Hiroshima University (1934), p. 93–98.

1. Introduction

The aim of the researches whose outline and principal results will be announced briefly in what follows³ is to extend the series of remarkable results of Messrs. WEIERSTRASS, STIELTJES, VITALI and MONTEL in the classical theory of convergence of singlevalued analytic functions of one complex variable first to multivalued analytic functions, and then, beyond that. However, the second generalisation is possible only up to the theorem of Mr. STIELTJES⁴.

3. The detailed proofs will be published later in the same journal.

4. For the other theorems, one must restrict oneself to characteristic surfaces “without points of the second kind.”

I. Normal Families of Characteristic Surfaces

2. The Function Space

In treating an analytic function $f(x)$ of a complex variable x , in general multi-valued, it will be more convenient, and more general, to look at it in the following 4-dimensional space:

Let y be a second complex variable, and let $x = x_1 + ix_2$, $y = y_1 + iy_2$, $i = \sqrt{-1}$. With these x, y , we shall always consider a space with rectangular coordinates (x_1, x_2, y_1, y_2) , and we shall represent them by (x, y) to abbreviate the notation.

In a given domain Δ in the space (x, y) , the set of points satisfying the equation $y = f(x)$ consists, if we look at it in the space (x, y) , of a surface of two (real) dimensions; we call it a *characteristic surface*.

3. Definition of Normal Families

Let (F) be a family of characteristic surfaces without points of the second kind in a domain A in (x, y) space.

The family (F) will be *normal at a point* of the domain if we can find a hypersphere (ω) about this point and so small that we can find adjoint holomorphic functions $F(x, y)$ in the hypersphere for the characteristics of the family in such a way that the family of functions $F(x, y)$ is normal⁵ in (ω) , and, in addition, the constant zero is not among the limits of the family.

It should be remarked that if a point of A does not belong to the limit points of the family of surfaces, the family is normal at this point because of the definition.

The family (F) will be said to be *normal in the domain* A if it is normal at every point of the domain.

A point of the domain will be said to belong to the (J) -*points* of the family if the family is not normal at this point.

5. In the sense of MONTEL.

4. Adjoint Functions

Let S be a characteristic surface without points of the second kind in a spatial domain A . Corresponding to any dicylinder in the interior of A , we can form, by the classical method of Mr. COUSIN, a certain adjoint holomorphic function in such a way that one can estimate, from both sides, bounds for the mean of the modulus of the function by means of the area of the portion of S in the dicylinder.

II. Sets of Class (H)

5. Definition

A set E of points of the space (x, y) will be said to be of *class* (H) in the spatial domain A to which E belongs, if it has the following three properties:

- 1°. E is closed in A .
- 2°. Let (a, b) be a point of A which does not belong to E ; we can then find a dicylinder with centre (a, b) such that the portion of E in this dicylinder consists of a finite number of pieces of characteristic surfaces, each of which can be represented in the form $y = f(x)$, where $f(x)$ is holomorphic on the corresponding portion of the dicylinder.
- 3°. The above properties are preserved by any biunique analytic transformation of (x, y) -space.

More precisely, if a domain (ω) in Δ is subjected to a biunique analytic transformation, together with the part of E in (ω) , then the transformed set still has properties (1°, 2°) on the new domain.

The importance of the above properties was noted for the first time by Mr. HARTOGS when he studied the set of singular points of an analytic function of two complex variables, and these sets give us, with a little modification, examples of sets of class (H) . Other examples will be found in the memoirs of

E.E. LEVI⁶ and Mr. G. JULIA⁷. And characteristic surfaces without points of the second kind form the simplest example.

The theory of sets of class (H) was developed by these mathematicians and others; naturally, their results were not proved abstractly. However, one will find that almost all these results are established using just the three properties given in the definition; for example, the results of No. 25 to No. 44 of the memoir of Mr. JULIA cited above.

6. Correspondences

Given an arbitrary set E in (x, y) -space, to a point x_0 of the x -plane there correspond points $(x_0, y_0), (x_0, y_1), \dots$ on E , which, as a set, can be empty, countable, or uncountable.

III. Generalisations of a Theorem of Mr. Hartsogs

8. Capacity of Sets

We shall first make some remarks on the capacity of sets of points in the plane with respect to logarithmic potentials, which will play a fundamental role in what follows. The theory of capacities, which notion was introduced by Mr. H. LEBESGUE, was developed later⁸. However, almost all research on the subject is devoted to three dimensions, and some words are necessary to give the basic notions in our case. I shall first give the definitions.

1°. The capacity of an open set $\mathcal{O}$ contained in a circle of radius smaller than $\frac{1}{2}$ is the upper bound of positive masses which one can distribute in the interior of $\mathcal{O}$ without the corresponding logarithmic potentials exceeding 1.

6. *Annali di Matematica pura ed applicata* (3) **17** (1910).

7. *Journal de Mathématiques pures et appliquées* (8) **8** (1928).

8. See, for example, O.D. KELLOGG, *Foundations of Potential Theory*, p. 150 ff.

That is to say, the function $\log d(x)$ is subharmonic in the sense of Mr. F. RIESZ⁹; however, in the present case, these functions can become “very infinite.”

From this lemma, one obtains the following theorems, which may be regarded as generalisations of a well-known theorem of Mr. HARTOGS.

Theorem 8.9. *If, for every point x in a set of non-zero capacity in the domain D , the set $\sigma(x)$ contains only a finite number of points of the y -plane (which can change with x), the same is true for every point of the domain D , and the points are such that the section $\sigma(x)$ consists of a finite number of algebroid functions.*

Theorem 8.10. *If, for every point x in a set of non-zero capacity in the domain D , the set $\sigma(x)$ contains only a countable set of points of the y -plane (which can change with x), the same is true for every point of the domain D ; the precise form is as follows:*

- 1°. *One can find a transfinite ordinal α of class at most II, independent of x , such that $\Phi^\alpha(x) = \emptyset$, where $\Phi^\alpha(x)$ represents the derived set of order α of the set of points $\sigma(x)$, x being supposed fixed.*
- 2°. *One can entirely cover the set E by a countable number of characteristic surfaces S which are respectively defined in hyperspheres (ω) and which have no points of the second kind in (ω) .*

9. *Acta Mathematica* **48** (1926), p. 329–343.

On Pseudoconvex Domains

SUR LES DOMAINES PSEUDOCONVEXES,
Proceedings of the Imperial Academy, Tokyo (1941), p. 7–10.

At the start of recent progress in the theory of analytic functions of several variables, F. HARTOGS discovered that every domain of holomorphy¹⁰ is a pseudoconvex domain. These two notions have become extremely important because of the development of the theory; however, the converse problem remains largely open even today. We shall treat this problem in the present Note, in which we shall place ourselves, for simplicity, in the space of 2 complex variables, but the conclusion applies, I think, to any number of variables.

1. Definitions

In the space of 2 complex variables x, y , let us consider a finite univalent domain D . We shall call it *pseudoconvex* if the set E complementary to D satisfies the continuity theorem in the neighbourhood of an arbitrary finite point P of E and if, in addition, this is true after any biunique pseudoconformal transformation in the neighbourhood of P . Here, the first condition means the following: if we draw a sufficiently small hypersphere around P and take, arbitrarily, a point (a, b)

10. A domain of holomorphy is a domain such that there exists a function holomorphic in the domain but not continuable beyond any boundary point.

in the hypersphere and a circumference of the form $x = a$, $|y - b| = r$, if (a, b) belongs to E without any point of the circumference doing so, one can find a positive number δ in such a way that to any x' in $|x - a| < \delta$, there corresponds at least one y' in $|y - b| < r$ such that (x', y') belongs to E .

Then, we have the

Theorem 8.11. *In the space of 2 complex variables, every finite, univalent, pseudoconvex domain is a domain of holomorphy.*

2. Construction

Let D be the given pseudoconvex domain. We are going to construct a function $X_j(x, y)$ ($j = 1, 2, \dots, \nu$) holomorphic in a subdomain D_3 of D , satisfying the following conditions:

- 1°. Suppose that the set of points of D_3 satisfying $|X_j(x, y)| < 1$ ($j = 1, 2, \dots, \nu$) does not have points in the neighbourhood of the intersection of the boundary of D with L .
- 2°. Suppose that for a sufficiently small positive number ε and for each j from $1, 2, \dots, \nu$, the set of points of D_3 satisfying $|X_j(x, y)| > 1 - \varepsilon$ does not have points in the neighbourhood of either L_1 or of L_2 .
- 3°. The points of D which either do not belong to D_3 or satisfy $|X_j(x, y)| < 1$ ($j = 1, 2, \dots, \nu$) constitute an open set because of the preceding hypothesis. Suppose that this set has a connected compound L_1 extending from the left of L_2 to the right of L_1 ¹¹.
- 4°. Suppose that we have, identically,

$$(X_j - X_j^0)R = (x - x_0)\Pi + (y - y_0)Q_j \quad (j = 1, 2, \dots, \nu)$$

when $(x, y) \in D_3$, $(x_0, y_0) \in D_3$, where X_j^0 stands for $X_j(x_0, y_0)$, Π , Q_j and R are holomorphic functions of

11. The precise meaning of this condition will be clear from the context.

the variables x, y, x_0, y_0 and R reduces to 1 for $x = x_0, y = y_0$.¹²

- 5°. Suppose that for each X_j , the number of points in the interior of D_3 satisfying

$$\frac{\partial X_j}{\partial y} = 0, \quad X_1 = a, \quad |X_j(x, y)| = 1$$

is at most finite.

- 6°. We shall denote the analytic variety $X_1 = a, |X_j(x, y)| = 1$ defined in D_3 by L_j .

Because of the preceding hypothesis, L_j is of dimension at most 2. Suppose that all intersections of the varieties L_j, L_k ($j \neq k$) are of dimension at most 1.

The domain Δ constructed thus is pseudoconvex. We shall denote by Δ_{11}, Δ_{12} and Δ_{13} respectively the parts of Δ to the left of L_1 , to the right of L_2 , and between L_1 and L_2 . Every connected component of any of the sets $\Delta_{11}, \Delta_{12}, \Delta_{13}$ is a domain of holomorphy.

3. The Integral Representation

Let S be the set of points consisting of the part of L in Δ_1 and its points of the boundary of D . Under these circumstances, let us consider the double integral taken over the two dimensional part of L :

$$I_j(x_0, y_0) = \frac{-1}{4\pi^2} \iint_L \psi_j(x, y; x_0, y_0) \varphi(x, y) dx dy \quad (j = 1, 2, \dots, \nu),$$

$$\psi_j = \frac{1}{(x - x_0)(X_j - X_j^0) - Q_j},^{13}$$

$\varphi(x, y)$ being an arbitrary holomorphic function on a certain open set containing S .

12. This condition ensures the functional independence of the X_j .

13. The precise form of the kernel depends on the geometric configuration.

When $(x, y) \in L$, ψ_j is holomorphic in $(x_0, y_0) \in \Delta_{13}$ except on L ; the same is therefore true of $I(x_0, y_0)$. Let $I_1(x_0, y_0)$ be the part of $I(x_0, y_0)$ to the left of L , and I_2 that on the right. We find easily, because of hypothesis 6, that I_1 and I_2 can be analytically continued a little across L in Δ_{13} , in such a way that $I_1 - I_2 = \varphi$.

We are going to modify ψ_j . Let L'_j be the part of L_j on the set $|X_p(x, y)| < 1$ ($p = 1, 2, \dots, \nu$; $p \neq j$); L'_j contains L_j . We construct an open set containing L'_j , sufficiently close to L'_j , and such that all its connected components are domains of holomorphy. $\mathcal{U}_j$ certainly exists because L'_j is contained in D_3 because of hypothesis 1.

Since the first COUSIN problem is always solvable on a univalent, finite, domain of holomorphy¹⁴, we can find, because of hypothesis 2, a meromorphic function $\Phi_j(x, y; x_0, y_0)$ in $(x, y) \in \mathcal{U}_j$, $(x_0, y_0) \in D_1$ in such a way as to have the same poles as ψ_j when $(x_0, y_0) \in D_3$ and be holomorphic when $(x_0, y_0) \notin D_3$.

$\Phi_j - \psi_j$ is holomorphic in $(x, y) \in \mathcal{U}_j$, $(x_0, y_0) \in D_3$. D_3 being convex with respect to functions holomorphic on D_1 , we can find, for a given positive number ε , a holomorphic function $\chi_j(x, y; x_0, y_0)$ in $(x, y) \in \mathcal{U}_j$, $(x_0, y_0) \in D_1$, such that $|\Phi_j - \psi_j - \chi_j| < \varepsilon$ on $(x, y) \in \mathcal{V}_1$, $(x_0, y_0) \in D$, where $\mathcal{V}_1$ is a suitable neighbourhood.

4. Fredholm Equation

Let us set

$$A_j = \Phi_j - \psi_j - \chi_j \quad \text{on } \mathcal{V}_1, \quad B_j = \Phi_j - \psi_j - \chi'_j \quad \text{on } \mathcal{V}_2.$$

We consider the following integrals instead of $I(x_0, y_0)$:

$$J_1(x_0, y_0) = \frac{-1}{4\pi^2} \iint_L (\psi_j + A_j) \varphi(x, y) dx dy,$$

$$J_2(x_0, y_0) = \frac{-1}{4\pi^2} \iint_L (\psi_j + B_j) \varphi(x, y) dx dy \quad (j = 1, 2, \dots, \nu).$$

14. This is a well-known result.

$J_1(x_0, y_0)$ is holomorphic on A_1 since this is the case for $\psi_j + A_j$ when $(x, y) \in \mathcal{U}_j$; and similarly $J_2(x_0, y_0)$ is holomorphic on A_2 . These functions can be continued analytically a little across L in A , and there satisfy the relation

$$J_1(x_0, y_0) - J_2(x_0, y_0) = \varphi(x_0, y_0) - \frac{1}{4\pi^2} \int_{\partial L} (A_j - B_j) \varphi(x, y) dx dy \quad (j = 1, 2, \dots, \nu).$$

Let us set $f = J_1 - J_2$; from this relation, it follows that f is holomorphic at every point of S .

We now consider f as given and φ as unknown in this relation, and we have a FREDHOLM integral equation of the second kind. Choosing ε sufficiently small, we therefore find the required $\varphi(x, y)$, holomorphic at every point of S .

The following assertion results from this:

Under the conditions of No. 2, given a function $f(x, y)$ single valued and holomorphic at every point of the common boundary S of the sets A_1 and A_2 , we can find 2 holomorphic functions $F_1(x, y)$ and $F_2(x, y)$ on A_1 and on A_2 respectively, such that we can continue them analytically a little across L in A , and such that we then have, identically $F_1(x, y) - F_2(x, y) = f(x, y)$.

This gives us our starting point.

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Additional Notes:

Note sur les familles de fonctions analytiques multiformes etc.

Journal of Science of the Hiroshima University **4** (1934), p. 93–98.

Sur les domaines pseudoconvexes

Proceedings of the Imperial Academy, Tokyo (1941), p. 7–10.

Added in Print

In 1980 Toshio NISHINO from Kyushu University, Japan, started editing Posthumous Papers of Kiyoshi OKA with the help of Akira TAKEUCHI, Kyoto University. To date seven volumes have appeared. Most of these papers were written in Japanese, however there are also some papers in French which we would like to list here:

1. **Fonctions algébriques permutables avec une fonction rationnelle non linéaire**
(Vol. 6; 87 pages; written about 1930)
2. **Sur les ensembles de points à 4 dimensions engendrés analytiquement**
(Vol. 7; 146 pages; written about 1934)
3. **Note sur les fonctions analytiques de plusieurs variables**
(Vol. 3; 7 pages; written about 1949)